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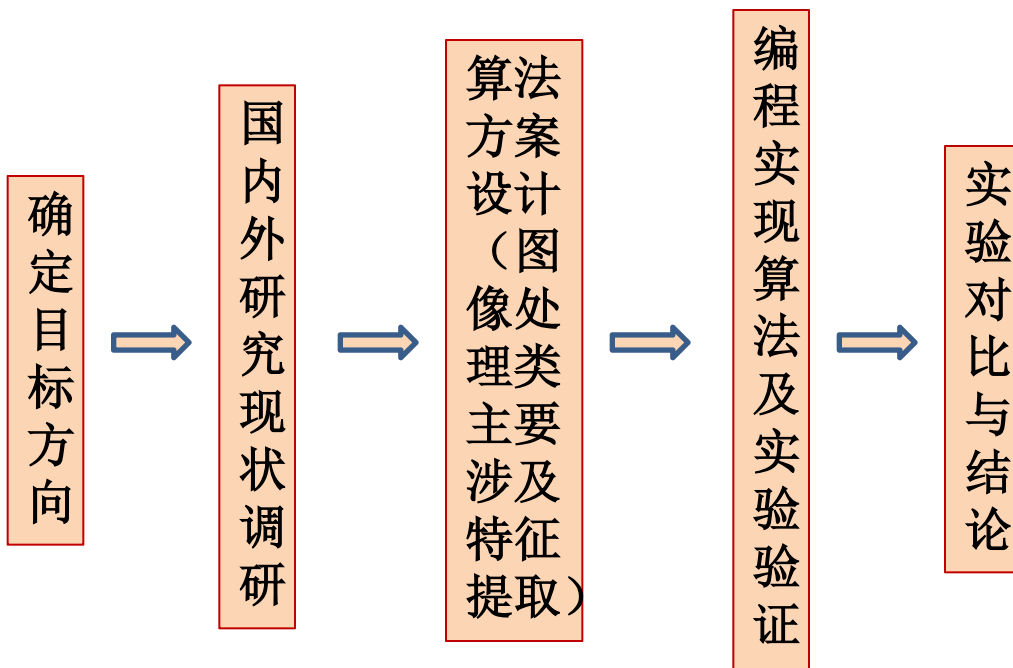


数字图像应用实例

- 数字图像取证
- 数字图像水印

天津大学电气自动化与信息工程学院

课程设计总体流程





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实例一

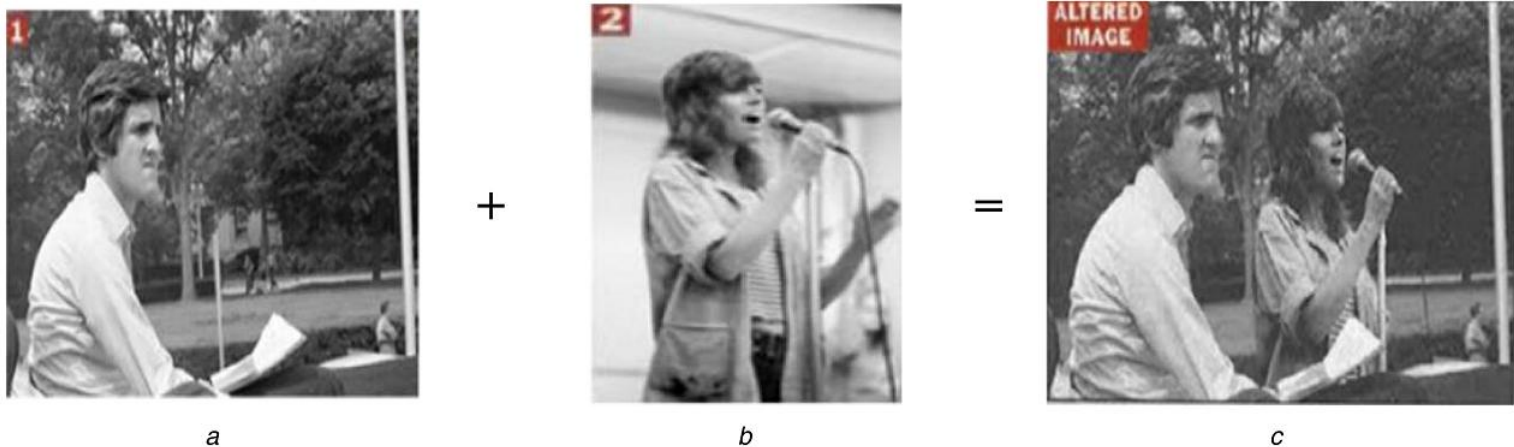
数字图像取证

(Digital Image Forensics/Image Forgery Detection)

图像篡改形式一: Splicing image forgery



Fig. 1 Two original authentic images and a spliced image
(a) Original image, (b) Original image, (c) Spliced image





IET Image Processing

Research Article

IET Journals
The Institution of
Engineering and Technology

Splicing image forgery detection using textural features based on the grey level co-occurrence matrices

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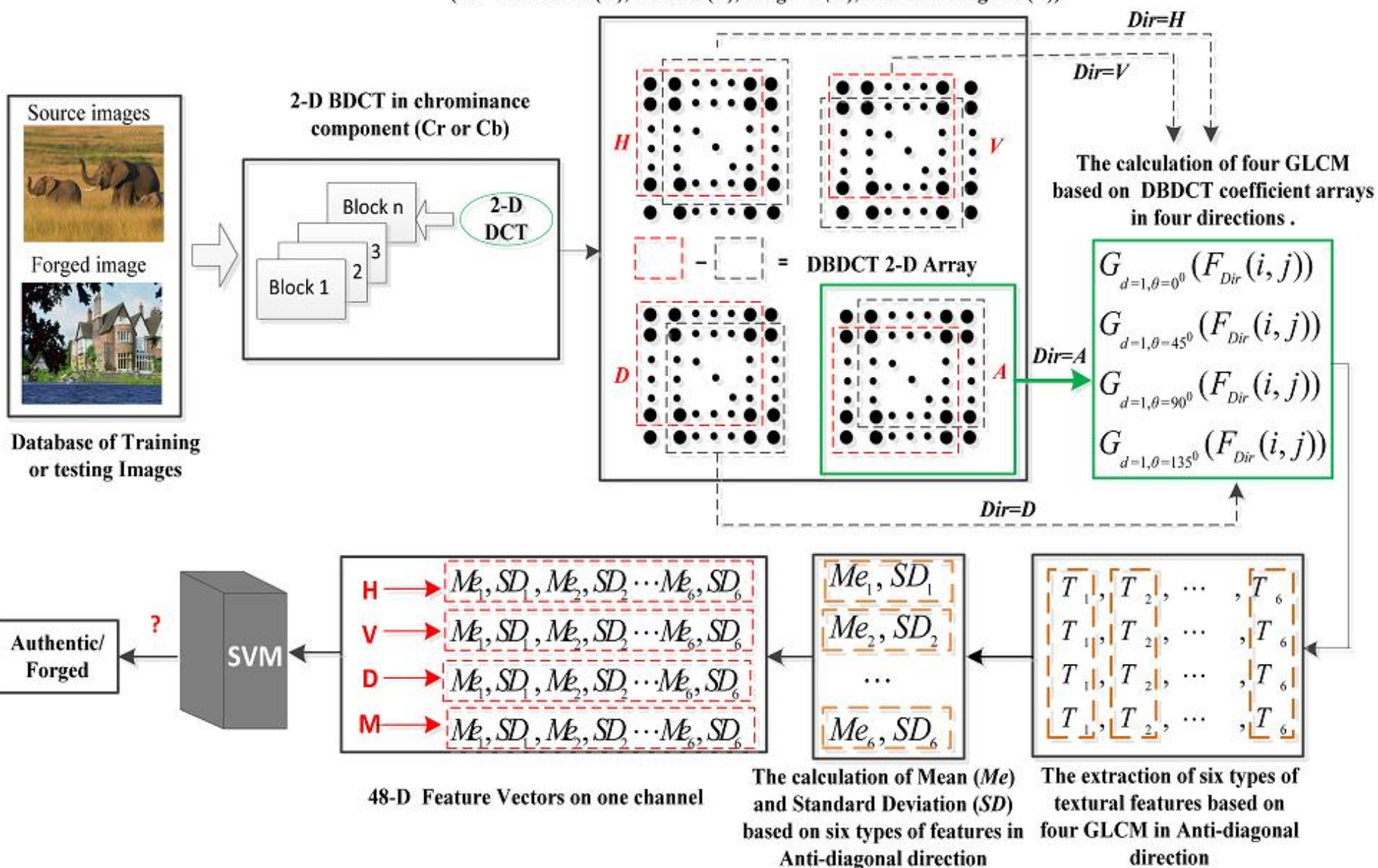
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Splicing image forgery detection 算法1



The calculation of DBDCT 2-D Array(denoted as F) in four directions ($Dir=Horizontal(H)$, Vertical(V), Diagonal(D), and Anti-diagonal(A)).



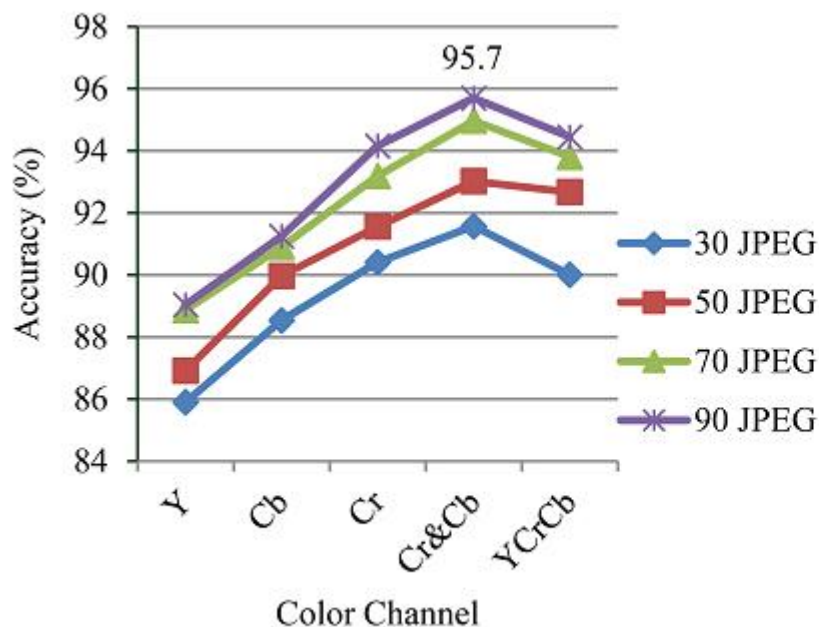
Splicing image forgery detection

算法1

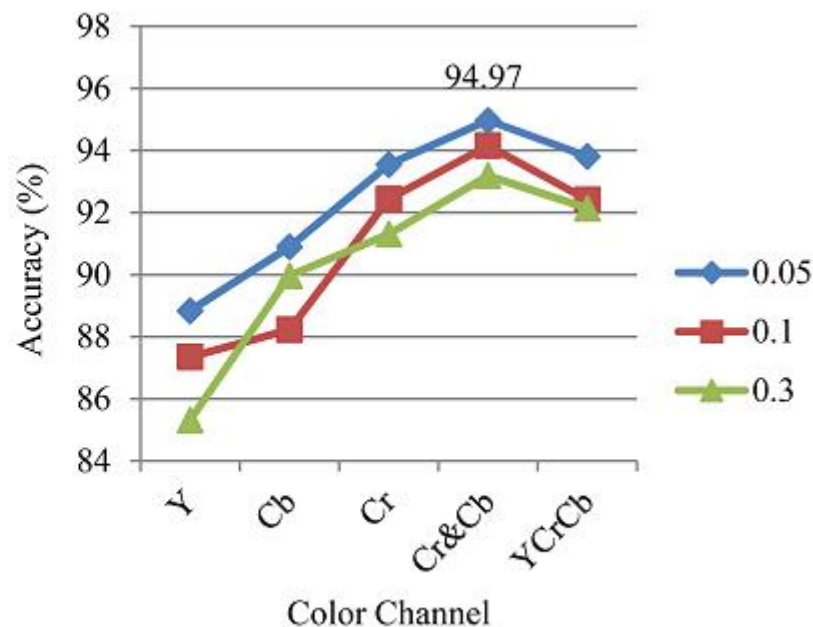


Table 4 Results of the comparison between the proposed and other algorithm on CASIA v1.0

Feature vector	Dimensionality	Accuracy, %
TF-GLCM	96	98.54
multi-resolution WLD [29]	640	93.33
SPT + LBP [9]	480	94.89
EF + LPQ [25]	256	95.41
GF + DCT [30]	300	96.13
DCT + LBP [8]	256	97



a



b

Fig. 9 Performance of proposed TF-GLCM

(a) Performance of the proposed TF-GLCM against JPEG compression, (b) Performance of the proposed TF-GLCM against Gaussian blur,

Illuminant-Based Transformed Spaces for Image Forensics

Tiago Carvalho, Fábio A. Faria, Hélio Pedrini, *Senior Member, IEEE*, Ricardo da S. Torres, *Member, IEEE*, and Anderson Rocha, *Senior Member, IEEE*

Abstract—In this paper, we explore transformed spaces, represented by image illuminant maps, to propose a methodology for selecting complementary forms of characterizing visual properties for an effective and automated detection of image forgeries. We combine statistical telltales provided by different image descriptors that explore color, shape, and texture features. We focus on detecting image forgeries containing people and present a method for locating the forgery, specifically, the face of a person in an image. Experiments performed on three different open-access data sets show the potential of the proposed method for pinpointing image forgeries containing people. In the two first data sets (DSO-1 and DSI-1), the proposed method achieved a classification accuracy of 94% and 84%, respectively, a remark-

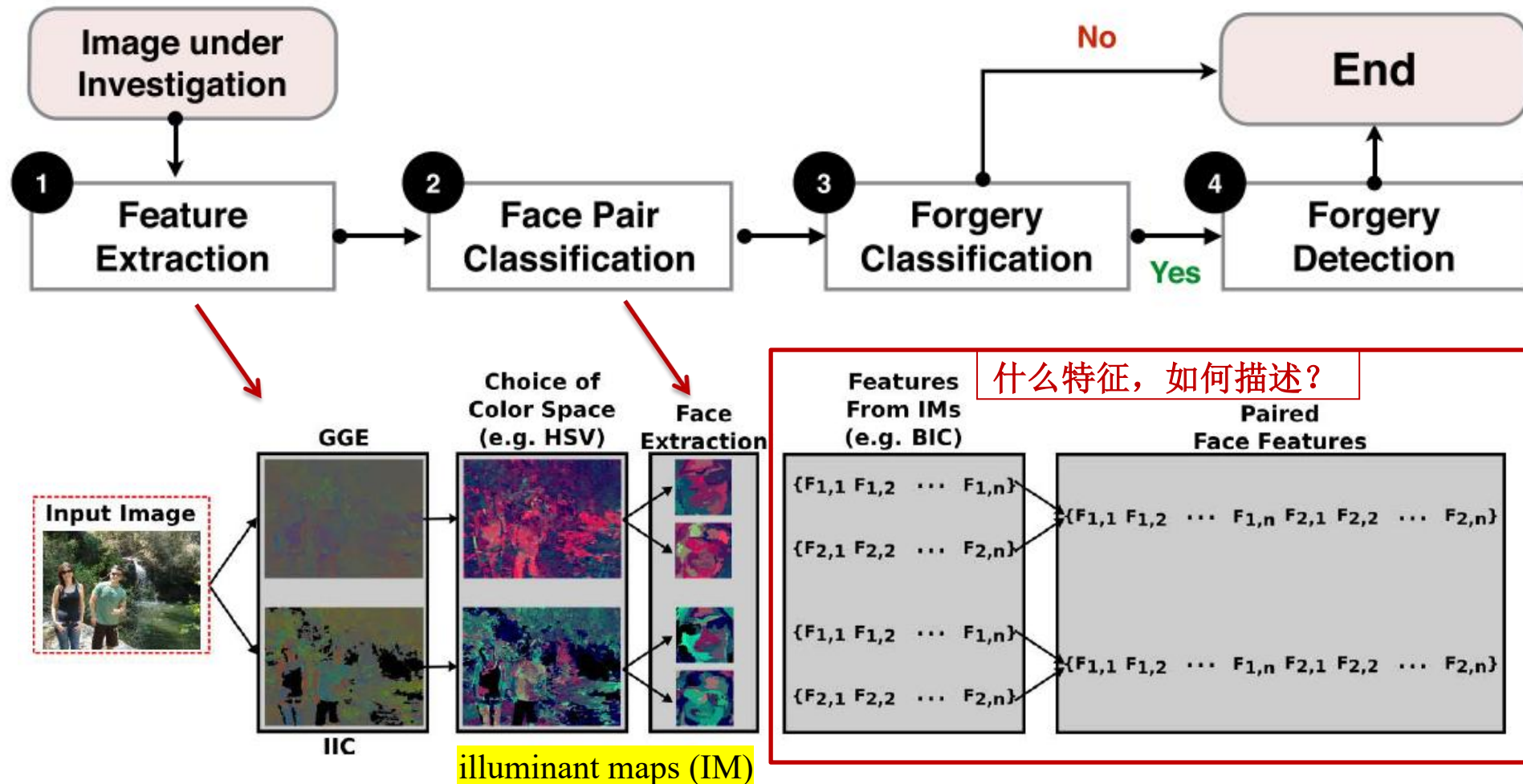


(a)



(b)

Fig. 1. Fake images created through splicing. (a) Conman Dimitri de Angelis (right) alongside US former president Bill Clinton (left). (b) Fake image of Brazil's former president (center) in a purported personal life moment with an investigated gang leader (left).



the generalized grayworld estimates algorithm (GGE) capturing **statistical-based** information; the inverse-intensity chromaticity space (IIC) capturing **physics-based** information

TABLE I

DIFFERENT DESCRIPTORS USED IN THIS WORK. EACH TABLE ROW REPRESENTS AN IMAGE DESCRIPTOR AND IT IS COMPOSED OF THE COMBINATION (TRIPLET) OF AN IM, A COLOR SPACE (ONTO WHICH IMs HAVE BEEN CONVERTED), AND THE DESCRIPTION TECHNIQUE USED TO EXTRACT THE DESIRED PROPERTY

IM Transf. Space	Color Space	Description Technique	Kind
GGE	Lab	ACC	Color
GGE	RGB	ACC	Color
GGE	YCbCr	ACC	Color
GGE	Lab	BIC	Color
GGE	RGB	BIC	Color
GGE	YCbCr	BIC	Color
GGE	Lab	CCV	Color
GGE	RGB	CCV	Color
GGE	YCbCr	CCV	Color
GGE	HSV	EOAC	Shape
GGE	Lab	EOAC	Shape
GGE	YCbCr	EOAC	Shape
GGE	HSV	LAS	Texture
GGE	Lab	LAS	Texture
GGE	YCbCr	LAS	Texture
GGE	Lab	LCH	Color
GGE	RGB	LCH	Color
GGE	YCbCr	LCH	Color

IM Transf. Space	Color Space	Description Technique	Kind
GGE	HSV	SASI	Texture
GGE	Lab	SASI	Texture
GGE	YCbCr	SASI	Texture
GGE	HSV	SPYTEC	Shape
GGE	Lab	SPYTEC	Shape
GGE	YCbCr	SPYTEC	Shape
GGE	HSV	UNSER	Texture
GGE	Lab	UNSER	Texture
GGE	YCbCr	UNSER	Texture
IIC	Lab	ACC	Color
IIC	RGB	ACC	Color
IIC	YCbCr	ACC	Color
IIC	Lab	BIC	Color
IIC	RGB	BIC	Color
IIC	YCbCr	BIC	Color
IIC	Lab	CCV	Color
IIC	RGB	CCV	Color
IIC	YCbCr	CCV	Color

IM Transf. Space	Color Space	Description Technique	Kind
IIC	HSV	EOAC	Shape
IIC	Lab	EOAC	Shape
IIC	YCbCr	EOAC	Shape
IIC	HSV	LAS	Texture
IIC	Lab	LAS	Texture
IIC	YCbCr	LAS	Texture
IIC	Lab	LCH	Color
IIC	RGB	LCH	Color
IIC	YCbCr	LCH	Color
IIC	HSV	SASI	Texture
IIC	Lab	SASI	Texture
IIC	YCbCr	SASI	Texture
IIC	HSV	SPYTEC	Shape
IIC	Lab	SPYTEC	Shape
IIC	YCbCr	SPYTEC	Shape
IIC	HSV	UNSER	Texture
IIC	Lab	UNSER	Texture
IIC	YCbCr	UNSER	Texture

IM Transformed Space, color space representation, and description technique

54 different image descriptors in total

How to automatically select the most important ones to keep and combine for an improved classification performance?

Splicing image forgery detection 算法 2

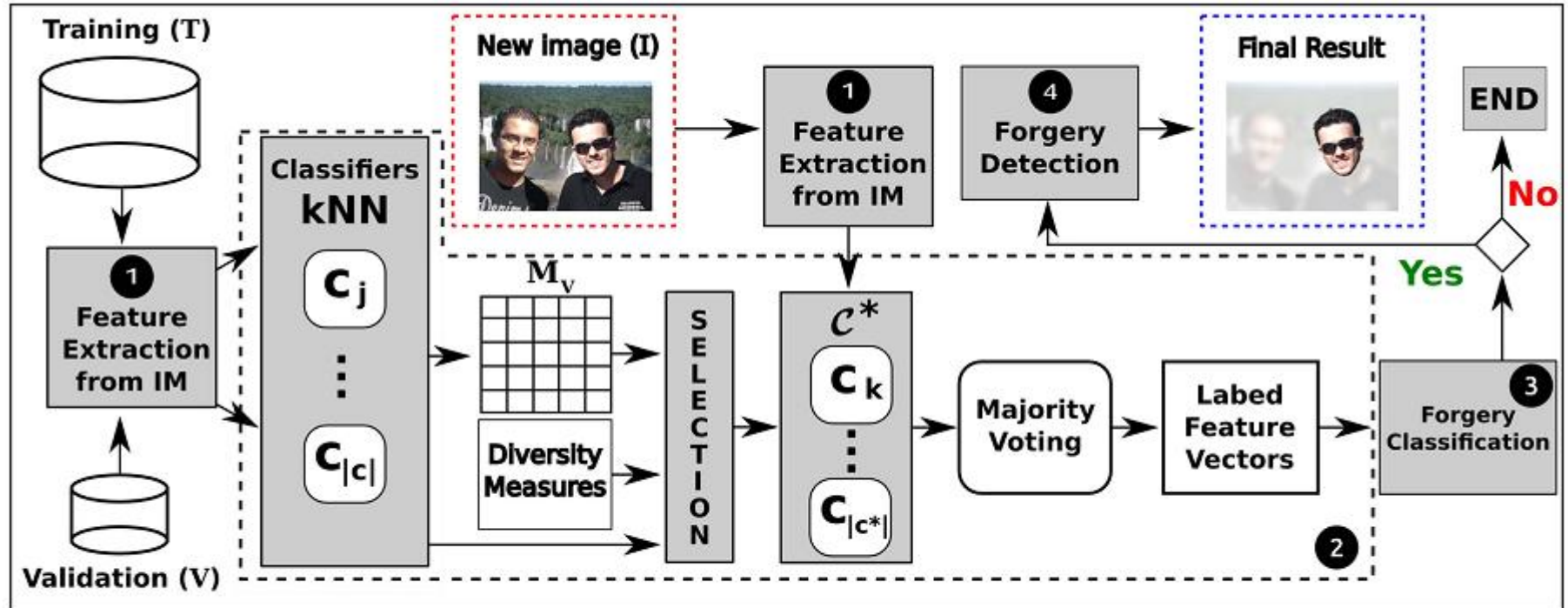


Fig. 4. Proposed framework for image splicing detection.

Initially, all classifiers $c_j \in \mathcal{C}$ are trained on the elements of a training set T . Next, the outcome of each classifier on the validation set V , different from T , is computed and stored into a matrix M_V , where $|M_V| = |V| \times |\mathcal{C}|$ and $|V|$ is the number of images in a validation set V and $|\mathcal{C}|$ is the number of classifiers. The actual training and validation data samples are known *a priori*.

In this selection process, five diversity measures (Correlation Coefficient ρ , Double-Fault Measure, Disagreement Measure, Interrater Agreement k , and Q-Statistic [32].³⁾ are computed to achieve the degree of agreement/disagreement between all available classifiers in $\mathcal{C}$. Finally, C^* , containing the most promising classifiers and satisfy a defined threshold $\mathcal{T}$, are

fake face定位:

$$fv_r = |f^{c_{IIC}} - f^{c_{GGE}}|$$

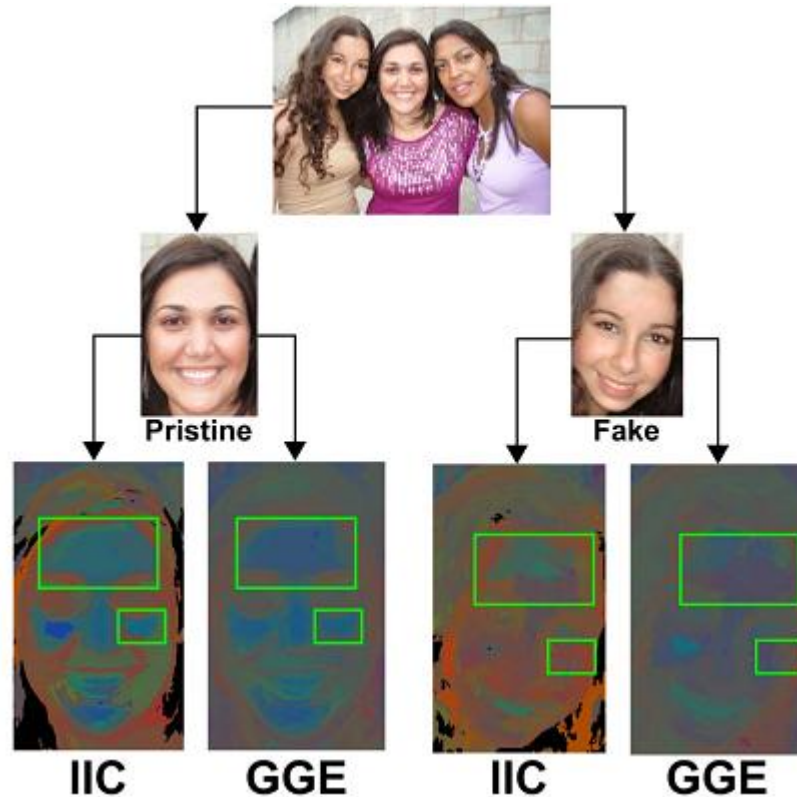


Fig. 5. Differences in ICC and GGE illuminant maps. The highlighted regions exemplify how the difference between ICC and GGE is increased on fake images. In special, the right cheek of the fake face varies from an orange color (in IIC) to a bluish color (in GGE). This is a drastic change wrt. a similar region of a normal face, which varies from an intensity of blue to another. This kind of variation can be captured through the difference between color descriptors, guiding us to a new way of identifying the most probable fake face.

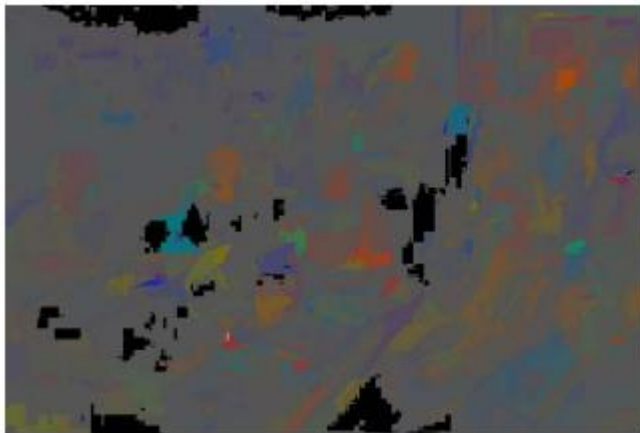


(a)

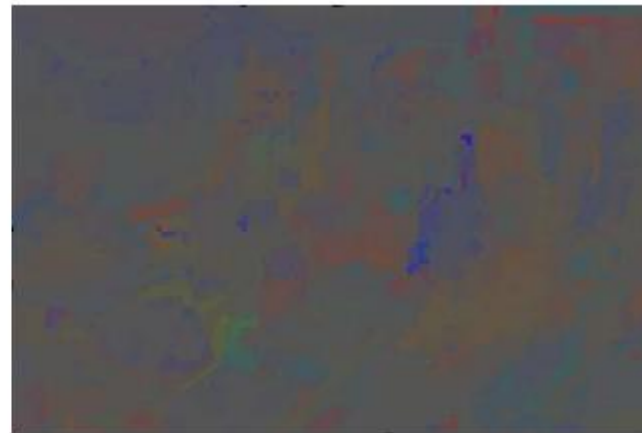


(b)

Fig. 14. The situation room images. (a) depicts the original image released by American government; (b) depicts one among many fake images broadcasted in the Internet. (a) Pristine. (b) Fake.



(a)



(b)

Fig. 15. IMs extracted from Figure 14(b). Successive JPEG compressions applied on the image make it almost impossible to detect a forgery by a visual analysis of IMs as proposed by Riess and Angelopoulou's method [6]. (a) IIC. (b) GGE.

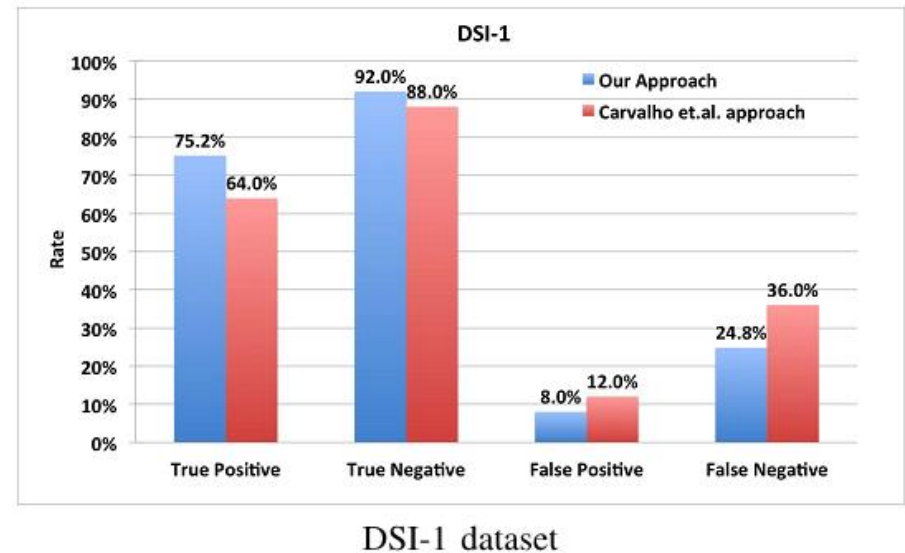
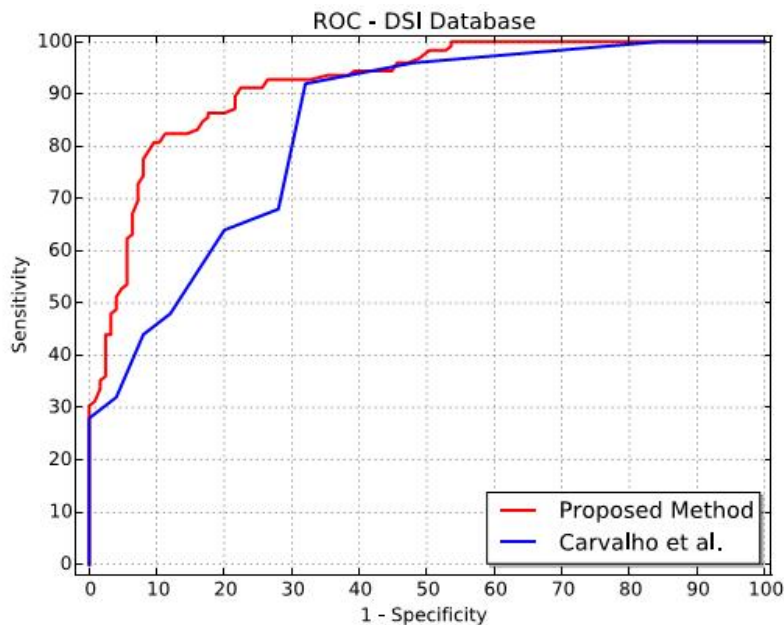


Fig. 12. Comparison between our approach and the one proposed by Carvalho et al. over the DSI-1 dataset.

Noise-Coded Illumination For Forensic And Photometric **Video** Analysis

算法3



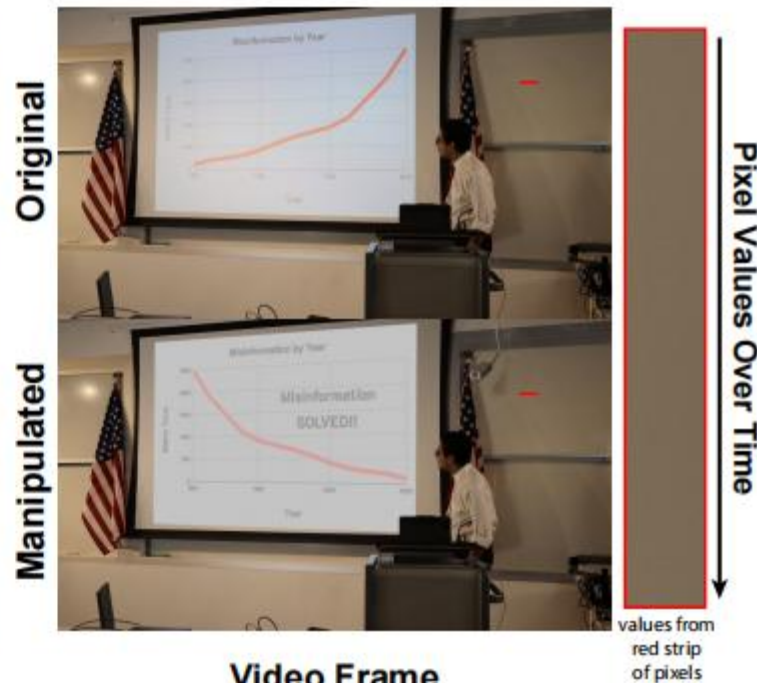
MICHAEL, ZEKUN HAO, SERGE BELONGIE, et al. Noise-Coded Illumination For Forensic And Photometric Video Analysis, <https://arxiv.org/abs/2507.23002v1>



Capture Setup

上图展示了一场讲座中放置编码光源的场景

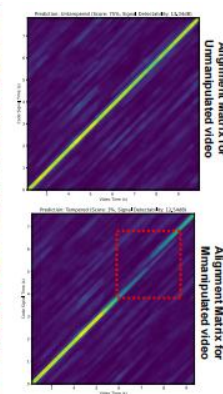
主动防御



Video Frame



Visualizing
Spatial Manipulation



Detecting
Temporal Manipulation

这是一种用于**光水印**的技术，能显著提升特定环境下视频时空操控的检测效率。

Original



Manipulated



Example Frame

Code Image 1

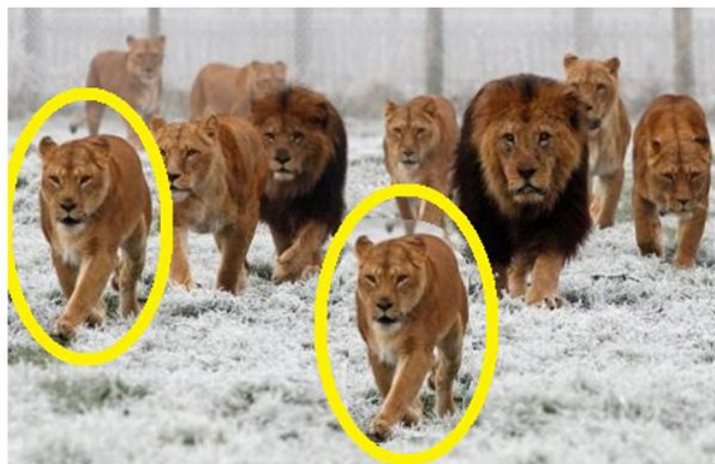
Code Image 2

光照代码承载着场景光照的视觉分解信息，即便攻击者破解了光照代码，仍难以伪造可信视频。这种隐写特性还为我们提供了时空维度的篡改检测手段。例如，我们可以精准定位视频中物体的插入或删除位置、剪辑间隔的时间长度，以及片段是否被加速、慢放或重新排序。这些细节不仅帮助我们判断是否存在篡改行为，更能揭示其潜在意图。

图像篡改形式二: copy-move forgery



a



b

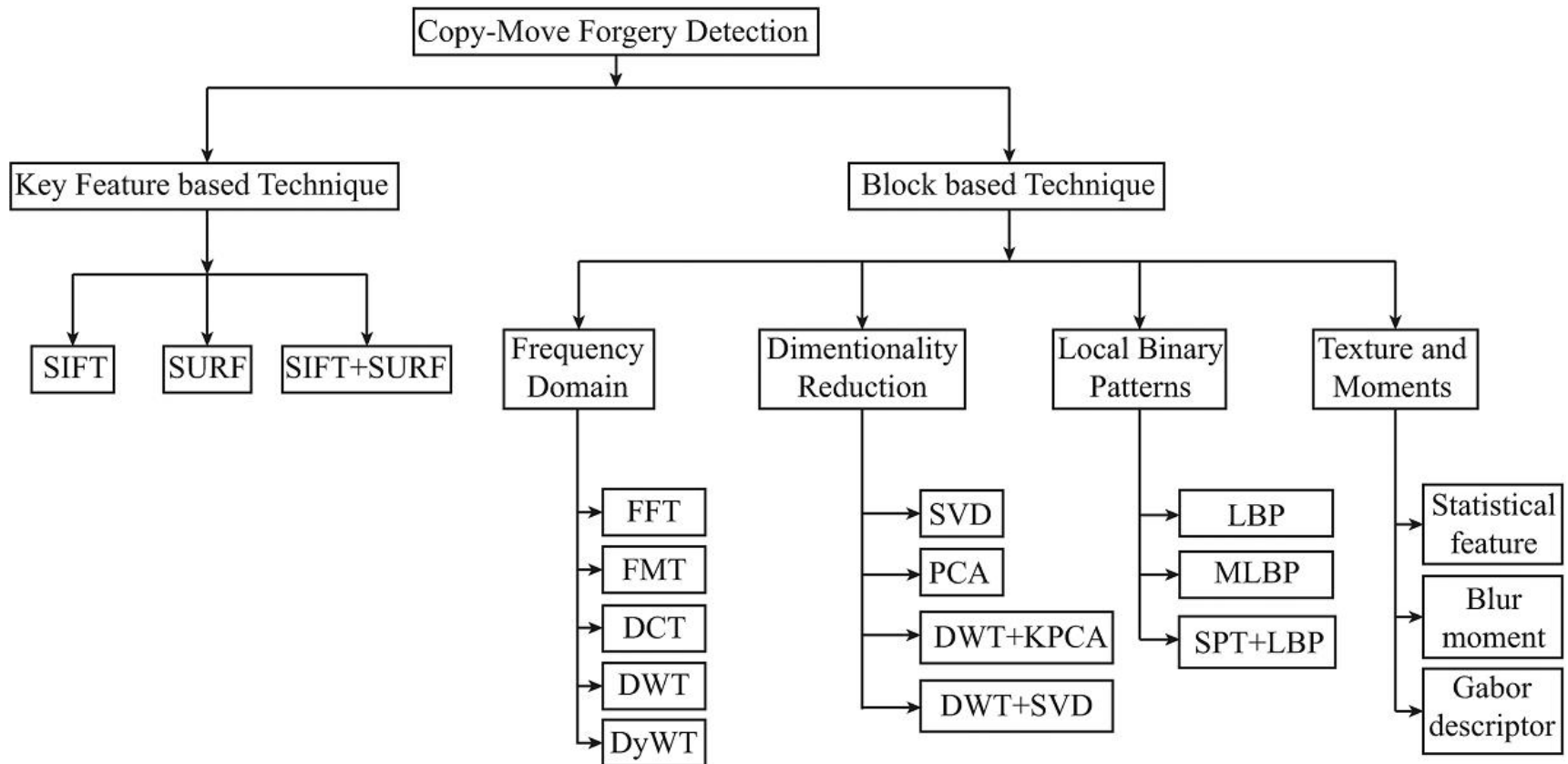
Fig. 1 *Example of copy-move forgery*

(a) Original image, (b) Forged image (duplicated object highlighted)



Fig. 1 *Example of image forgery (appeared in press The New York Times in July 2008)*

copy-move forgery detection



copy-move forgery detection

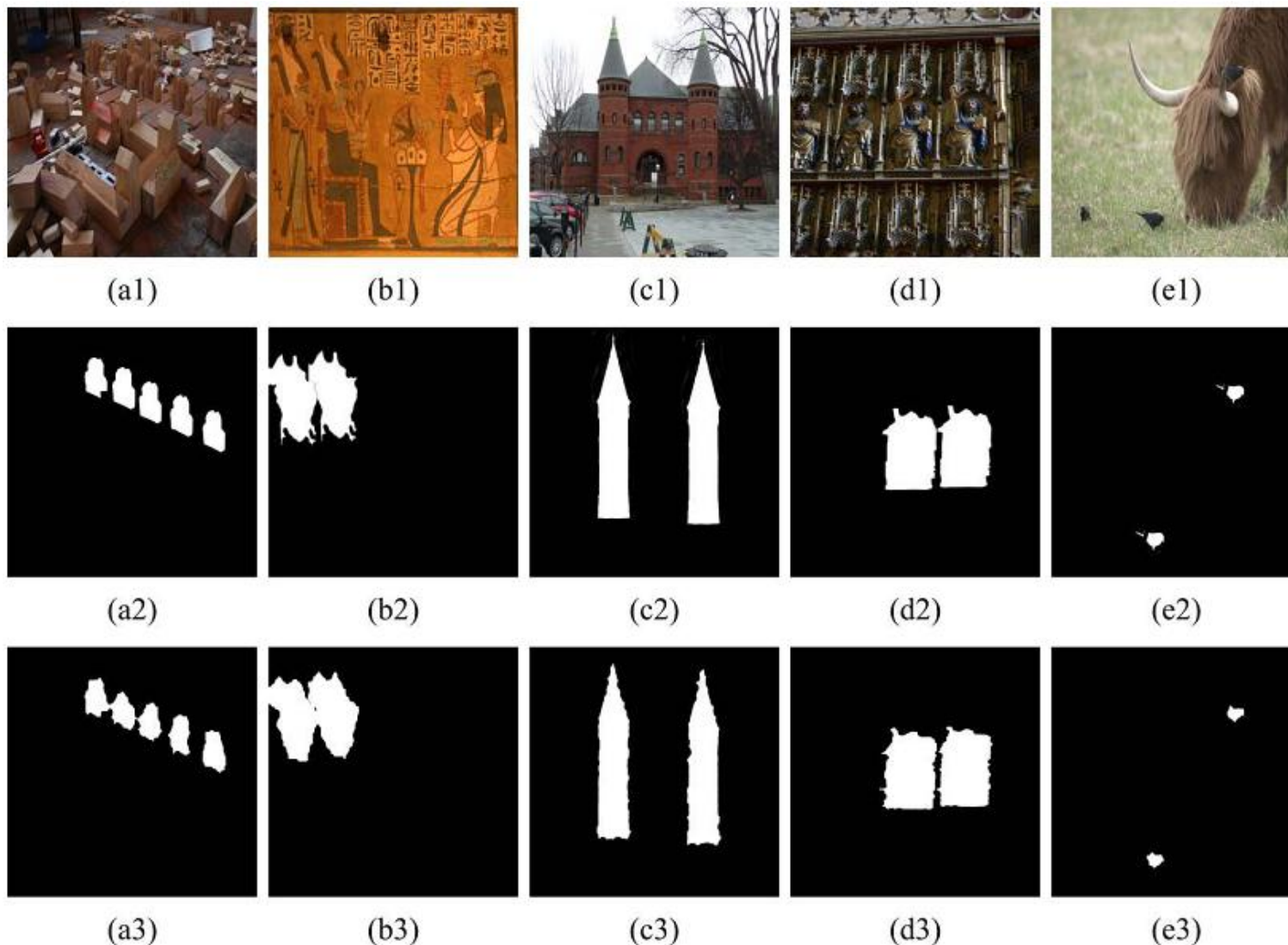


Fig. 7. The copy-move forgery detection results of the proposed scheme (a1) ~ (e1). The five host images from the dataset; (a2) ~ (e2). The ground-truth forgery regions of the corresponding host images; (a3) ~ (e3). The detected forgery regions of the corresponding images, using the proposed forgery detection scheme.

图像篡改形式三: Fake Colorized Image

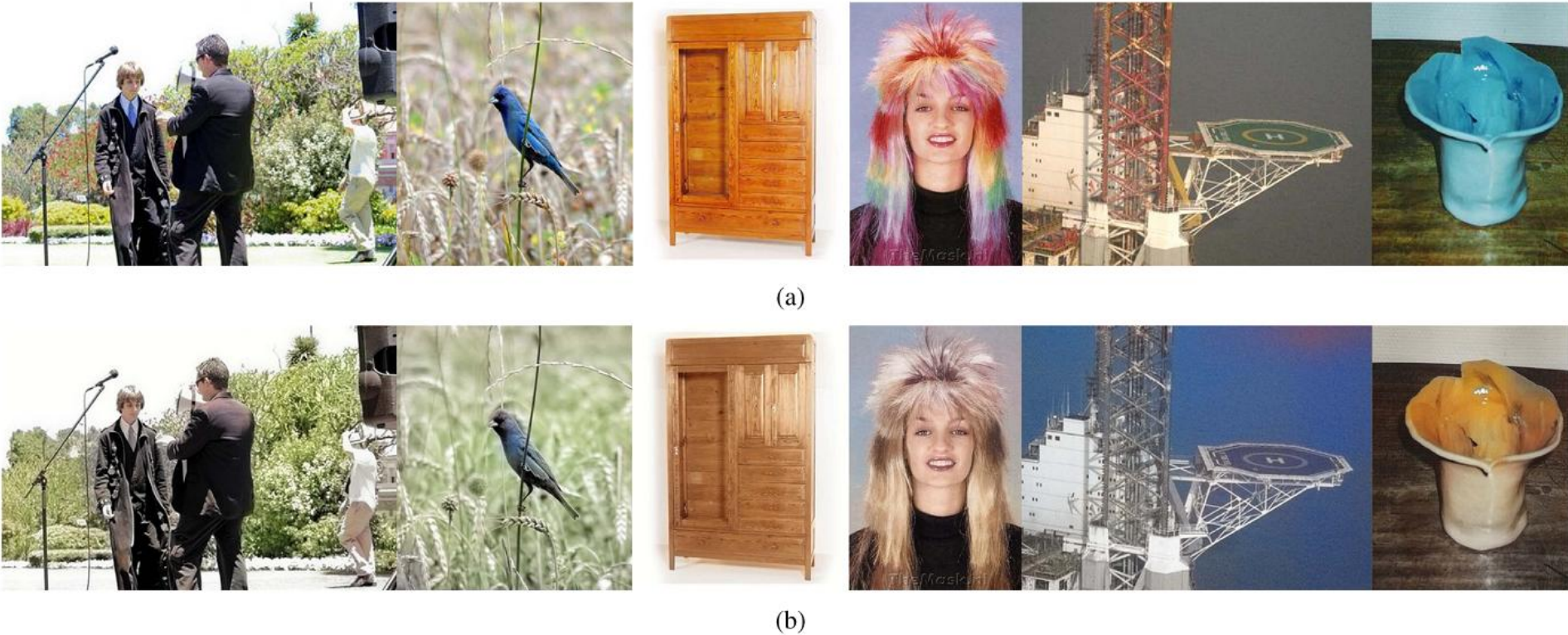
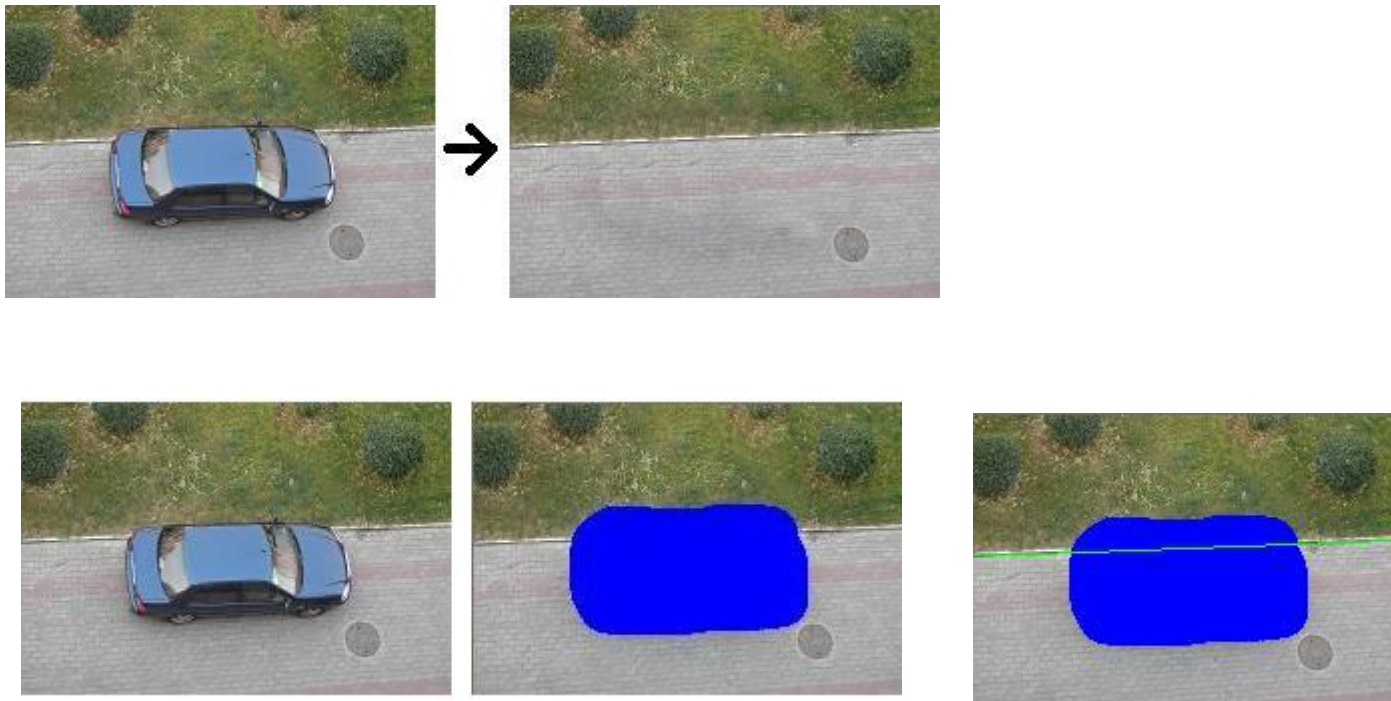


Fig. 1. (a) Real images. (b) Fake colorized images.

图像篡改形式四：用image inpainting 移除目标物体



智能图像处理软件：目标移除、图像修复(Image Completion, Inpainting)

http://blog.sina.com.cn/s/blog_c190cb710102vslx.html

AI based inpainting examples



Benedetta Tondi, Andrea Costanzo, Mauro Barni. OF-SemWat: ROBUST SEMANTIC TEXT WATERMARKING OF AI-GENERATED IMAGES OF ARBITRARY RESOLUTION,

<https://arxiv.org/abs/2509.24823v1> 图像的语义描述——也可能对应于输入的文本提示——被作为水印嵌入到图像中。



An *airplane flies low* near modern skyscrapers in a dense city. The glass buildings reflect sunlight, and the sky is partly cloudy. The perspective emphasizes the dramatic proximity of the *plane* and towers.

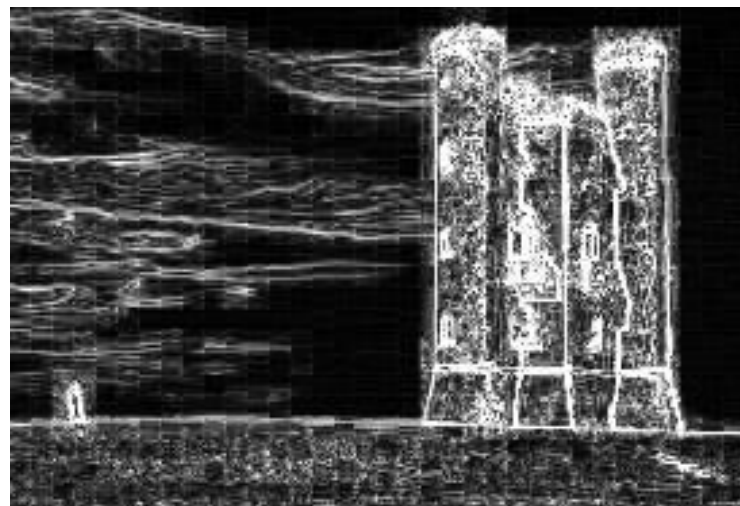


A serene young woman in a flowing beige gown, sitting and cradling a *newborn baby* in her lap. Warm sunlight filters through, casting a soft glow, evoking classic Madonna and *Child* iconography.

从左到右：原始图像、带水印的图像、AI编辑后的带水印图像及篡改检测定位结果

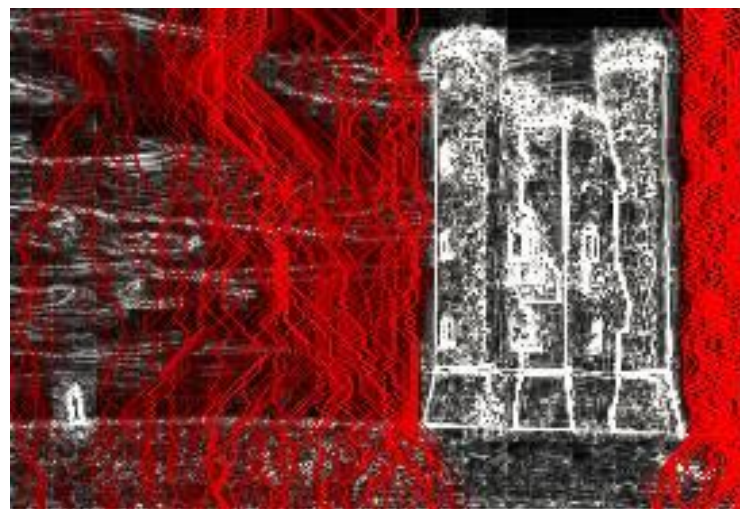
图像篡改形式五： 用seam carving 改变图像尺寸 (Retargeting)

输入图像



计算梯度画出能量图

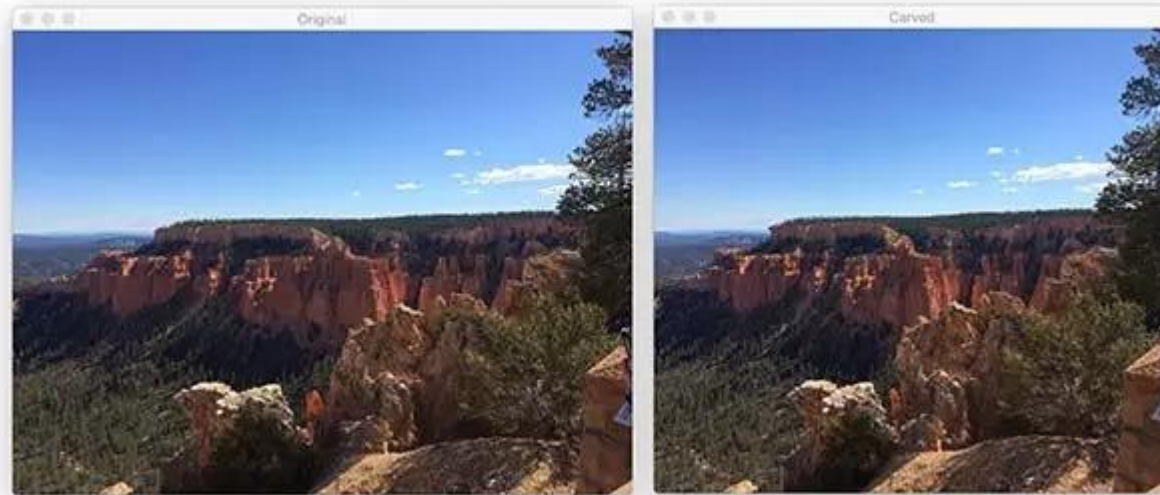
要调整图像大小，可去除低能量接缝来对图像降采样，或复制低能量接缝进行升采样。**seam carving**的目的是在调整图像本身的大小的同时保留图像的最突出（即“有趣”）区域。



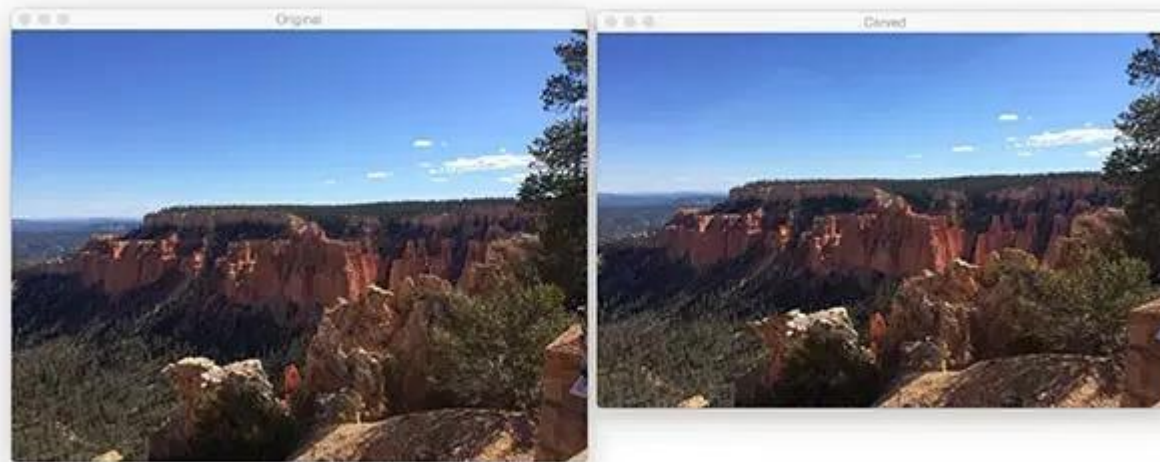
使用能量图，生成一组缝隙，从左到右或从上到下跨越整个图像

这些接缝通过动态规划方法有效地计算，并根据它们的能量进行排序。具有低能量的接缝被放置在列表的前面，而高能量接缝被放置在列表的后面。

图像篡改形式五：用seam carving 改变图像大小



原始图与处理后图（横向压缩）



原始图与处理后图（纵向压缩）

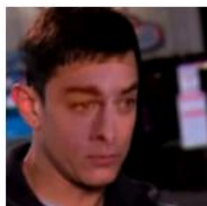
图像篡改形式五：用seam carving 移除目标物体

Seam Carving GUI: Windows 下的图像处理工具



也可用openCV中的Grabcut来实现

深度学习下的图像篡改：Deepfakes (DFs)



FaceSwap
(2020)



FaceDancer
(2023)



Wav2Lip
(2020)



NST
(2016)

Face Swapping 换脸



FOMM
(2019)



HyperReenact
(2023)

Face Reenactment 人脸重演



SadTalker
(2023)

Audio-driven Talking-face Video 音视频合成



Midjourney
(2023)

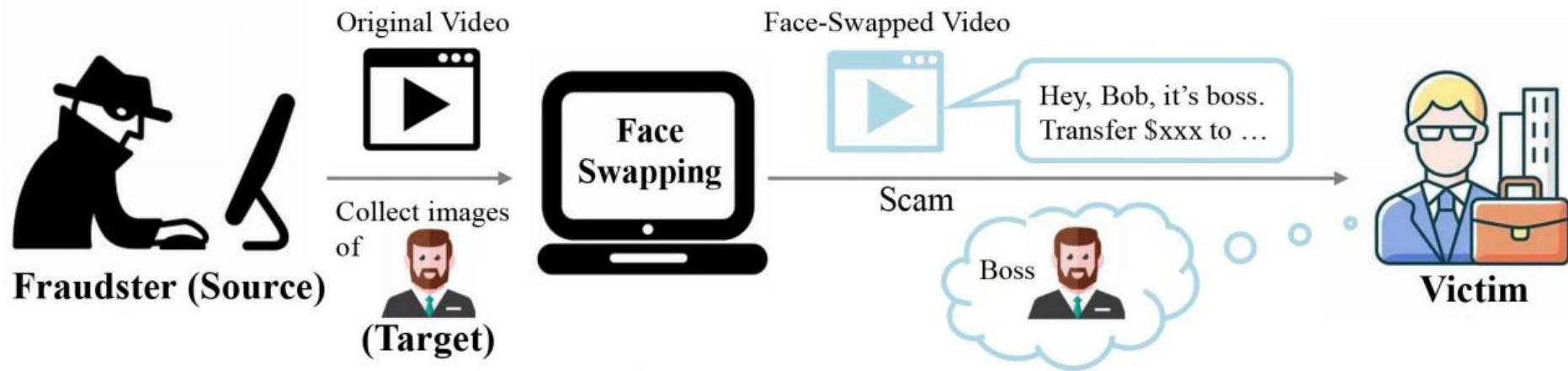
Artists' style mimicry 艺术风格模仿

Examples of various DF categories including face swapping (FaceSwap , 2020; FaceDancer, 2023), face reenactment (FOMM , 2019; HyperReenact , 2023), audio-driven talking-face video synthesis (Wav2Lip, 2020; SadTalker, 2023), and artists' style mimicry (NST, 2016; Midjourney, 2023).

近年来，伪造的视听内容被用于冒充国家领导人、名人等公众人物的案例屡见不鲜，严重损害公众信任并影响社会舆论。例如，2025年3月8日，一段特朗普总统与埃隆·马斯克的深度伪造视频曾短暂出现在美国住房和城市发展部的显示屏上。

Deepfakes (DFs): Face Swapping

Zhongyi Zhang , Jie Zhang , Wenbo Zhou, et al. **FACETRACER**: Unveiling Source Identities From Swapped Face Images and Videos for Fraud Prevention, IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 47, NO. 12, DECEMBER 2025, 12021-12037

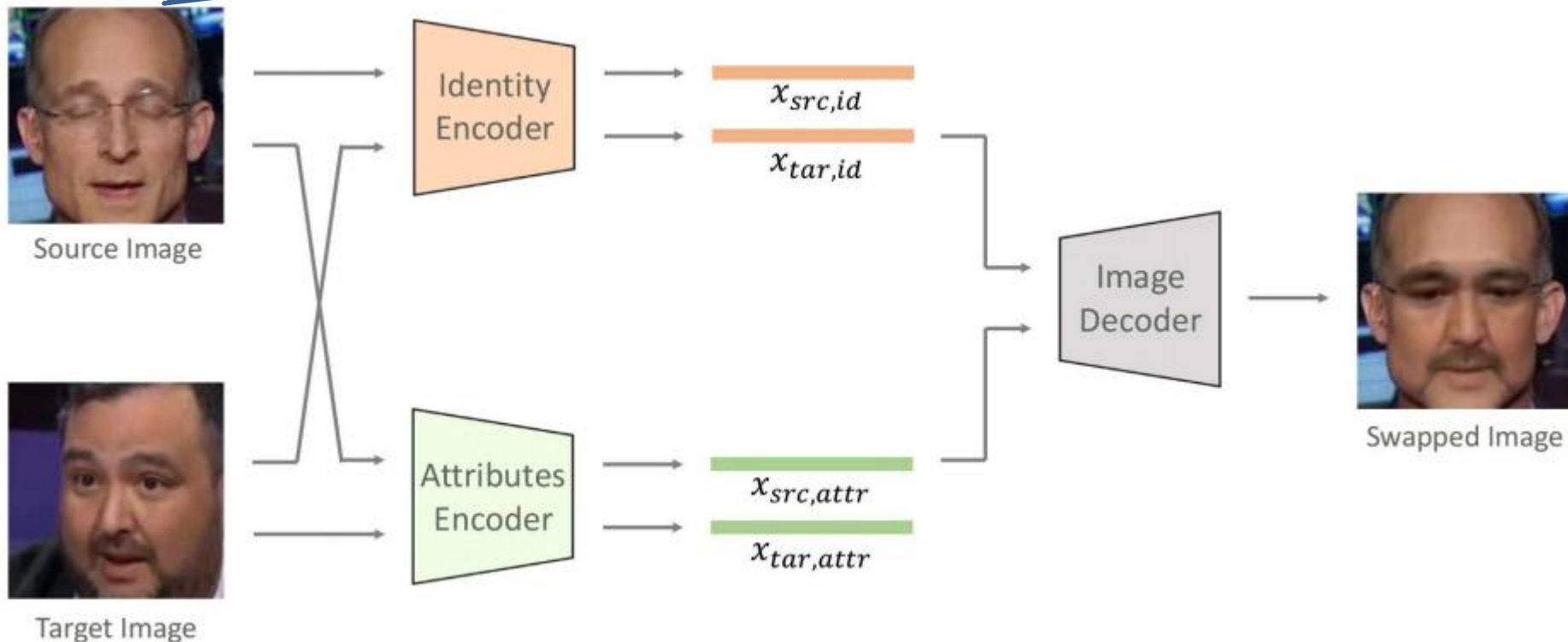


若伪造的假脸能够被还原为原始人脸，将会提供更可靠的证据，可协助追踪潜在犯罪者，为追责和维权提供有力支持。

Zhongyi Zhang , Jie Zhang , Wenbo Zhou, et al. **FACETRACER: Unveiling Source Identities From Swapped Face Images and Videos for Fraud Prevention**, IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 47, NO. 12, DECEMBER 2025, 12021-12037



诈骗人自己的图像或视频



在换脸后的图像中，面部呈现出目标人物的身份特征，同时保留了源图像的关键属性，如姿态、表情和光照背景等特征。

The goal of face swapping

$\leftarrow \text{---} \text{---} \rightarrow$ *Be Away*

$\rightarrow \text{---} \text{---} \leftarrow$ *Be Close*

$\longrightarrow$ Identity Vector

$\text{---} \longrightarrow$ Decomposed Vector

 Unit Circle of Identity Information

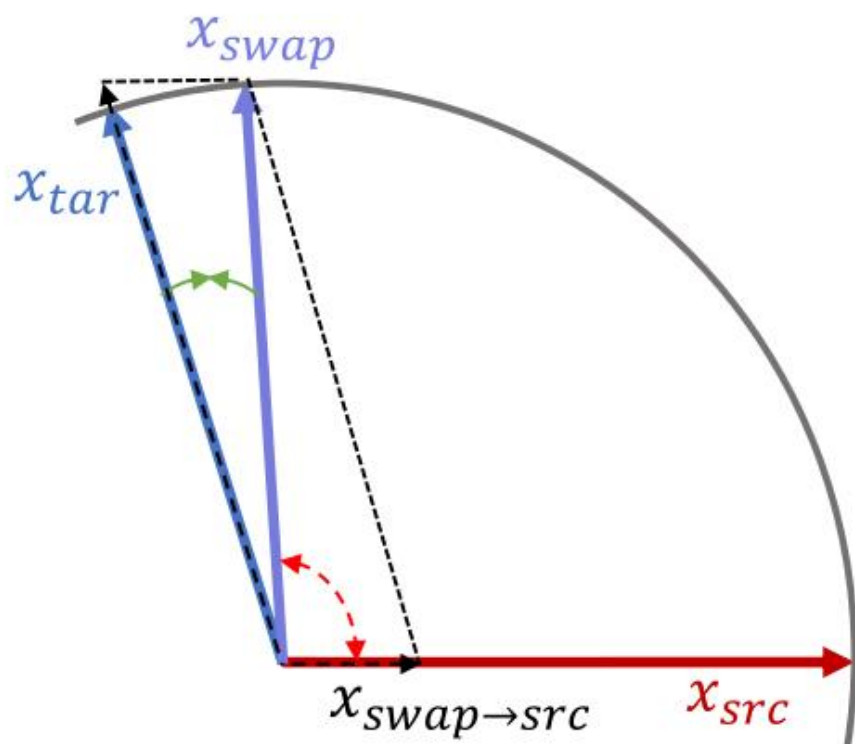


Fig. 4. Relationships between identity information during face-swapping. All identity information is normalized and located on the unit circle. The identity information of the swapped face image is dominated by the target identity information and still contains some source identity information.

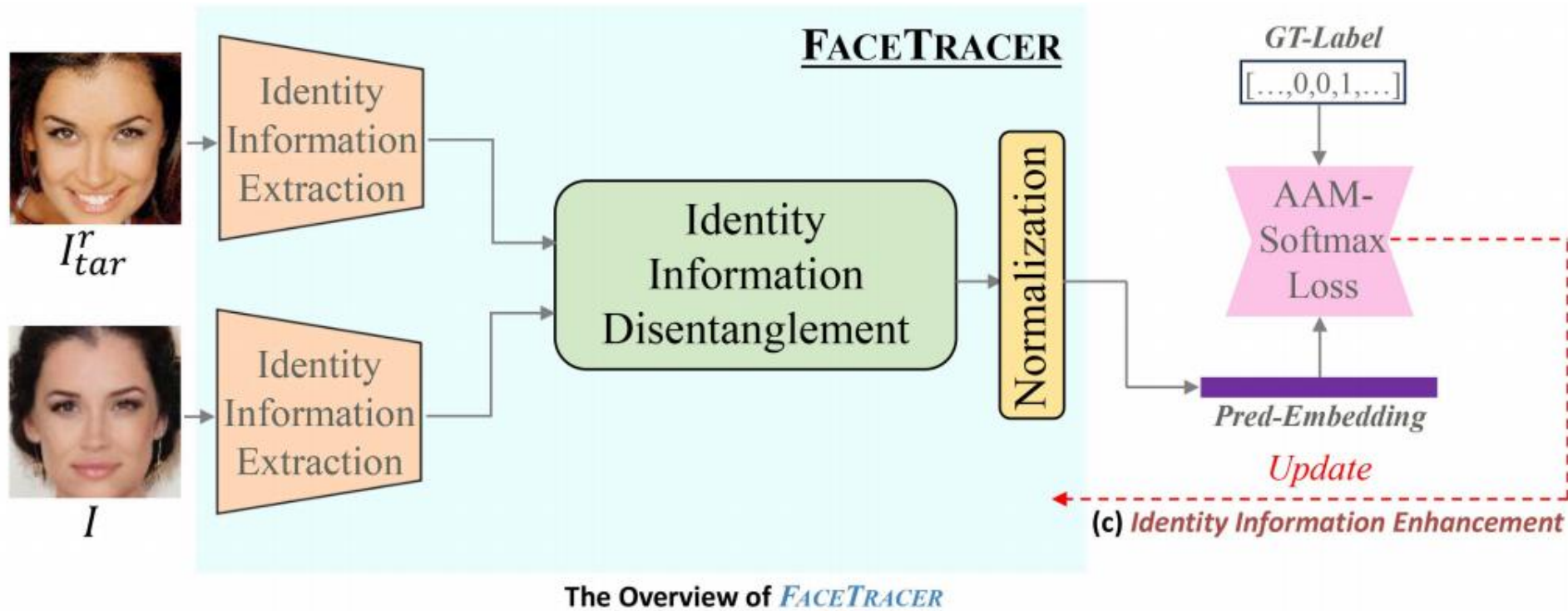
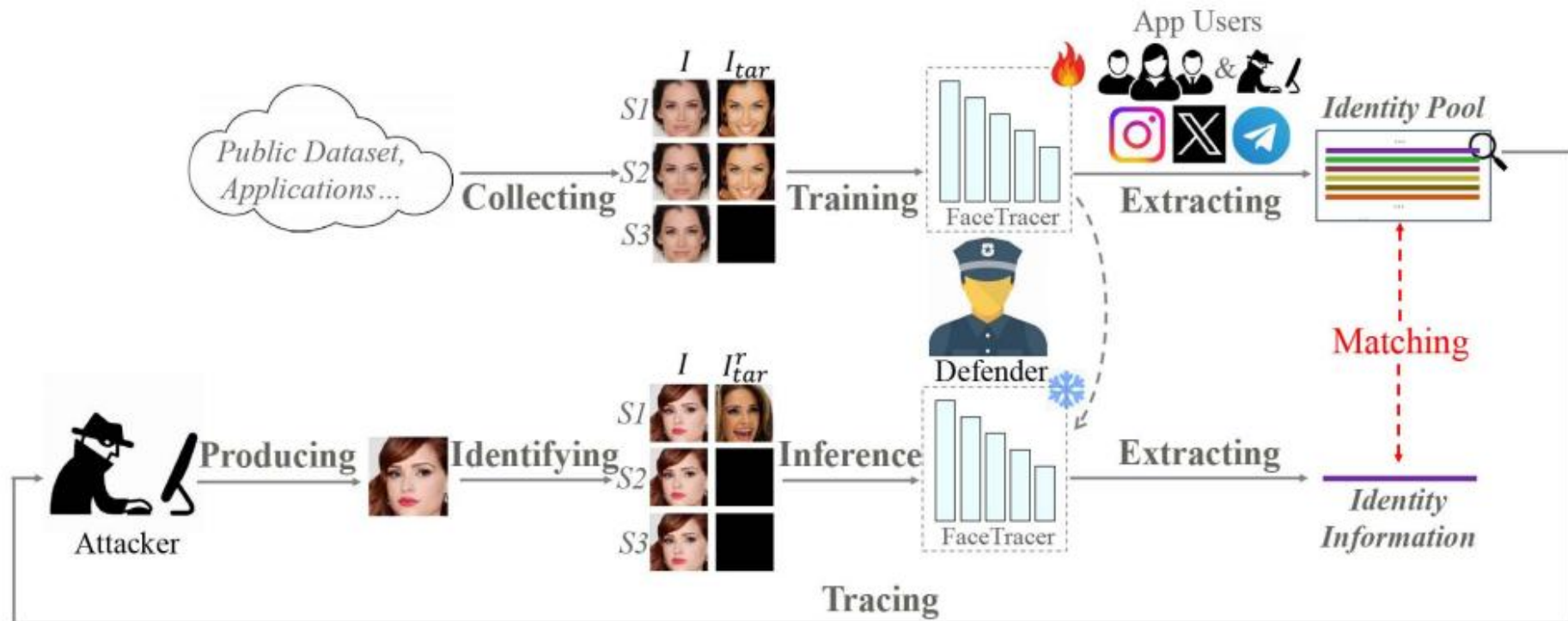
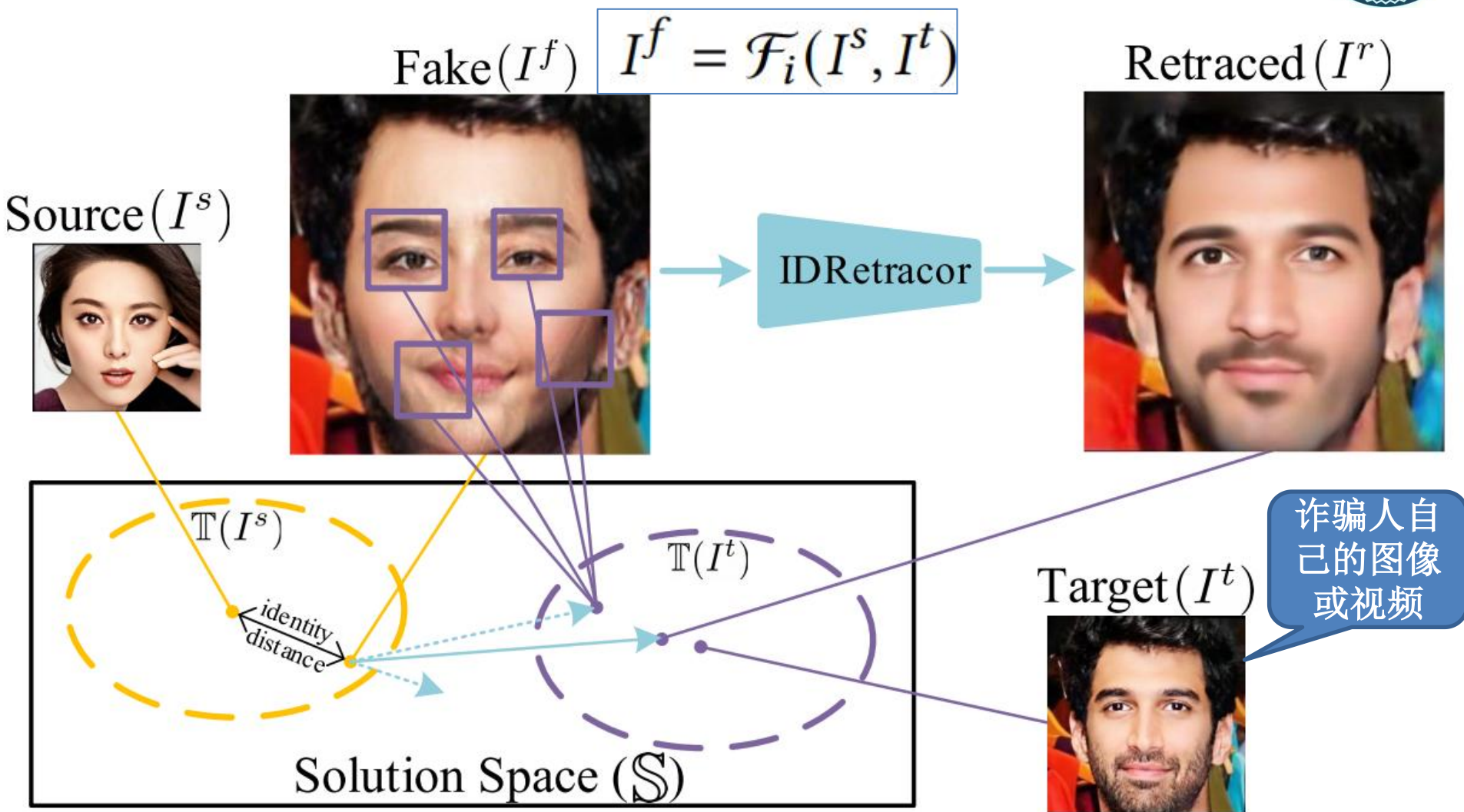


Fig. 6. FACETRACER consists of three designs: a) *identity information extraction* module aims to extract crude identity embedding from the input image, b) *identity information disentanglement* module eliminates the influence of the identity information from the target person, and c) *identity information enhancement* module further enhances the discriminative capability. The red dotted line indicates the optimization of FACETRACER with the identity information enhancement

Identity information disentanglement module, targeting at retaining only the source-related identity information while removing target-related identity information; Identity information enhancement module aims to better separate the extracted identity information of different source person on the unit circle.

Zhongyi Zhang , Jie Zhang , Wenbo Zhou, et al. **FACETRACER**: Unveiling Source Identities From Swapped Face Images and Videos for Fraud Prevention, IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 47, NO. 12, DECEMBER 2025, 12021-12037





$$\mathbb{T}(I) = \{x | \text{Sim}(x, I) > \tau\}.$$

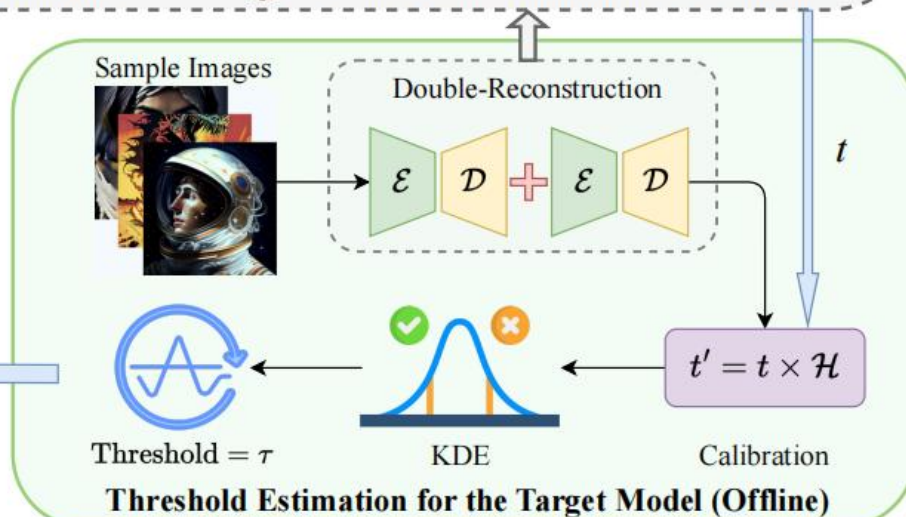
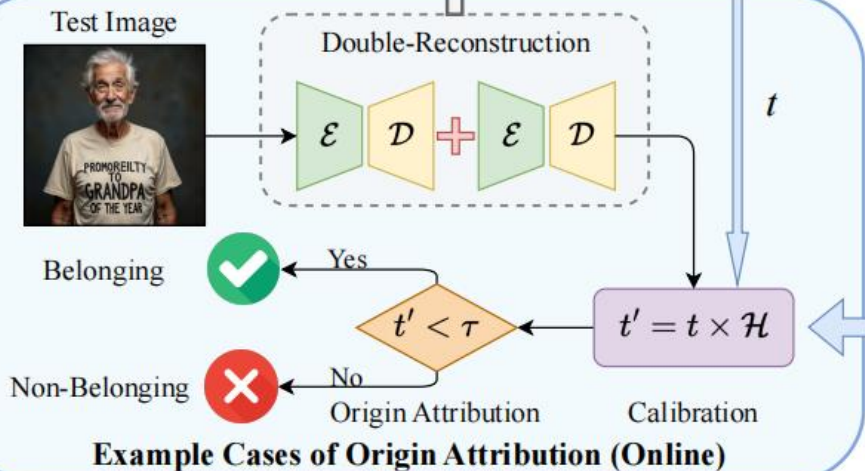
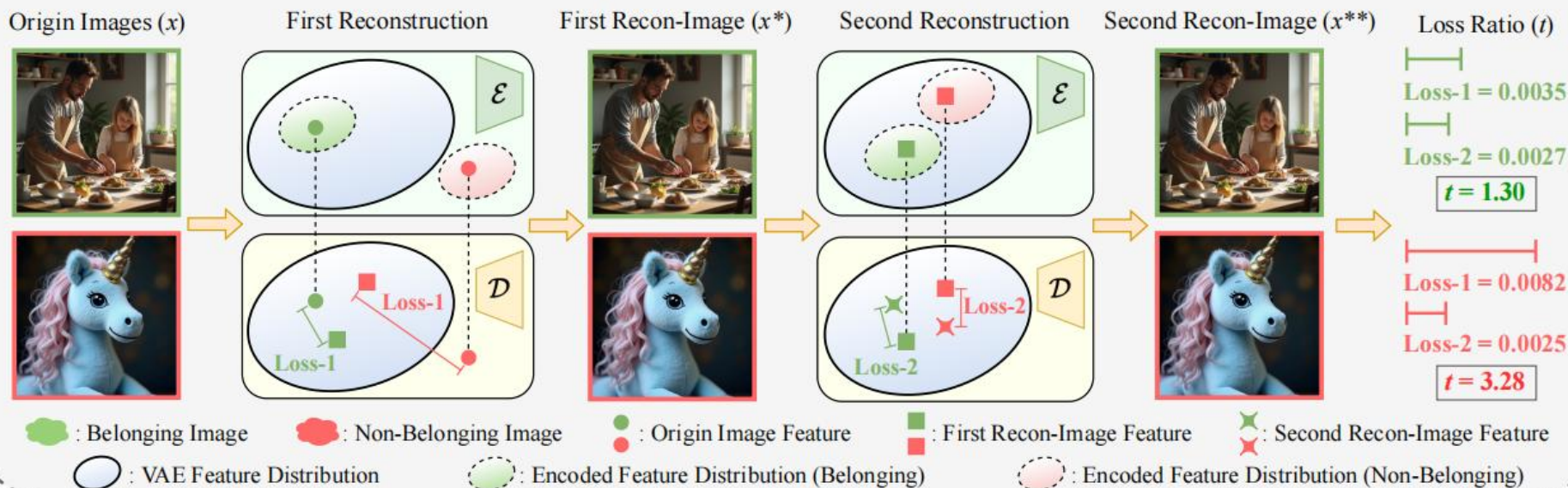
注意：这里的“目标图像”与“源图像”与 FACETRACER 一文中定义的正相反！

自然图像与AI生成图像的自动识别



C. Wang, Z. Yang, Y. Wang, W. Zhang, and K. Chen, "AEDR: Training-Free AI-Generated Image Attribution via Autoencoder DoubleReconstruction," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 12, 2026, pp. 9675–9683.

Double-Reconstruction Detailed Process



latent feature consistency across reconstructions

KDE (kernel density estimation)

C. Wang, Z. Yang, Y. Wang, W. Zhang, and K. Chen, “AEDR: Training-Free AI-Generated Image Attribution via Autoencoder DoubleReconstruction,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 12, 2026, pp. 9675–9683.

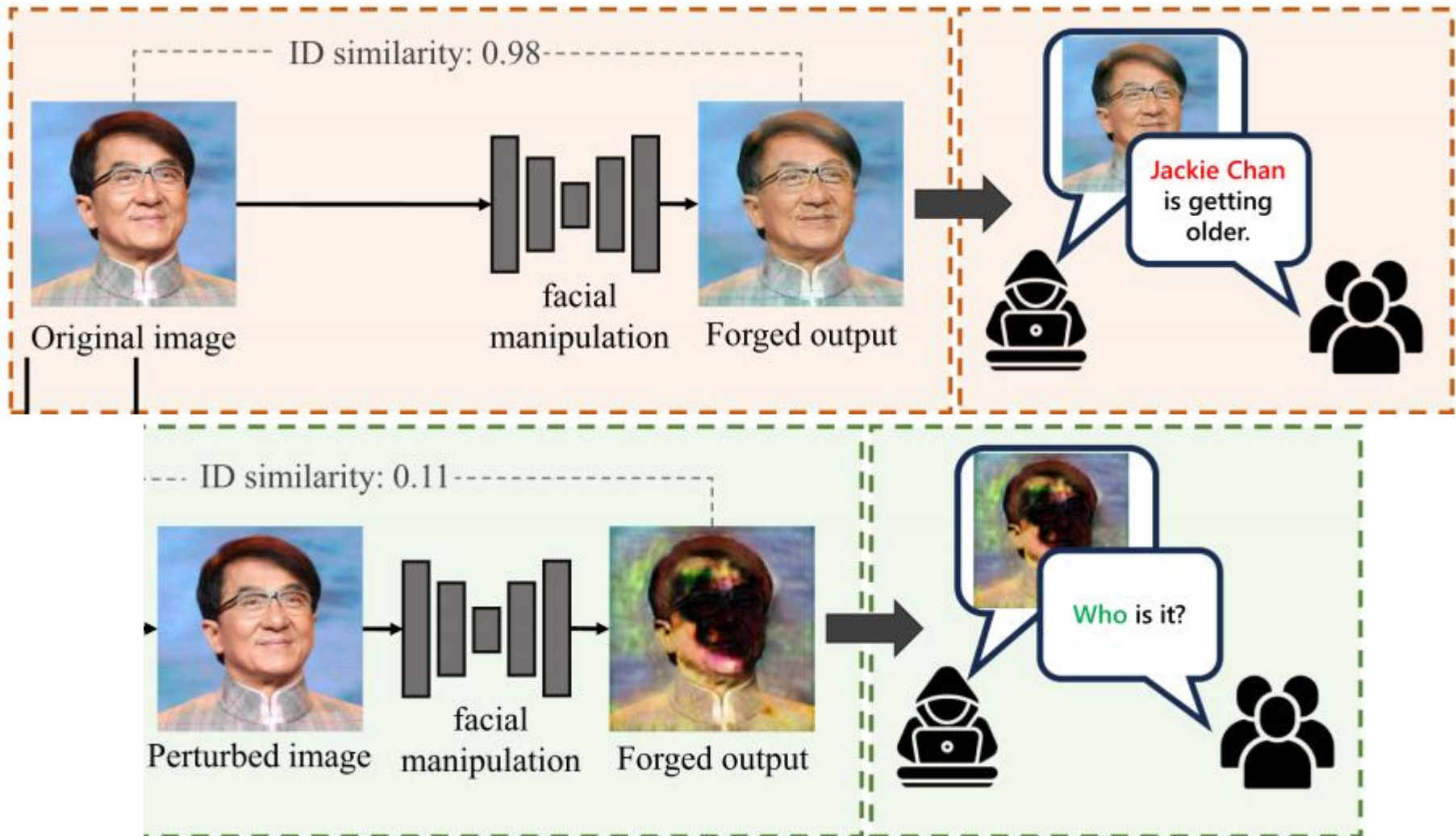
相比之下，包含复杂纹理或动态背景图像往往会导致这些损失值出现显著波动。为解决这一固有偏差，引入了一种考虑图像均匀性的损失校准机制：基于**灰度共生矩阵进行校准**。

$$\mathcal{H} = \sum_{i=0}^{\ell-1} \sum_{j=0}^{\ell-1} \frac{P(i, j)}{1 + |i - j|}$$

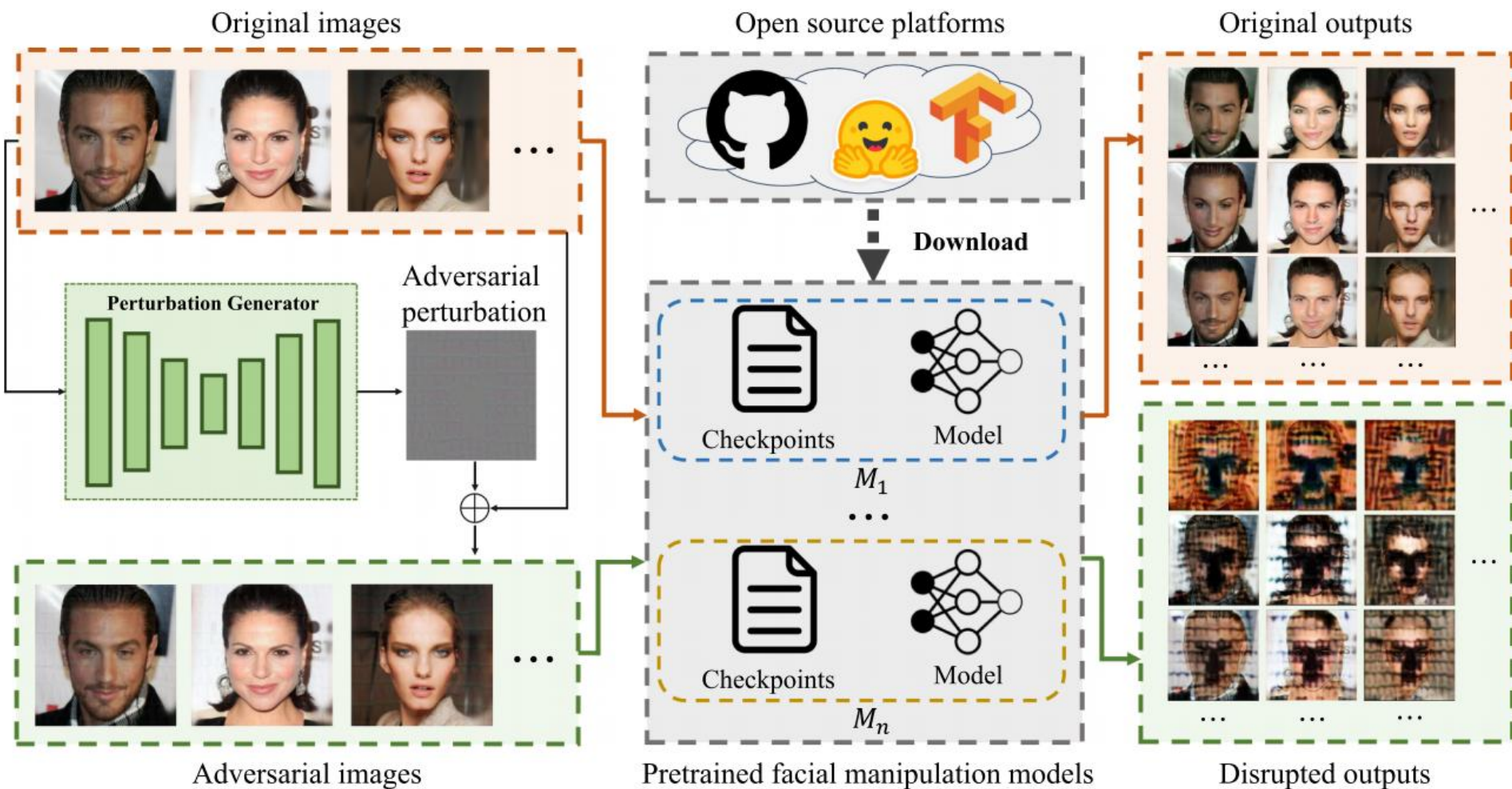
where $P(i, j)$ denotes the co-occurrence probability of grayscale levels i and j at a specified orientation and offset. The parameter ℓ represents the total number of grayscale levels (default value of 32 to ensure computational efficiency), and $|i - j|$ indicates the absolute difference in intensity between the corresponding pixel pairs. Based on this, the calibrated attribution signal is defined as follows:

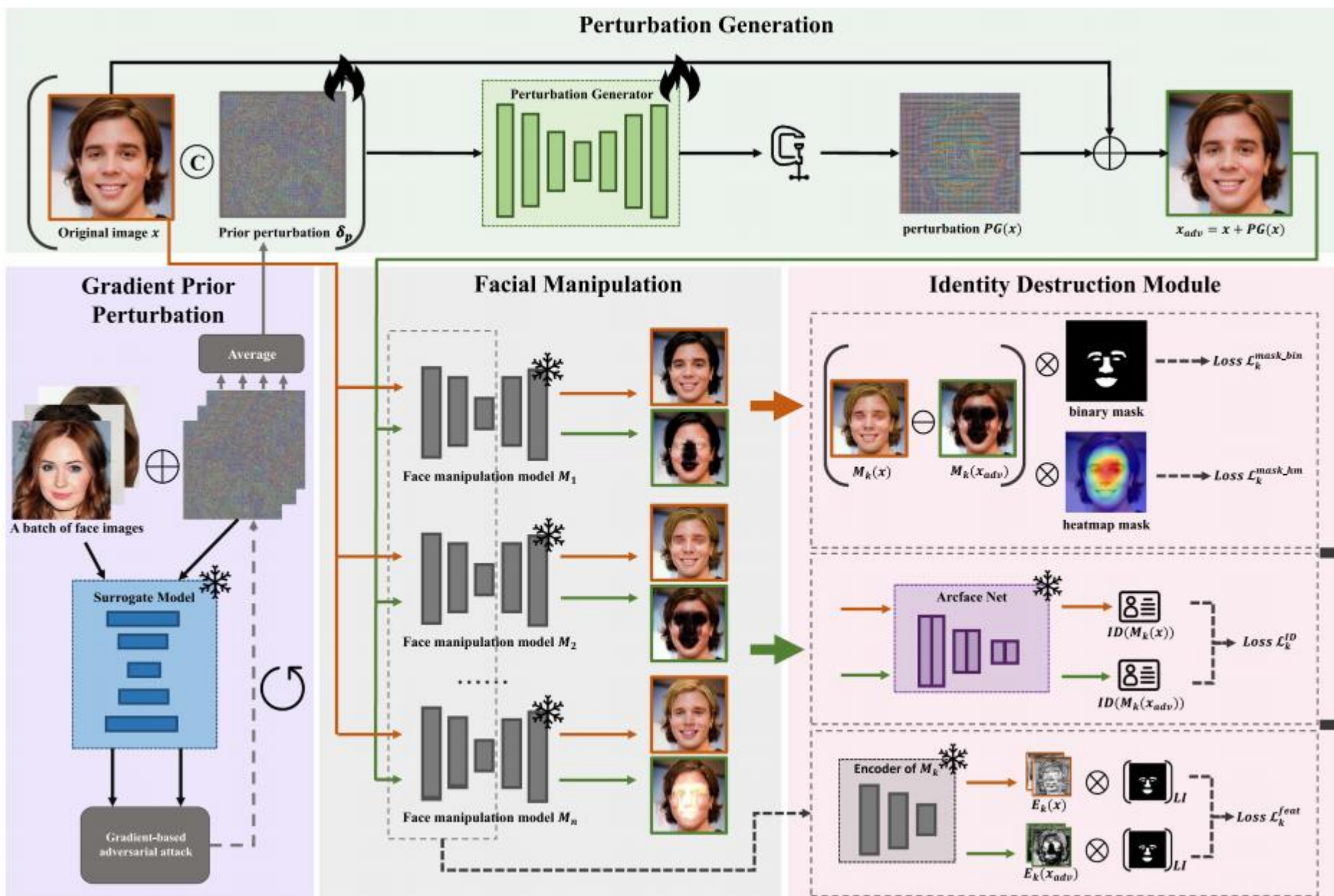
$$t' = t \times \mathcal{H} = \frac{\mathcal{L}_1 \times \mathcal{H}}{\mathcal{L}_2}.$$

Proactive Deepfake Defense: Disruption



Zuomin Qu , Wei Lu , Xiangyang Luo, et al. ID-Guard: A Universal Framework for Combating Facial Manipulation via Breaking Identification, IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 48, NO. 2, FEBRUARY 2026, 1720-1735







天津大学

TianJin University

Founded 1895



实例二

数字图像水印

(Digital Image Watermarking)

数字水印的基本概念

数字水印技术是将特定的标记（水印）嵌入到某一媒体信息中，以此实现对该媒体信息进行某种程度的保护或监控。

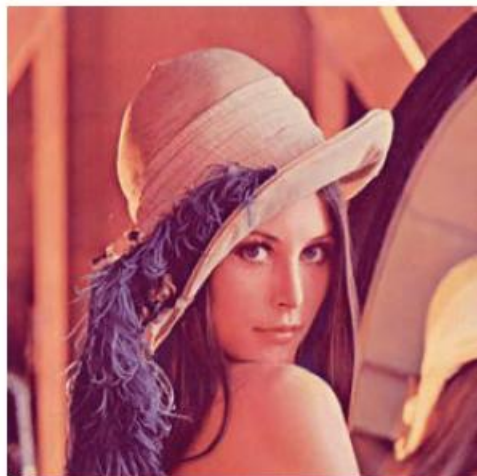


直接标识版权

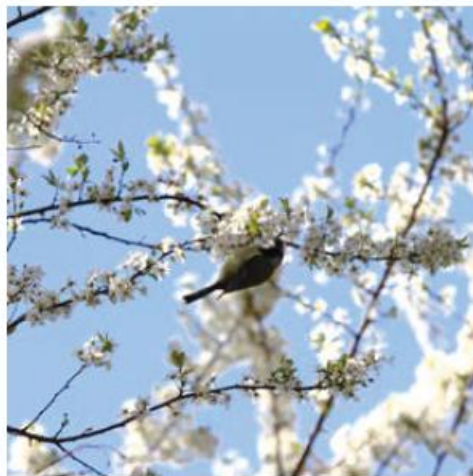


可见水印标识版权

数字水印的基本概念



(a)



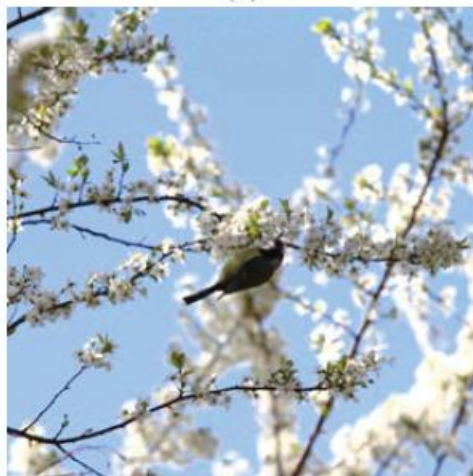
(b)



(c)



(d)



(e)

不可见水印



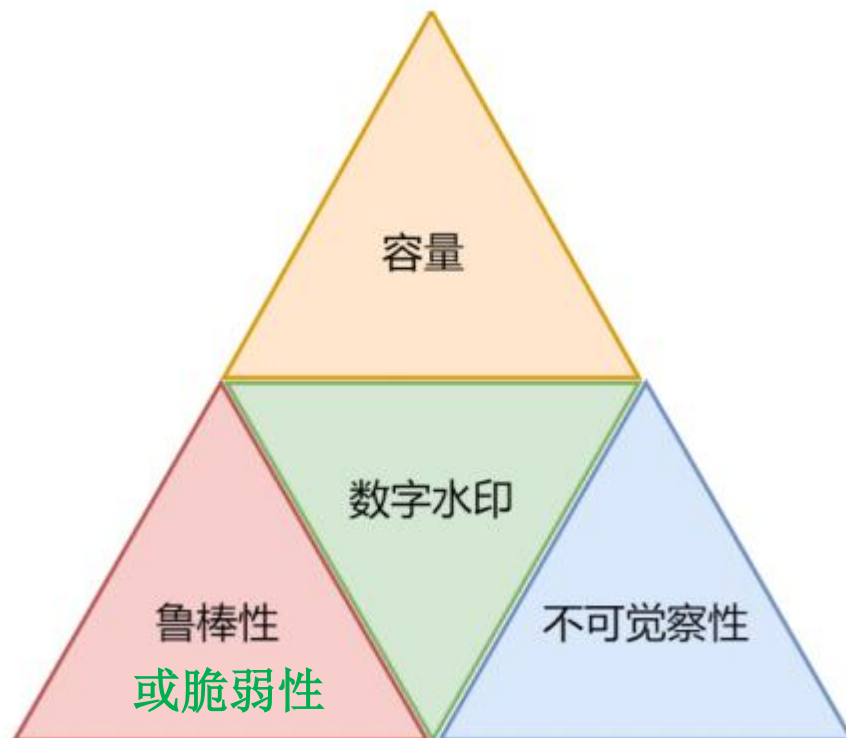
(f)

Fig. 8: (a), (b), (c) Original image, (d), (e), (f) Proposed template and watermark-embedded image.

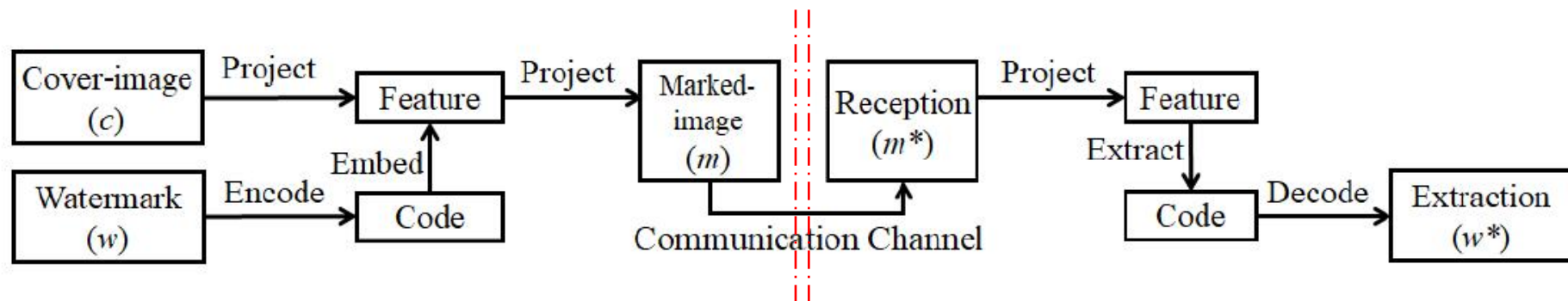
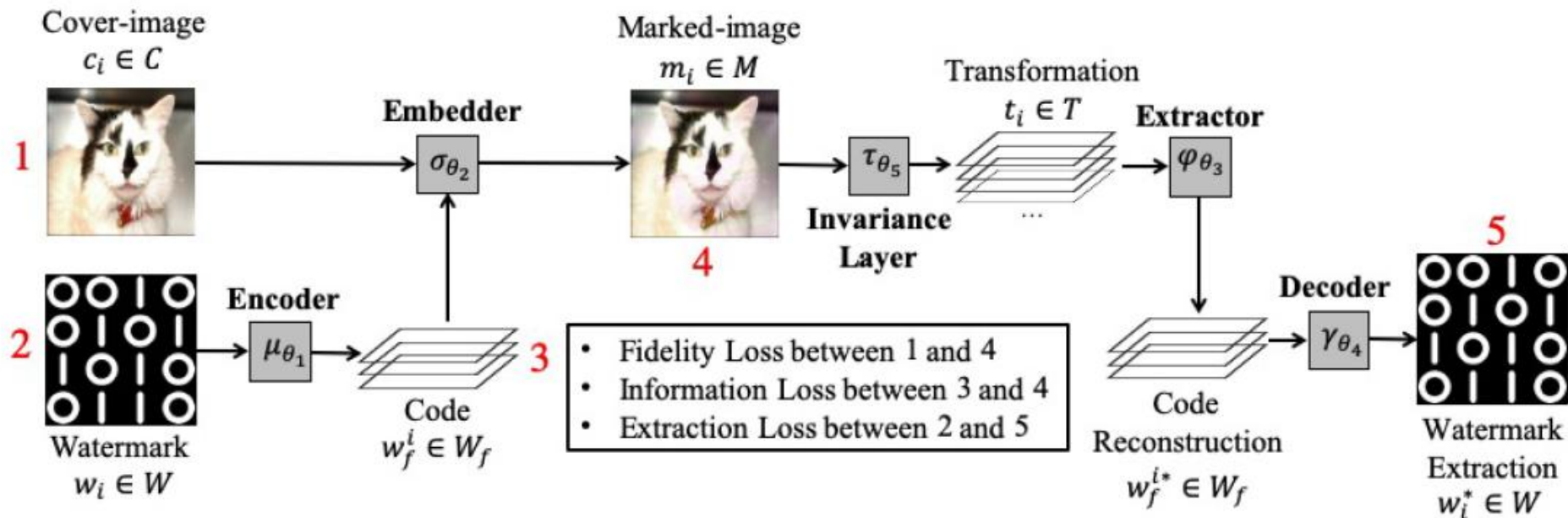
W.-H. Kim, J.-U. Hou, S.-M. Mun, and H.-K. Lee, "Convolutional neural network architecture for recovering watermark synchronization," *arXiv preprint arXiv:1805.06199*, 2018.

数字水印的分类和评价指标

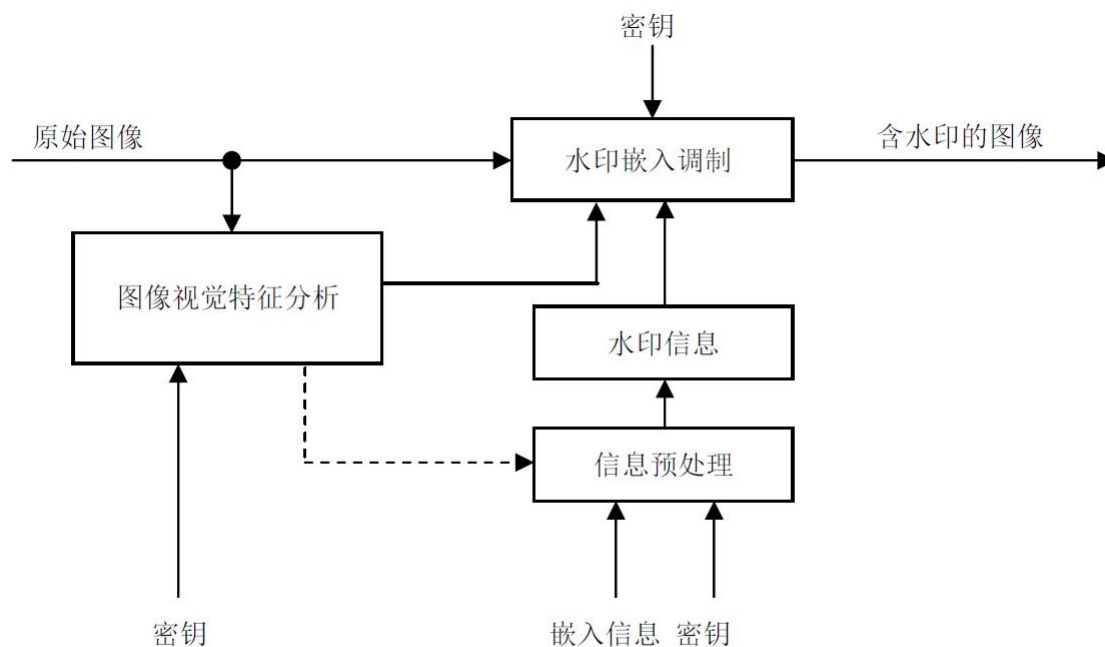
- ◆ 鲁棒性水印（主要用于版权保护）
- ◆ （半）脆弱性水印（主要用于完整性认证）



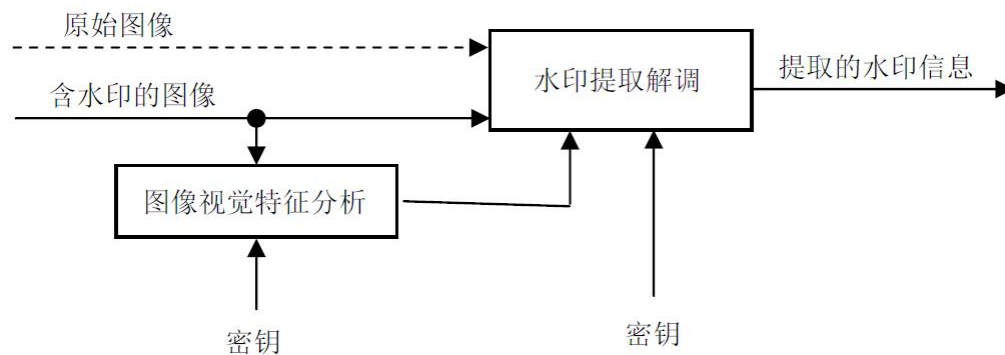
数字图像水印系统框图



数字图像水印系统框图

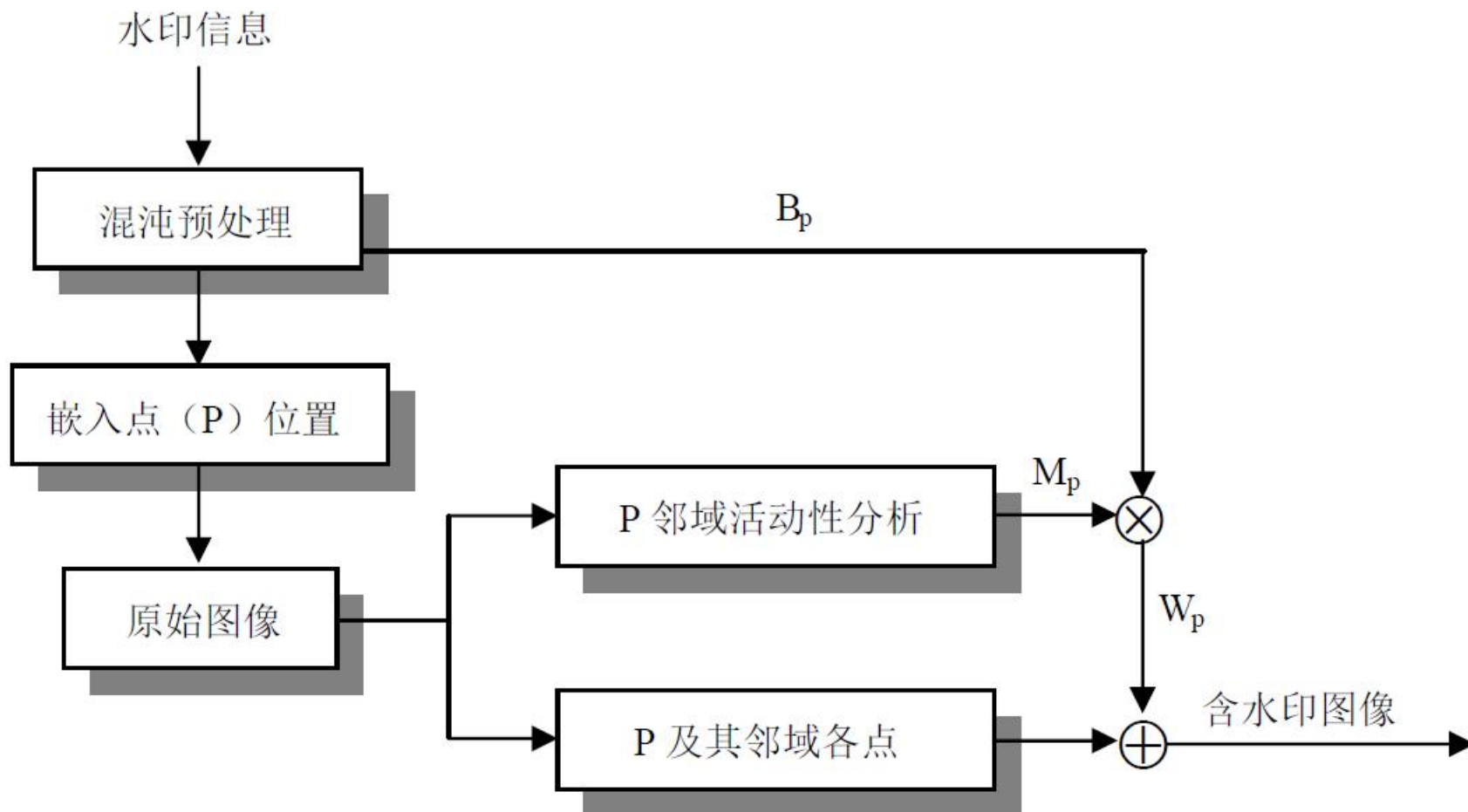


(a) 水印嵌入过程

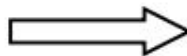


(b) 水印提取过程

空间域鲁棒性图像水印算法

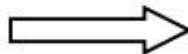
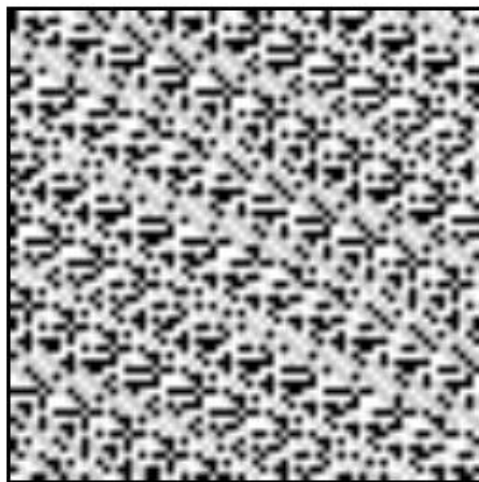
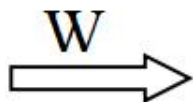


空间域鲁棒性图像水印算法——水印信息混沌预处理



(a)消息向量 B

(b) 方阵 W



(c)方阵 W'

(d) W' 在 I 中的位置

空间域鲁棒性图像水印算法——水印嵌入

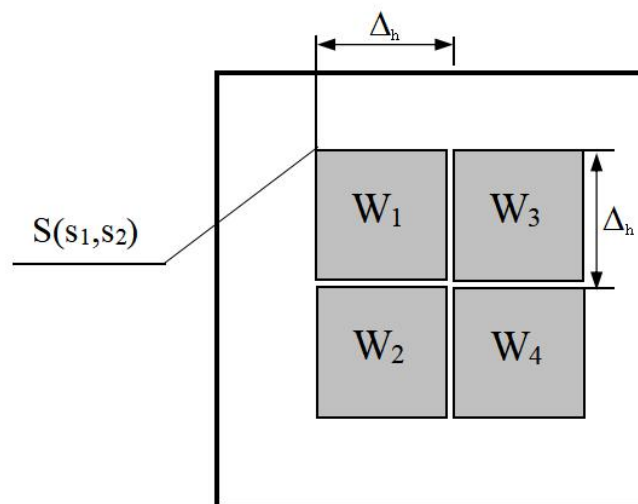


使水印具有低通特性并利用人眼视觉掩盖特性

$$\bar{I}_p = \frac{1}{8} \sum_{j=1}^8 I_j$$

$$\begin{bmatrix} I'_1 & I'_2 & I'_3 \\ I'_4 & I'_p & I'_5 \\ I'_6 & I'_7 & I'_8 \end{bmatrix} = \begin{bmatrix} I_1 & I_2 & I_3 \\ I_4 & \bar{I}_p & I_5 \\ I_6 & I_7 & I_8 \end{bmatrix} + \begin{bmatrix} \beta_1 & \beta_2 & \beta_3 \\ \beta_4 & \alpha_p & \beta_5 \\ \beta_6 & \beta_7 & \beta_8 \end{bmatrix} \bullet B_p$$

布局水印嵌入位置
以实现几何校准



空间域鲁棒性图像水印算法——水印检测



设信息位 b_i 共被嵌入 L 次，水印检测时，根据位置映射关系找到对应于 b_i 的 L 个嵌入点位置，计算出每一点灰度值 I'_p 与其预测值 $\bar{I}'_p$ 的差值 δ ，再求出 L 个差值的平均值 $\bar{\delta}$ ，最后根据 $\bar{\delta}$ 的符号判断出 b_i 的数值，水印检测算法可用公式描述如下：

$$\bar{I}'_p = \frac{1}{8} \sum_{j=1}^8 I'_j$$

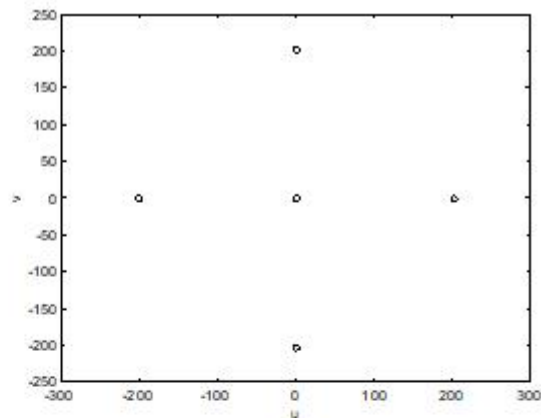
$$\delta = I'_p - \bar{I}'_p$$

$$b_i = \begin{cases} 0 & \text{if } \bar{\delta} < 0 \\ 1 & \text{if } \bar{\delta} > 0 \end{cases}$$

空间域鲁棒性图像水印算法——几何校准



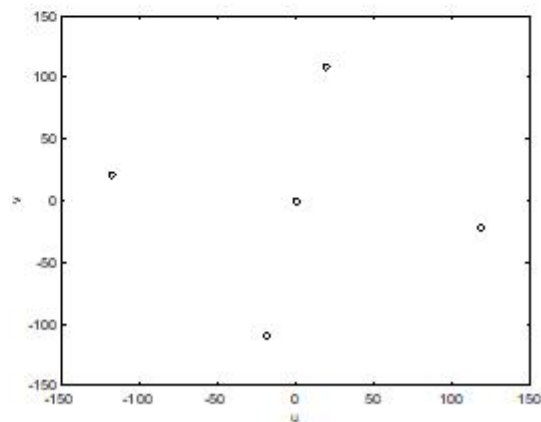
(a) 含水印的图像 I'



(b) 对应于 I' 的 MX



(c) I' 旋转后的图像 R



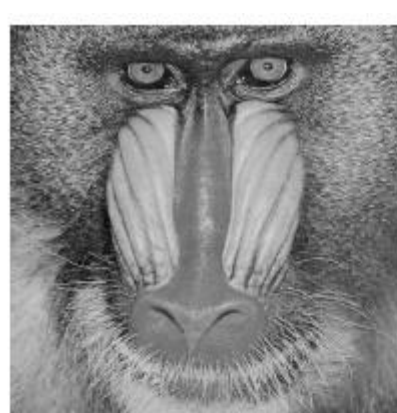
(d) 对应于 R 的 MX



(a) 原始图像 Lena



(b) 原始图像 Boat



(c) 原始图像 Baboon



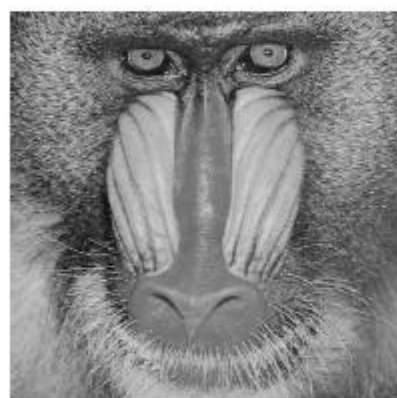
(d) 原始图像 Pepper



(a') 含水印图像 Lena



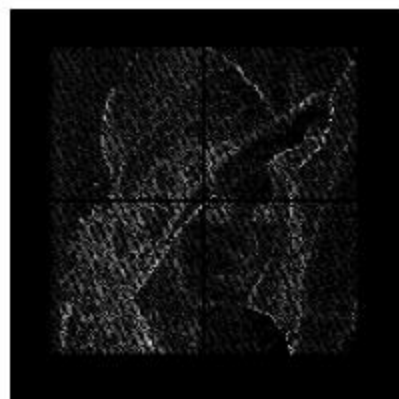
(b') 含水印图像 Boat



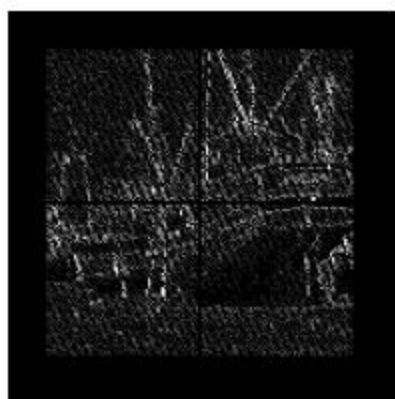
(c') 含水印图像 Baboon



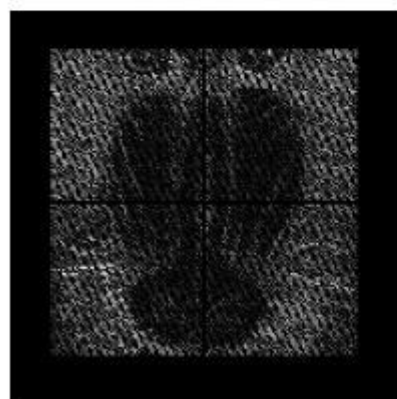
(d') 含水印图像 Pepper



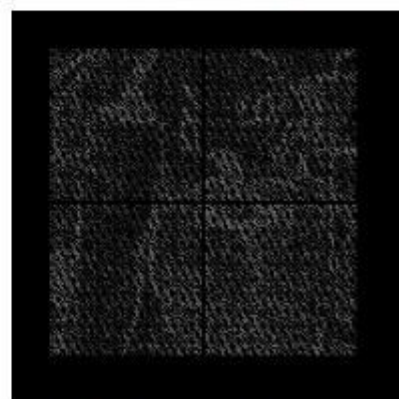
(a'') 原始图像与水印图像的差值图像
(对应像素灰度差值的绝对值 $\times 20$)



(b'') 原始图像与水印图像的差值图像
(对应像素灰度差值的绝对值 $\times 20$)



(c'') 原始图像与水印图像的差值图像
(对应像素灰度差值的绝对值 $\times 20$)



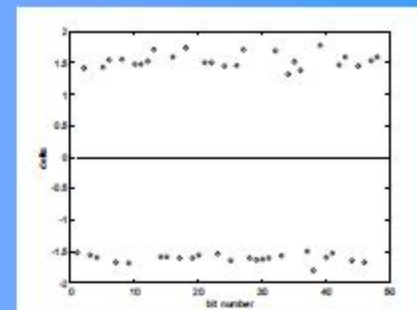
(d'') 原始图像与水印图像的差值图像
(对应像素灰度差值的绝对值 $\times 20$)



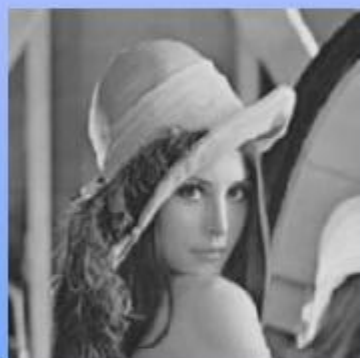
(a0)原始图像I



(a) 嵌入水印的图像I'



(a') 对应于I'的(delta)



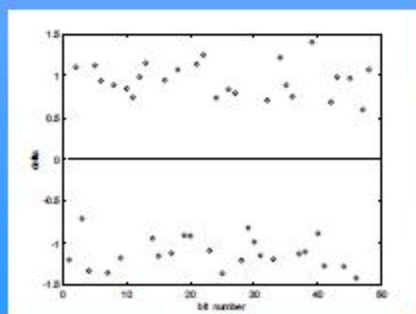
(b) 中值滤波后的水印图像I₁'



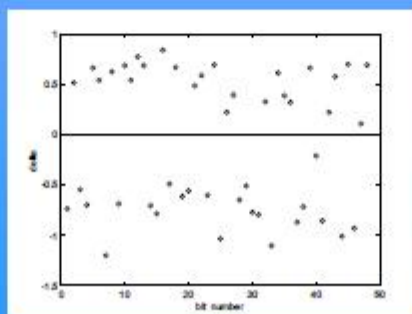
(c) 低通滤波后的水印图像I₂'



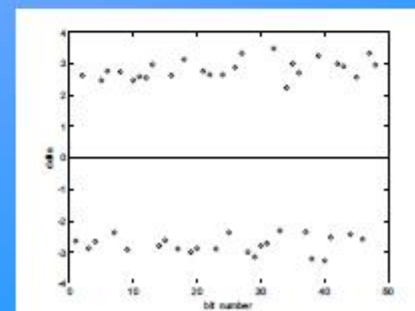
(d) 直方图均衡后的水印图像I₃'



(b') 对应于I₁'的(delta)



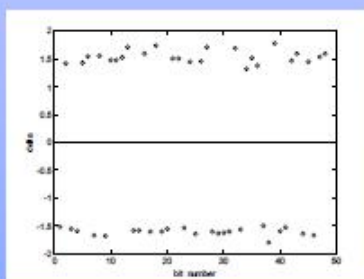
(c') 对应于I₂'的(delta)



(d') 对应于I₃'的(delta)



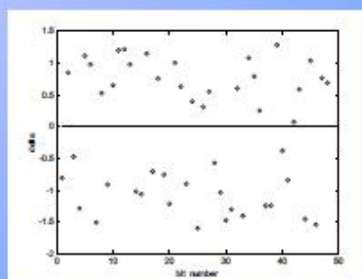
(e) 剪切后的水印图像 I_4'



(e') 对应于 I_4' 的(delta)



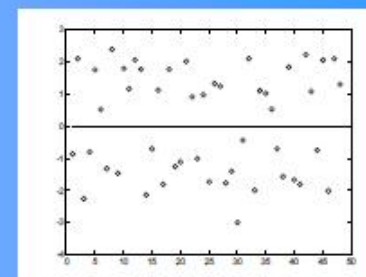
(f) 经JPEG压缩(Q=50%)
后的水印图像 I_5'



(f') 对应于 I_5' 的(delta)



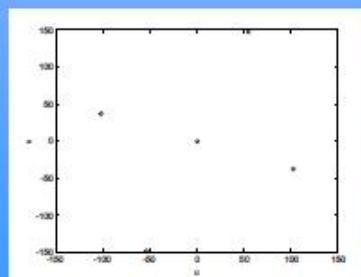
(g) 加入噪声(10%)后的水印图像 I_6'



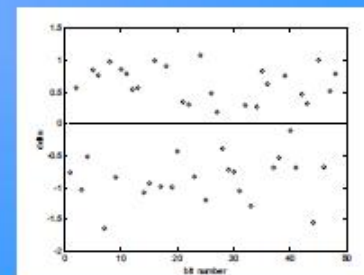
(g') 对应于 I_6' 的(delta)



(h) 经JPEG压缩(Q=75%)、旋转
和缩小后的水印图像 I_7'



(h') I_7' 旋转角度追踪结果



(h'') I_7' 反几何变换后检测到
的(delta)

Hidden: Hiding data with deep networks

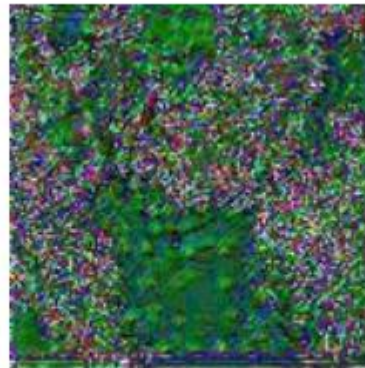


Jiren Zhu, Russell Kaplan, Justin Johnson, and Li Fei-Fei. Hidden: Hiding data with deep networks. In *ECCV*, 2018.



Cover Image

+

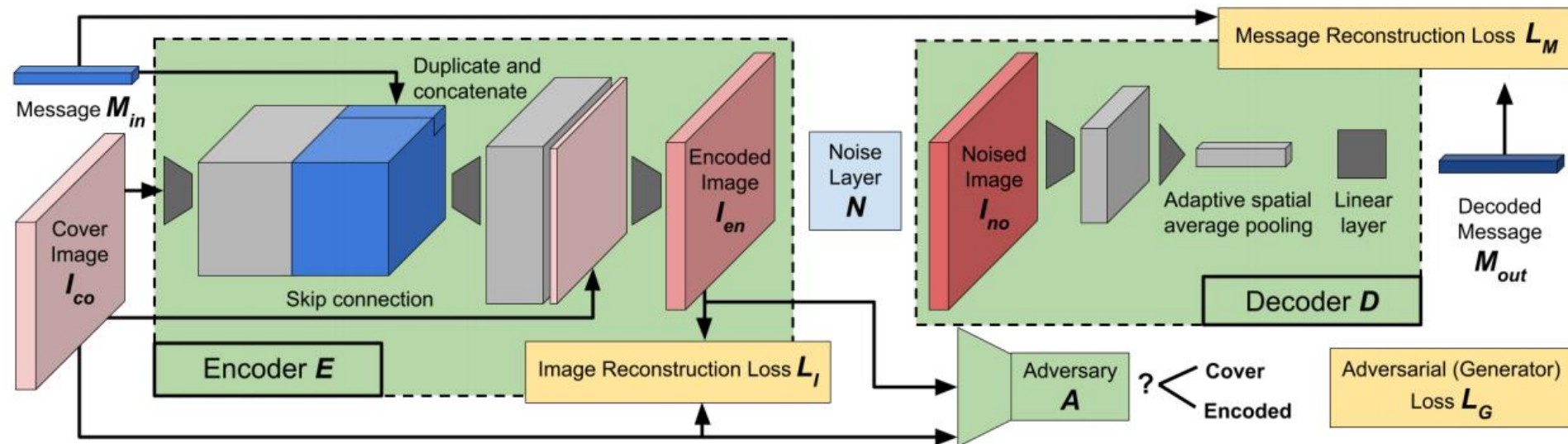


=



"Copyright ID: 1337"

HiDDeN
Perturbation



Hidden: Hiding data with deep networks

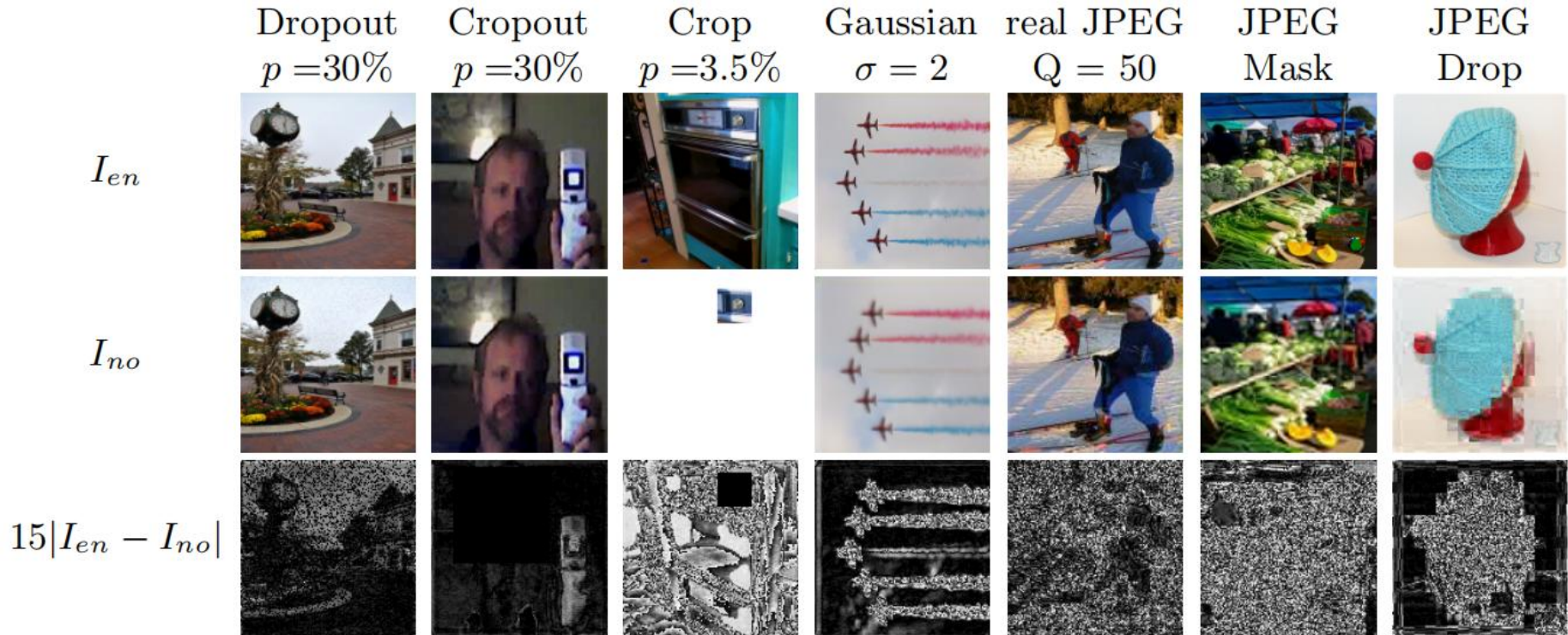


Fig. 3. Illustration of non-identity noise layers. JPEG-Mask and JPEG-Drop are differentiable approximations of the JPEG compression (see Figure 4). **Top:** Encoded image I_{en} . **Middle:** Noised image I_{no} . **Bottom:** Magnified difference $|I_{en} - I_{no}|$. Even under heavy distortion, such as a Crop layer which retains only 3.5% of the original image, our model still learns to recover the watermark with high accuracy (see Section 4).

Hidden: Hiding data with deep networks



Jiren Zhu, Russell Kaplan, Justin Johnson, and Li Fei-Fei. Hidden: Hiding data with deep networks. In *ECCV*, 2018.

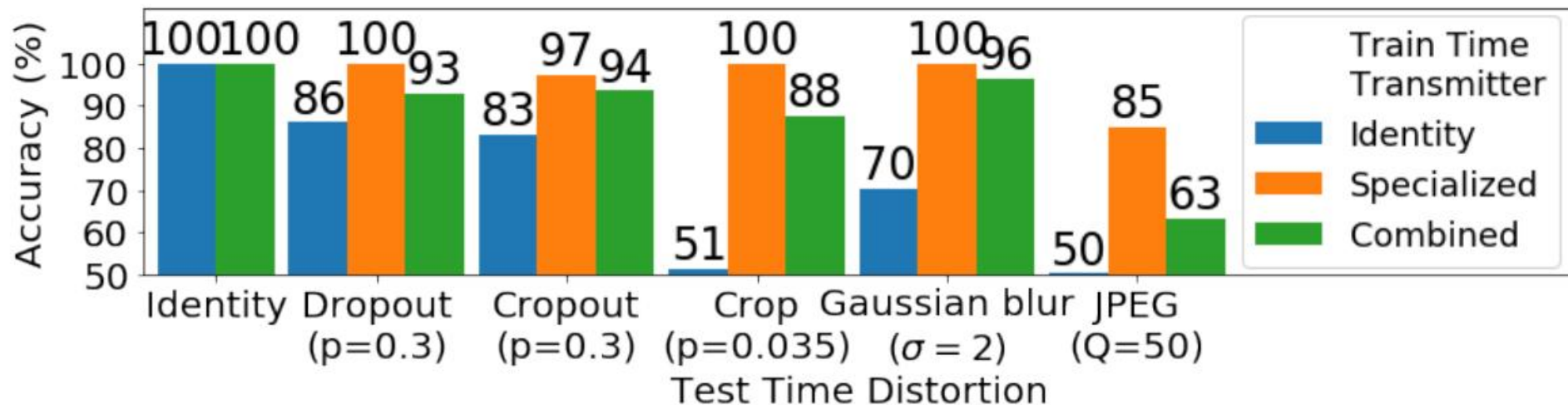


Fig. 9. Robustness of our models against different test time distortions. Each cluster uses a different test time distortion. Identity (blue) is trained with no image distortion; Specialized (orange) is trained on the same type of distortion used during testing; Combined (green) is trained on all types of distortions.

Convolutional Neural Network Architecture for Recovering Watermark Synchronization



W.-H. Kim, J.-U. Hou, S.-M. Mun, and H.-K. Lee, "Convolutional neural network architecture for recovering watermark synchronization," *arXiv preprint arXiv:1805.06199*, 2018.

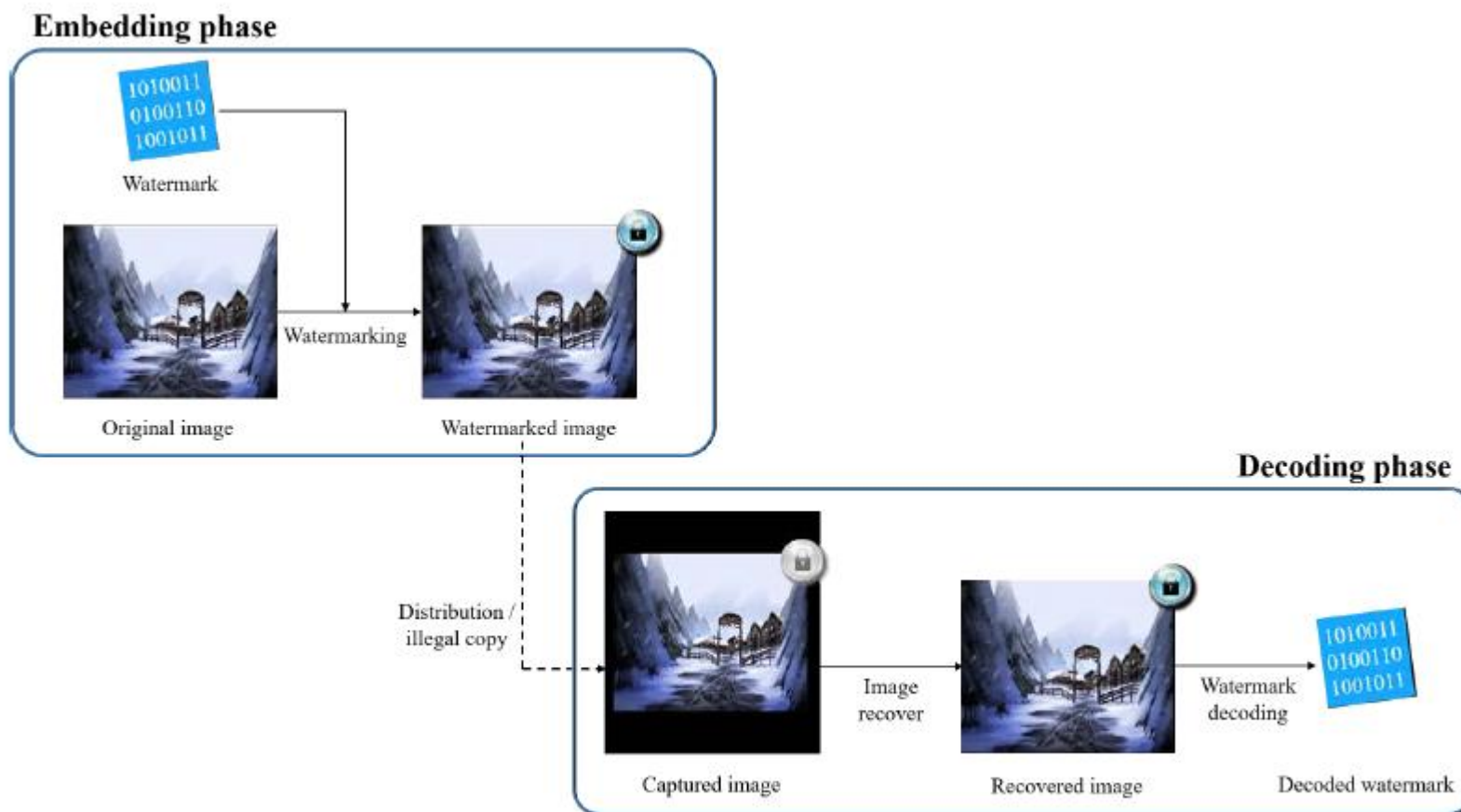


Fig. 1: Illegal distribution scenario for real-time web-comics and watermark extraction process suitable for this scenario. Illegal distribution is frequently caused by screen capturing on Internet browsers and smartphone applications. Scaling and translation occur in this process and these distortions must be corrected before watermark decoding.

Convolutional Neural Network Architecture for Recovering Watermark Synchronization

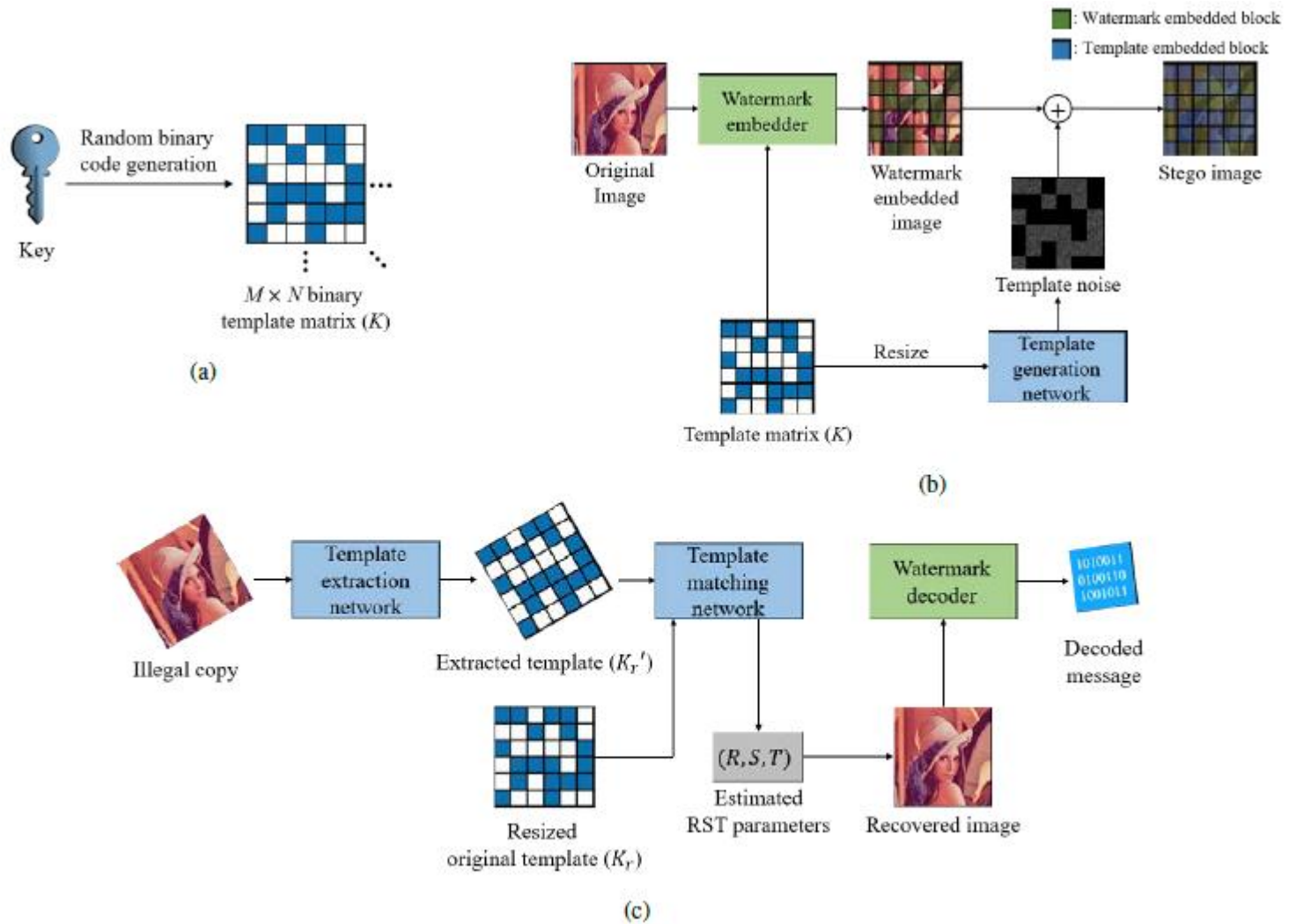


Fig. 2: An overview of the proposed method. (a) preparation step, (b) insertion step, and (c) decoding step. A Stego image is an image in which both a watermark and a template are inserted.

StegaStamp: Invisible Hyperlinks in Physical Photographs

M. Tancik, B. Mildenhall, and R. Ng, “Stegastamp: Invisible hyperlinks in physical photographs,” *arXiv preprint arXiv:1904.05343*, 2019.

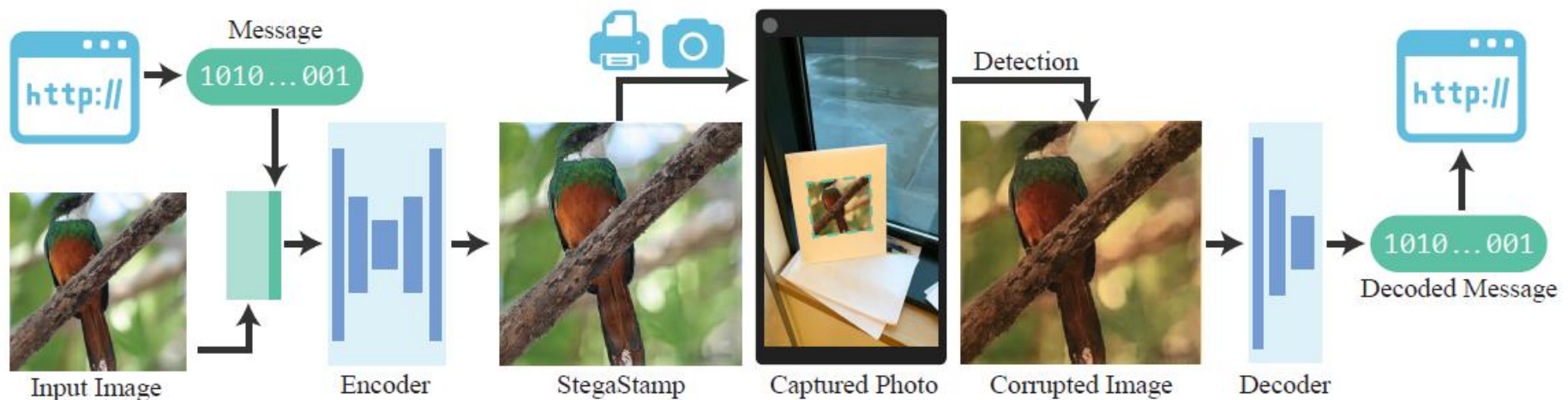


Figure 1: Our deep learning system is trained to hide hyperlinks in images. First, an encoder network processes the input image and hyperlink bitstring into a StegaStamp (encoded image). The StegaStamp is then printed and captured by a camera. A detection network localizes and rectifies the StegaStamp before passing it to the decoder network. After the bits are recovered and error corrected, the user can follow the hyperlink. To train the encoder and decoder networks, we simulate the corruptions caused by printing, reimaging, and detecting the StegaStamp with a set of differentiable image augmentations.

StegaStamp: Invisible Hyperlinks in Physical Photographs

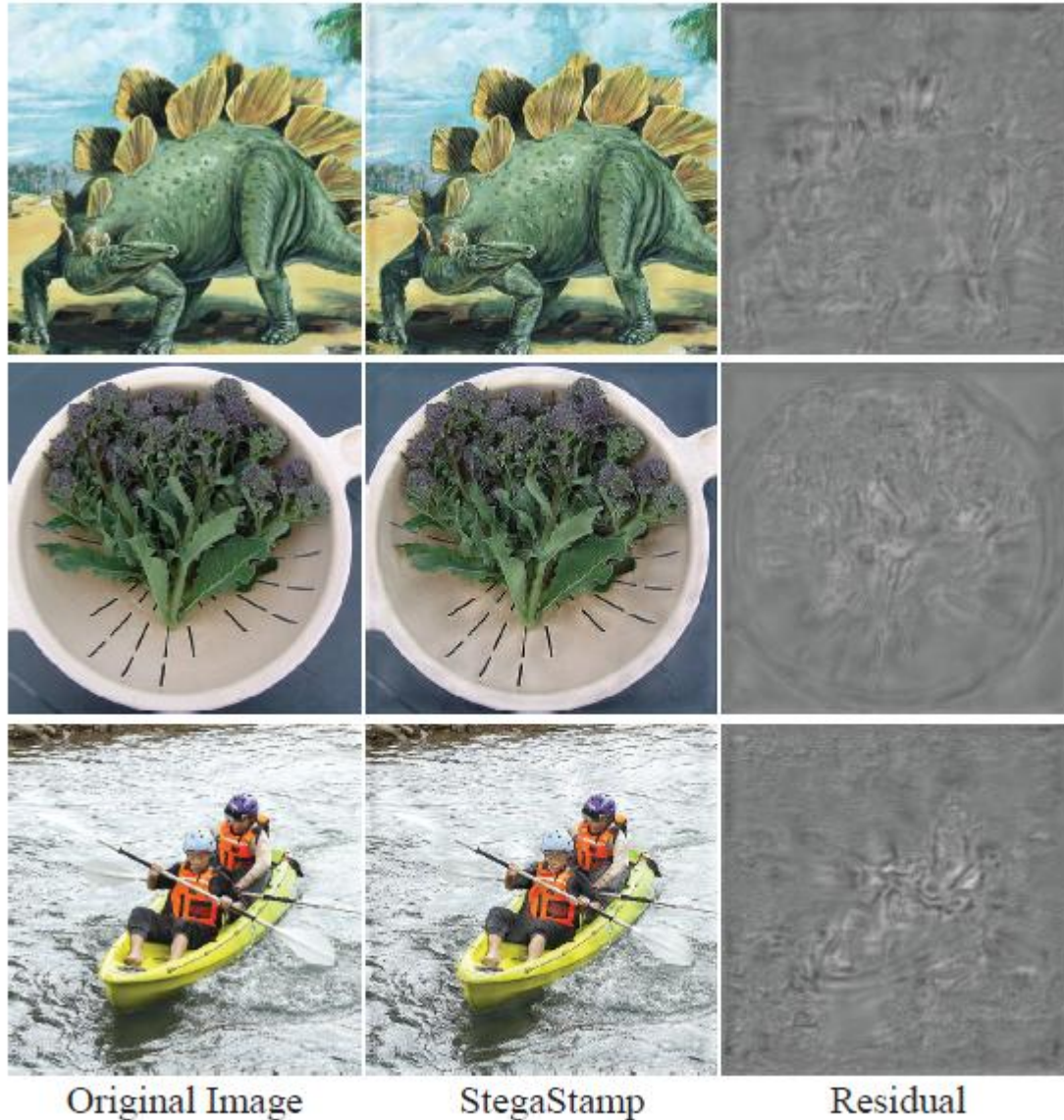
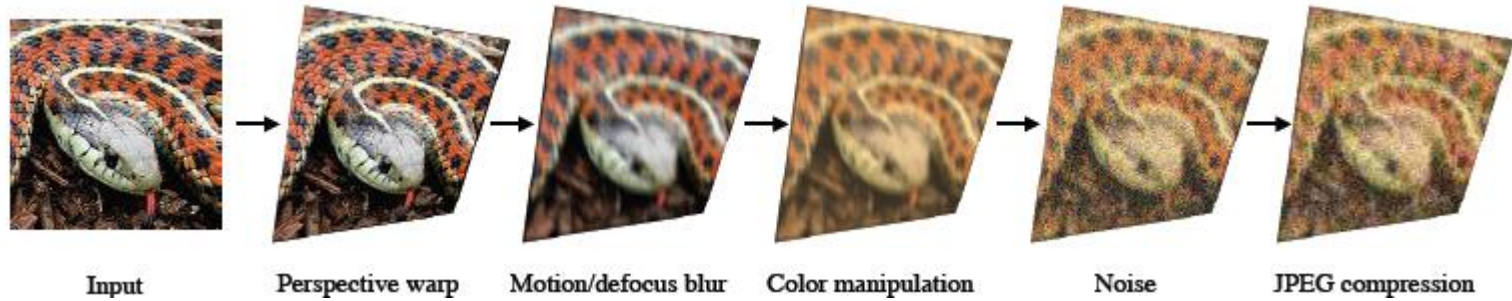


Figure 2: Examples of encoded images. The residual is calculated by the encoder network and added back to the original image to produce the encoded StegaStamp. These examples have 100 bit encoded messages and are robust to the image perturbations that occur through the printing and imaging pipelines.

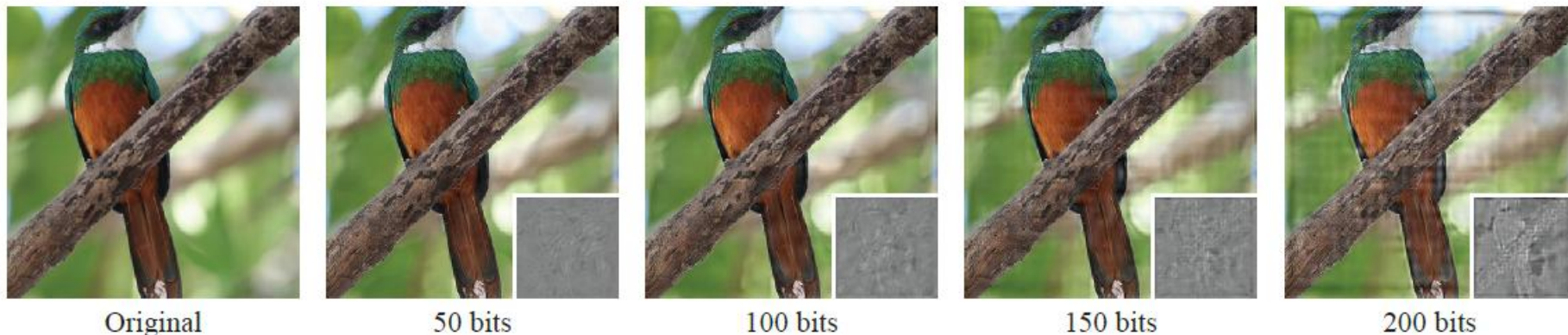
StegaStamp: Invisible Hyperlinks in Physical Photographs



StegaStamp: Invisible Hyperlinks in Physical Photographs

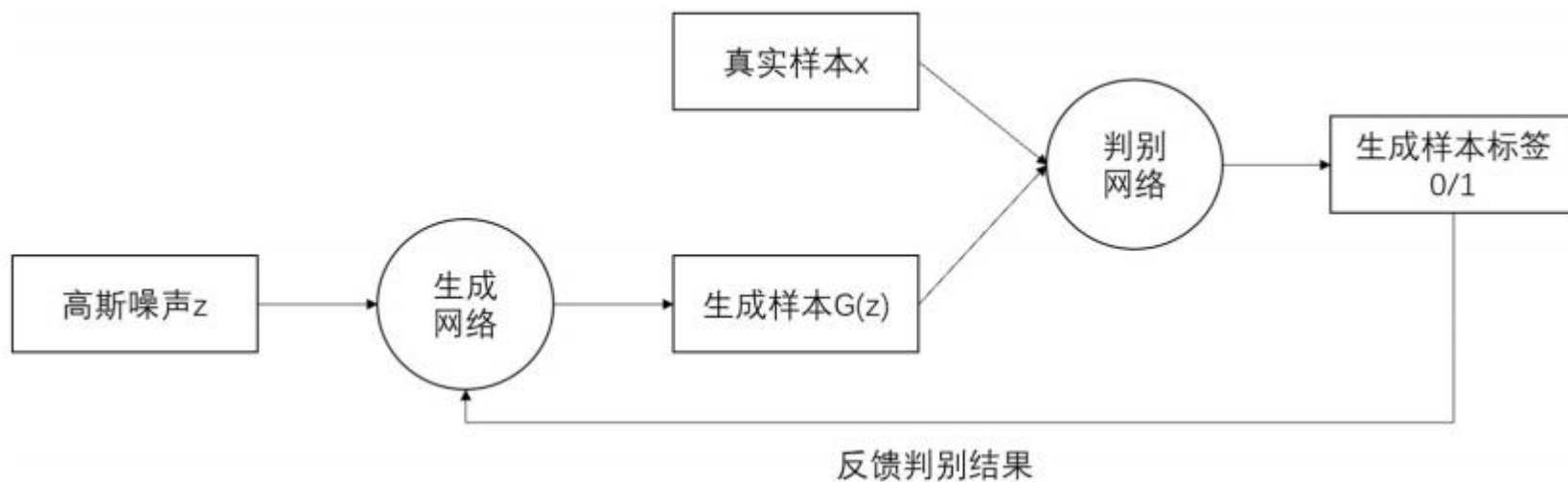


Figure 5: Despite not explicitly training the method to be robust to occlusion, we find that our decoder can handle partial erasures gracefully, maintaining high accuracy.



给生成图像添加水印——生成模型GAN

从生成的图像中可以提取出水印，水印既可作为“指纹”或“版权标识”追踪生成模型，还可以其为依据对图像进行真伪鉴别（到底是生成图像还是自然图像）



生成对抗网络（**Generative Adversarial Network, GAN**）原理框图

Artificial Fingerprinting for Generative Models: Rooting Deepfake Attribution in Training Data



Ning Yu^{1,2*} Vladislav Skripniuk^{3*} Sahar Abdelnabi³ Mario Fritz³

¹University of Maryland ²Max Planck Institute for Informatics

³CISPA Helmholtz Center for Information Security

{ningyu, vladislav}@mpi-inf.mpg.de

{sahar.abdelnabi, fritz}@cispa.saarland

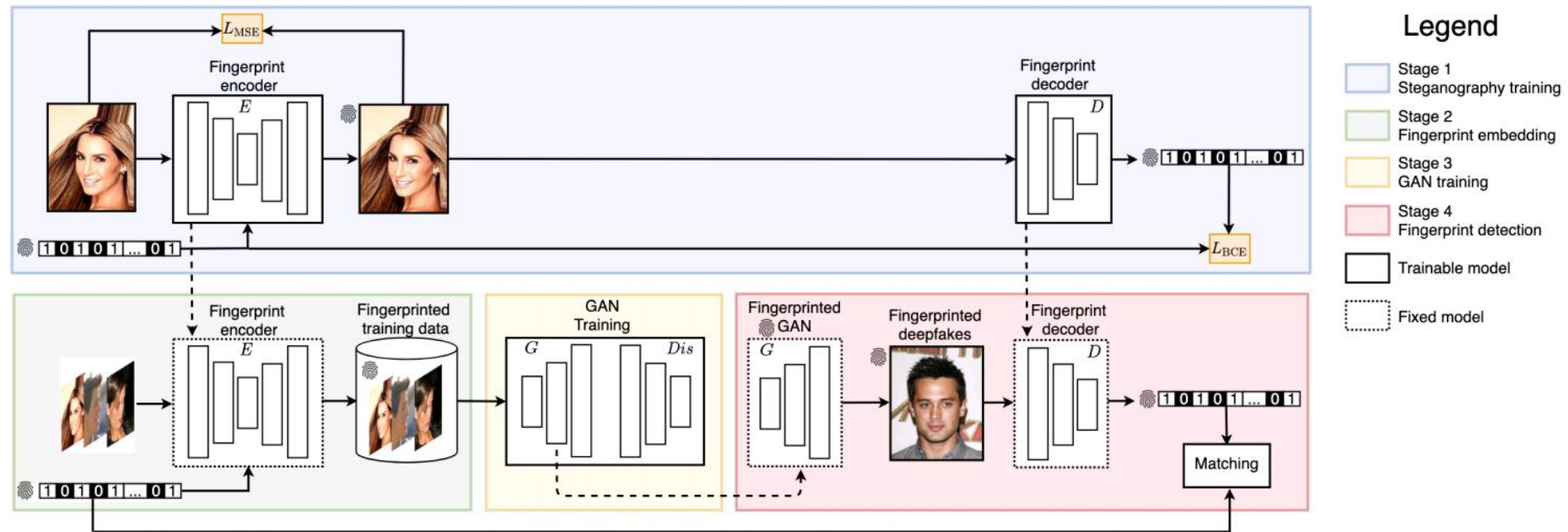
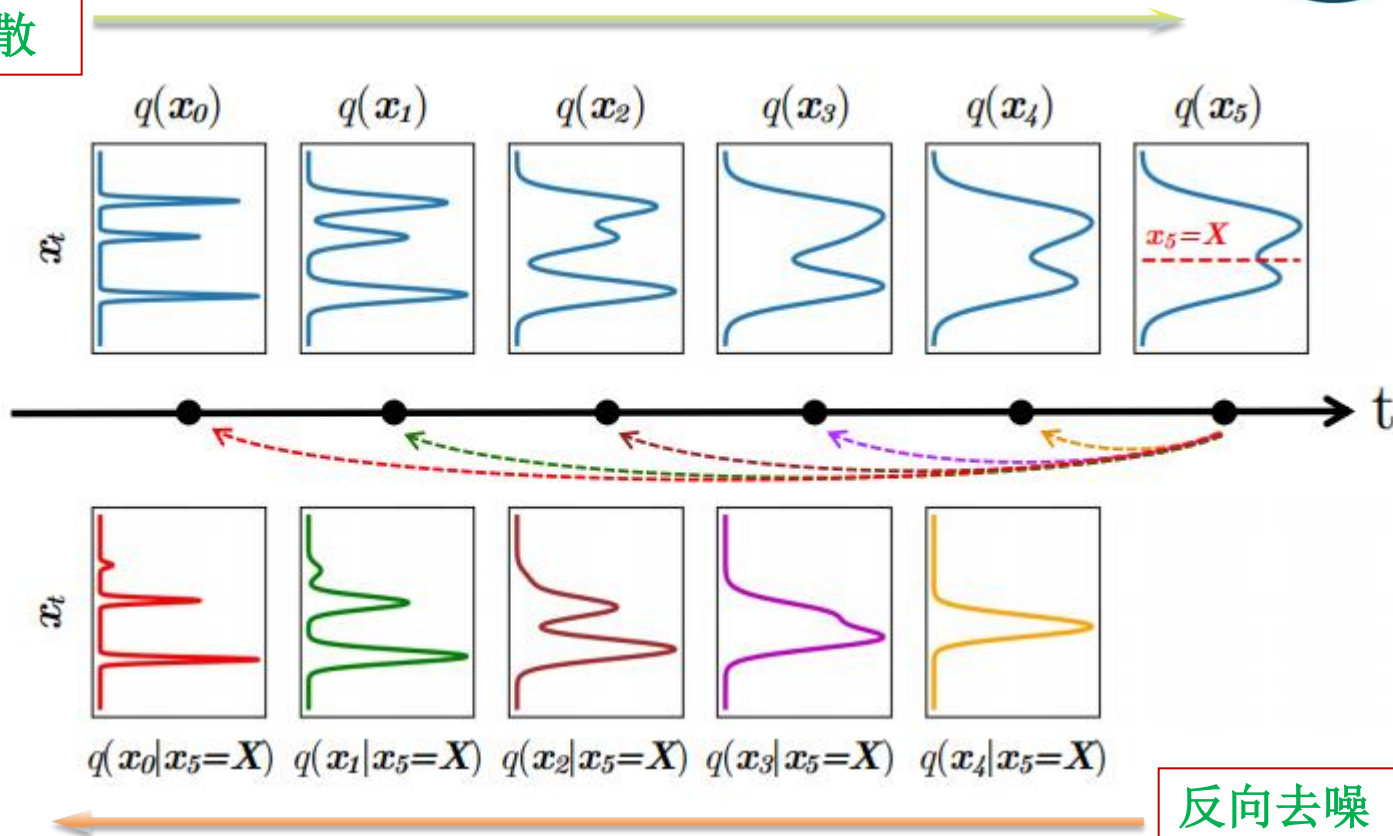


Figure 1: Our solution pipeline consists of four stages. We first train an image steganography encoder and decoder. Then we use the encoder to embed artificial fingerprints into the training data. After that, we train a generative model with its original protocol. Finally, we decode the fingerprints from the generated deepfakes.

给生成图像添加水印——基于扩散 (Diffusion)模型的生成

前向扩散

Marginal Diffused
Data Distributions



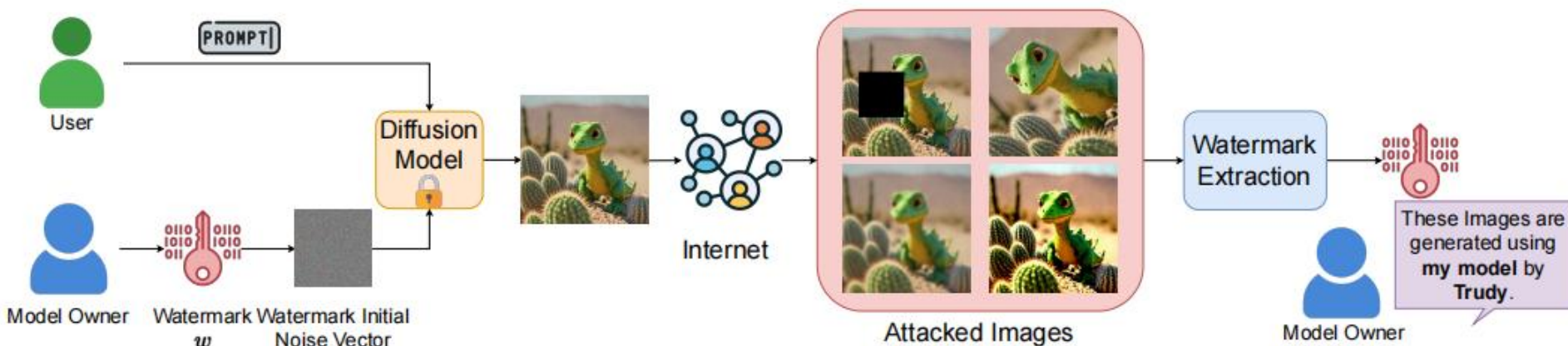
反向去噪

Figure 2: **Top:** The evolution of 1D data distribution $q(\mathbf{x}_0)$ through the diffusion process. **Bottom:**, The visualization of the true denoising distribution for varying step sizes conditioned on a fixed $\mathbf{x}_5$. The true denoising distribution for a small step size (i.e., $q(\mathbf{x}_4 | \mathbf{x}_5 = X)$) is close to a Gaussian distribution. However, it becomes more complex and multimodal as the step size increases.

给生成图像添加水印——基于扩散 (Diffusion) 模型的生成

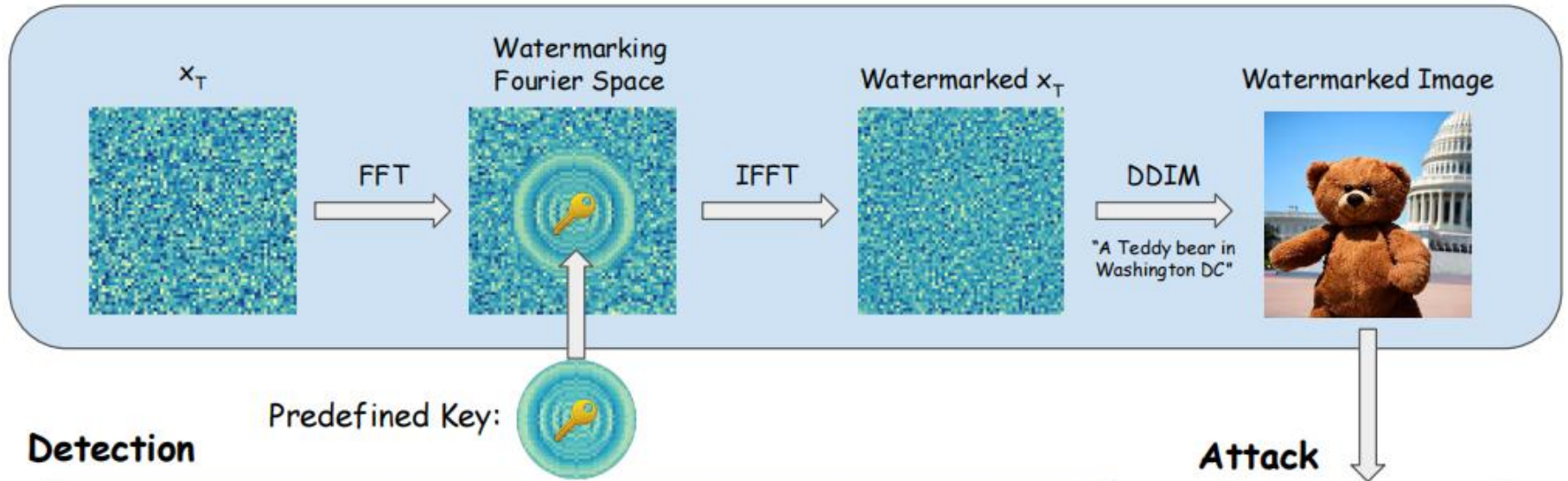
Sequentially noising an image in a diffusion forward process

前向扩散

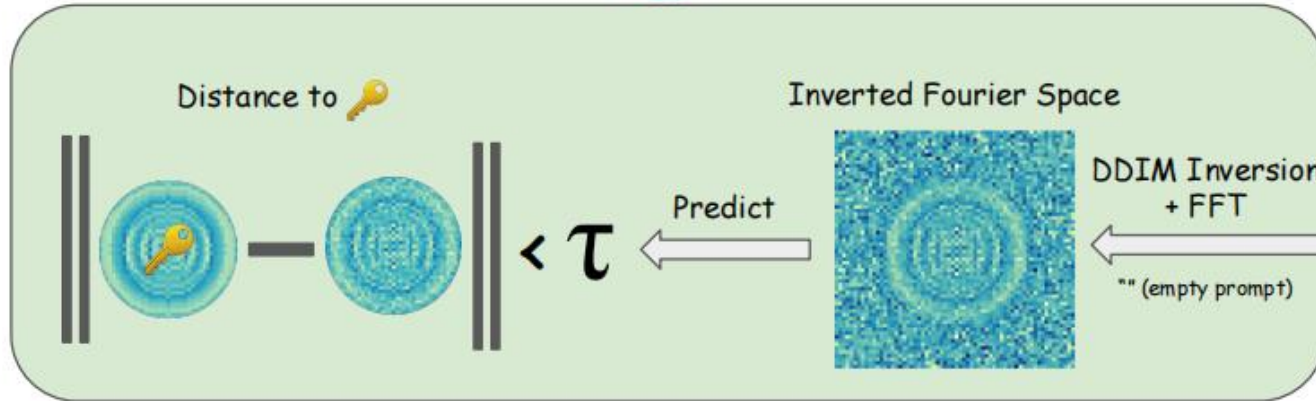


一般水印流程

Generation



Detection



Attack



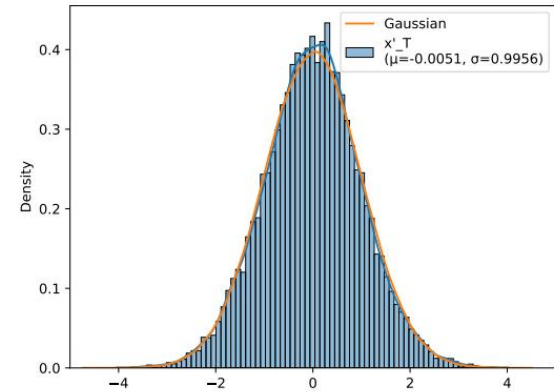
Figure 1: Pipeline for *Tree-Ring Watermarking*. A diffusion model generation is watermarked and later detected through ring-patterns in the Fourier space of the initial noise vector.

For the reverse diffusion process, **DDIM** [Song and Ermon, 2020] is an efficient deterministic sampling strategy, mapping from a Gaussian vector $x_T \sim \mathcal{N}(0, 1)$ to an image $x_0 \in q(x)$. For each

水印的嵌入

$$\mathcal{F}(x_T)_i \sim \begin{cases} k_i^* & \text{if } i \in M \\ \mathcal{N}(0, 1) & \text{otherwise.} \end{cases}$$

水印的统计检测



Histogram of the array x'_T obtained for a natural image, which is Gaussian.

For any test image x'_0 , we compute the approximate initial vector x'_T and then set $y = \mathcal{F}(x'_T)$. We then define the following null hypothesis

$$H_0 : y \text{ is drawn from a Gaussian distribution } \mathcal{N}(\mathbf{0}, \sigma^2 I_C). \quad (4)$$

Here, σ^2 is an unknown variance, which we estimate for each image² using the formula $\sigma^2 = \frac{1}{M} \sum_{i \in M} |y_i|^2$. To test this hypothesis, we define the score

$$\eta = \frac{1}{\sigma^2} \sum_{i \in M} |k_i^* - y|^2. \quad (5)$$

Our formula for η is closely related to Equation (3), but we switch to a sum-of-squares metric and remove the variance from y to simplify statistical analysis. When H_0 is true, the distribution of η is exactly a noncentral χ^2 distribution [Patnaik, 1949], with $|M|$ degrees of freedom and non-centrality parameter $\lambda = \frac{1}{\sigma^2} \sum_i |k_i^*|^2$.

We declare an image to be watermarked if the value of η is too small to occur by random chance. The probability of observing a value as small as η is given by the cumulative distribution function Φ_{χ^2} of the noncentral χ^2 distribution:

$$p = \Pr \left(\chi^2_{|M|, \lambda} \leq \eta \middle| H_0 \right) = \Phi_{\chi^2}(z). \quad (6)$$

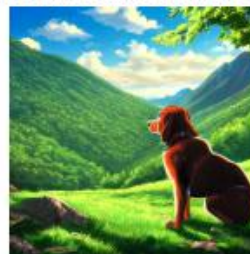
Φ_{χ^2} is a standard statistical function [Glasserman, 2003], available in `scipy` and many other statistics libraries.



“Anime art of a dog in Shenandoah National Park”



P-value = 0.27



3.73e-60



7.41e-16

“Synthwave style artwork of a person is kayaking in Acadia National Park”



0.15



1.38e-19



1.51e-8

“An astronaut riding a horse in Zion National Park”



0.91



9.91e-51



2.90e-05

“A painting of Yosemite National Park in Van Gogh style”



0.41



1.22e-35



9.46e-07

Figure 3: The qualitative results show three types of images: non-watermarked, *Tree-Rings* watermarked, and attacked watermarked images. A P-value is provided below each image, which corresponds to the probability of the detected watermark structure occurring by random chance. From top to bottom, the watermarked images are attacked by color jitter with a brightness factor of 6, Gaussian blur with an 8×8 filter size, Gaussian noise with $\sigma = 0.1$, and a 180° rotation, respectively.

Tree-ring watermarks: Fingerprints for diffusion images that are invisible and robust

We employ **Stable Diffusion-v2** [Rombach et al., 2022], an open-source, state-of-the-art latent text-to-image diffusion model, along with a 256×256 **ImageNet diffusion model**.

Table 1: Main Results. T@1%F represents TPR@1%FPR. We evaluate watermark accuracy in both benign and adversarial settings. Adversarial here refers to average performance over a battery of image manipulations. An extended version with additional details and standard error estimates can be found in Supplementary Material.

Model	Method	AUC/T@1%F (Clean)	AUC/T@1%F (Adversarial)	FID ↓	CLIP Score ↑
Stable Diff. FID = 25.29 CLIP Score = 0.363	DwtDct	0.974 / 0.624	0.574 / 0.092	25.10 _{.09}	0.362 _{.000}
	DwtDctSvd	1.000 / 1.000	0.702 / 0.262	25.01 _{.09}	0.359 _{.000}
	RivaGAN	0.999 / 0.999	0.854 / 0.448	24.51_{.17}	0.361 _{.000}
	Tree-Ring _{Zeros}	0.999 / 0.999	0.963 / 0.715	26.56 _{.07}	0.356 _{.000}
	Tree-Ring _{Rand}	1.000 / 1.000	0.918 / 0.702	25.47 _{.05}	0.363 _{.001}
	Tree-Ring _{Rings}	1.000 / 1.000	0.975 / 0.694	25.93 _{.13}	0.364_{.000}
ImageNet FID = 17.73	DwtDct	0.899 / 0.244	0.536 / 0.037	17.77 _{.01}	-
	DwtDctSvd	1.000 / 1.000	0.713 / 0.187	18.55 _{.02}	-
	RivaGAN	1.000 / 1.000	0.882 / 0.509	18.70 _{.02}	-
	Tree-Ring _{Zeros}	0.999 / 1.000	0.921 / 0.476	18.78 _{.00}	-
	Tree-Ring _{Rand}	0.999 / 1.000	0.940 / 0.585	18.68 _{.09}	-
	Tree-Ring _{Rings}	0.999 / 0.999	0.966 / 0.603	17.68_{.16}	-

Tree-Ring_{Zeros}: We choose the mask to be a circular region to preserve invariance to rotations in image space. The key is chosen to be an array of zeros, which creates invariance to shifts, crops, and dilations. This key is invariant to manipulations, but at the cost of departing severely from the Gaussian distribution. It also prevents multiple keys from being used to distinguish between models.

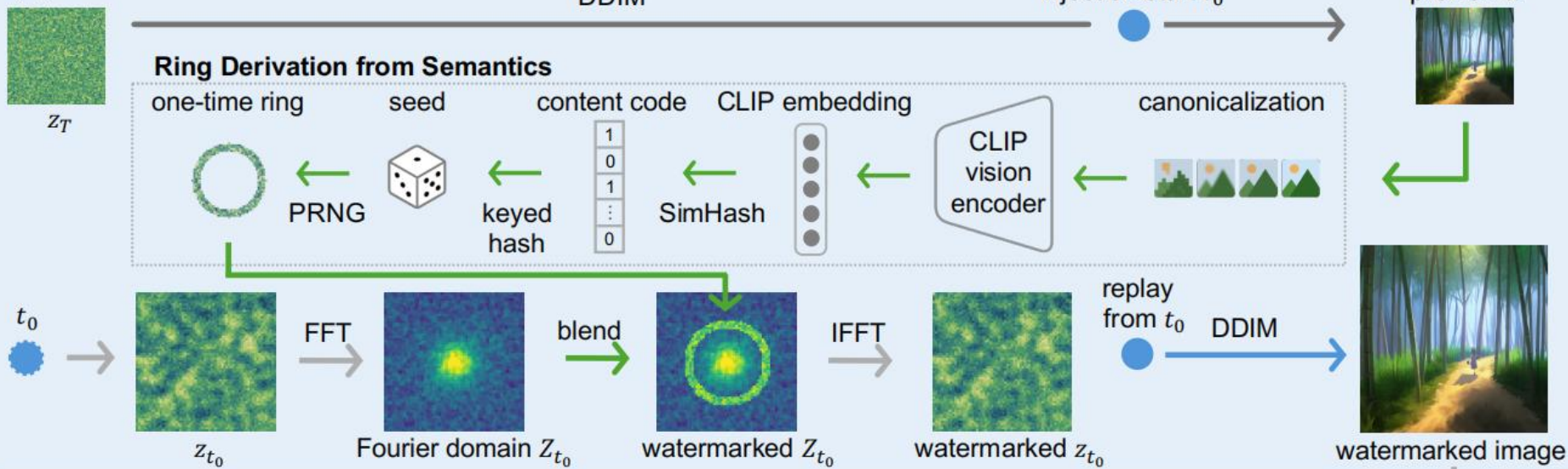
Tree-Ring_{Rand}: We draw the a fixed key k^* from a Gaussian distribution. The key has the same iid Gaussian nature as the original Fourier modes of the noise array, and so we anticipate this strategy will have the least impact on generation quality. This method also offers the flexibility for the model owner to possess multiple keys. However, it is not invariant to make image manipulations.

Tree-Ring_{Rings}: We introduce a pattern comprised of *multiple rings*, and constant value along each ring. This makes the watermark invariant to rotations. We choose the constant ring values from a Gaussian distribution. This provides some invariance to multiple types of image transforms, while also ensuring that the overall distribution is only minimally shifted from an isotropic Gaussian.

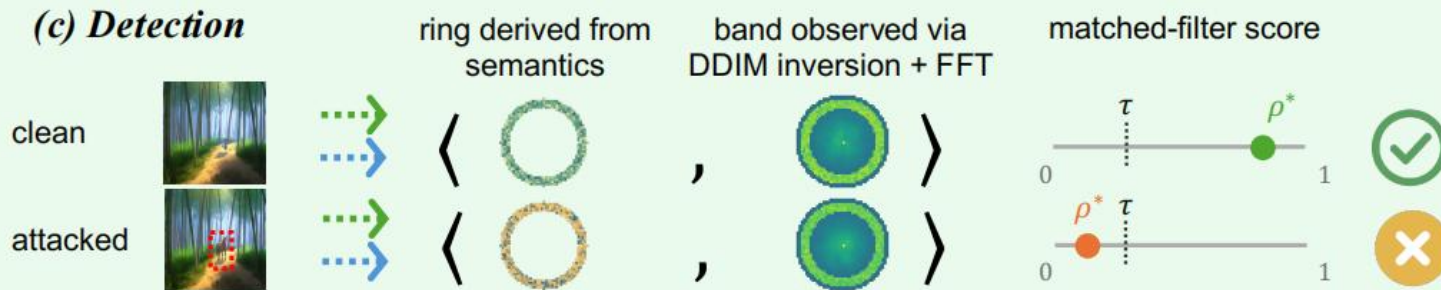
Table 2: AUC under each Attack for Stable Diffusion, showing the effectiveness of *Tree-Ring_{Rings}* over a number of augmentations. Cr. & Sc. refers to random cropping and rescaling. Additional results for the ImageNet model can be found in Supplementary Material.

Method	Clean	Rotation	JPEG	Cr. & Sc.	Blurring	Noise	Color Jitter	Avg
DwtDct	0.974	0.596	0.492	0.640	0.503	0.293	0.519	0.574
DwtDctSvd	1.000	0.431	0.753	0.511	0.979	0.706	0.517	0.702
RivaGan	0.999	0.173	0.981	0.999	0.974	0.888	0.963	0.854
<i>T-R_{Zeros}</i>	0.999	0.994	0.984	0.999	0.977	0.877	0.907	0.963
<i>T-R_{Rand}</i>	1.000	0.486	0.999	0.971	0.999	0.972	0.994	0.918
<i>T-R_{Rings}</i>	1.000	0.935	0.999	0.961	0.999	0.944	0.983	0.975

(a) Generation



(c) Detection



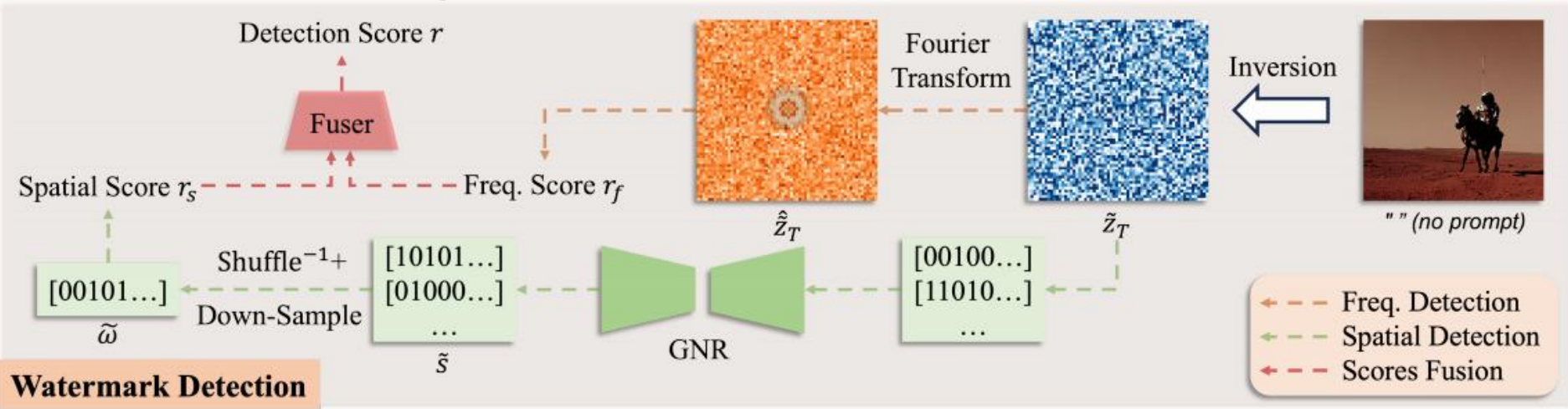
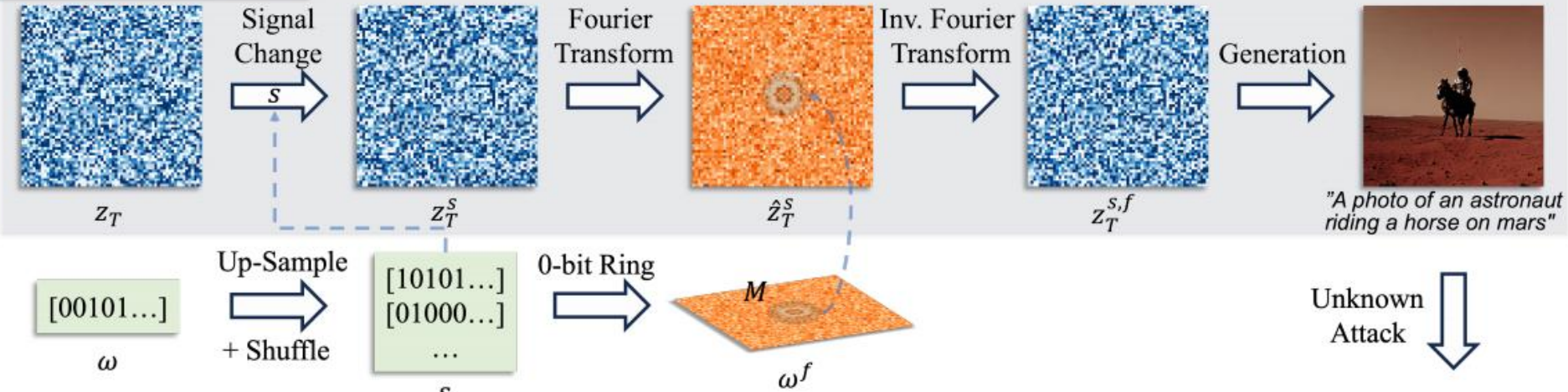
(b) Attack



Figure 1: **The IRIS framework.** (a) Generation turns the canonicalized CLIP embedding of the preview into a keyed one-time ring and blends it into the late DDIM steps of the same trajectory. (b) An attack replaces part of the content. (c) Detection correlates the ring derived from the query’s semantics against the observed band. The replacement moves the recomputed code, the keyed hash maps it to a different ring, and the score falls below τ , while the clean image verifies.

Li, K., Huang, Z., Hou, X., and Hong, C. **Gaussmarker: Robust dual-domain watermark for diffusion models**. In *ICML, 2025a. Proceedings of the 42 nd International Conference on Machine Learning*, Vancouver, Canada. PMLR 267, 2025. 引入高斯噪声恢复器（GNR），对旋转和裁剪攻击表现优异。

Watermark Injection



Watermark Detection

Figure 1. Overview of GaussMarker. The l -bits watermark ω is up-sampled and shuffled into a signal map s for l -bits spatial-domain watermark. s is used to sample a Fourier map ω^f for zero-bit frequency-domain watermark. Both s and ω^f are fixed during the injection and detection. (1) Watermark Injection. We inject a multi-bit watermark and a zero-bit watermark into the spatial domain and frequency domain of Gaussian noise map z_T respectively. (2) Watermark Detection. We extract the frequency and spatial detection scores simultaneously with GNR enhancing the robustness, and fuse two scores to make more robust detection.

Two popular **subject-driven image synthesis** models



Figure 6: Images synthesized based on the original artwork from Edward Hopper (top) by Textual Inversion (middle) and DreamBooth (bottom). The prompt content is “A painting of the reader in the style of [V]”.



Two popular subject-driven image synthesis models

Textual Inversion. Textual Inversion learns the embedding of the subject “[V]” while freezing the parameters of the text-to-image model, here, Latent Diffusion. The training takes input pairs of “[V]” and images. During synthesis, if we want to synthesize a painting of a daisy in the same style of an artist [V], the new prompt could be something like “*A painting of daisy in the style of [V]*”.

DreamBooth. Unlike textual inversion, DreamBooth is based on fine-tuning model parameters and uses Stable Diffusion as the backbone. Specifically, the prompt is designed to contain an additional term denoting the category of the subject. In this case, the above prompt example becomes “*A painting of a daisy in the style of [V] painting*”.

The Stable Signature: Rooting Watermarks in Latent Diffusion Models

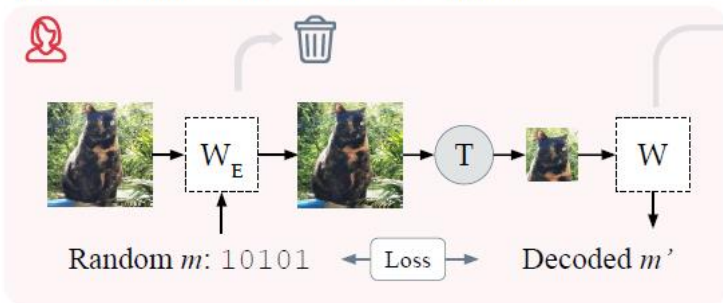
Pierre Fernandez^{1,2} Guillaume Couairon^{1,3} Hervé Jégou¹ Matthijs Douze¹ Teddy Furon²

¹Meta AI

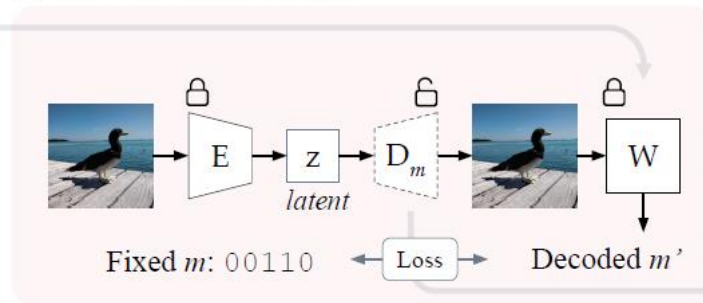
²Centre Inria de l'Université de Rennes

³Sorbonne University

(a) Pre-train watermark encoder/extractor



(b) Fine-tune LDM decoder



(c) Generate

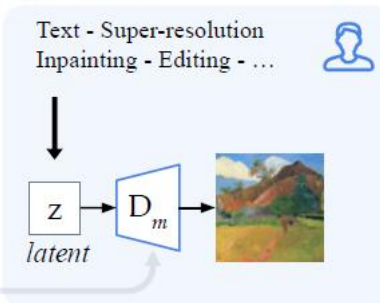


Figure 2. Steps of the method. (a) We pre-train a watermark encoder $\mathcal{W}_E$ and extractor $\mathcal{W}$, to extract binary messages. (b) We fine-tune the decoder $\mathcal{D}$ of the LDM's auto-encoder with a fixed signature m such that all the generated images (c) lead to m through $\mathcal{W}$.

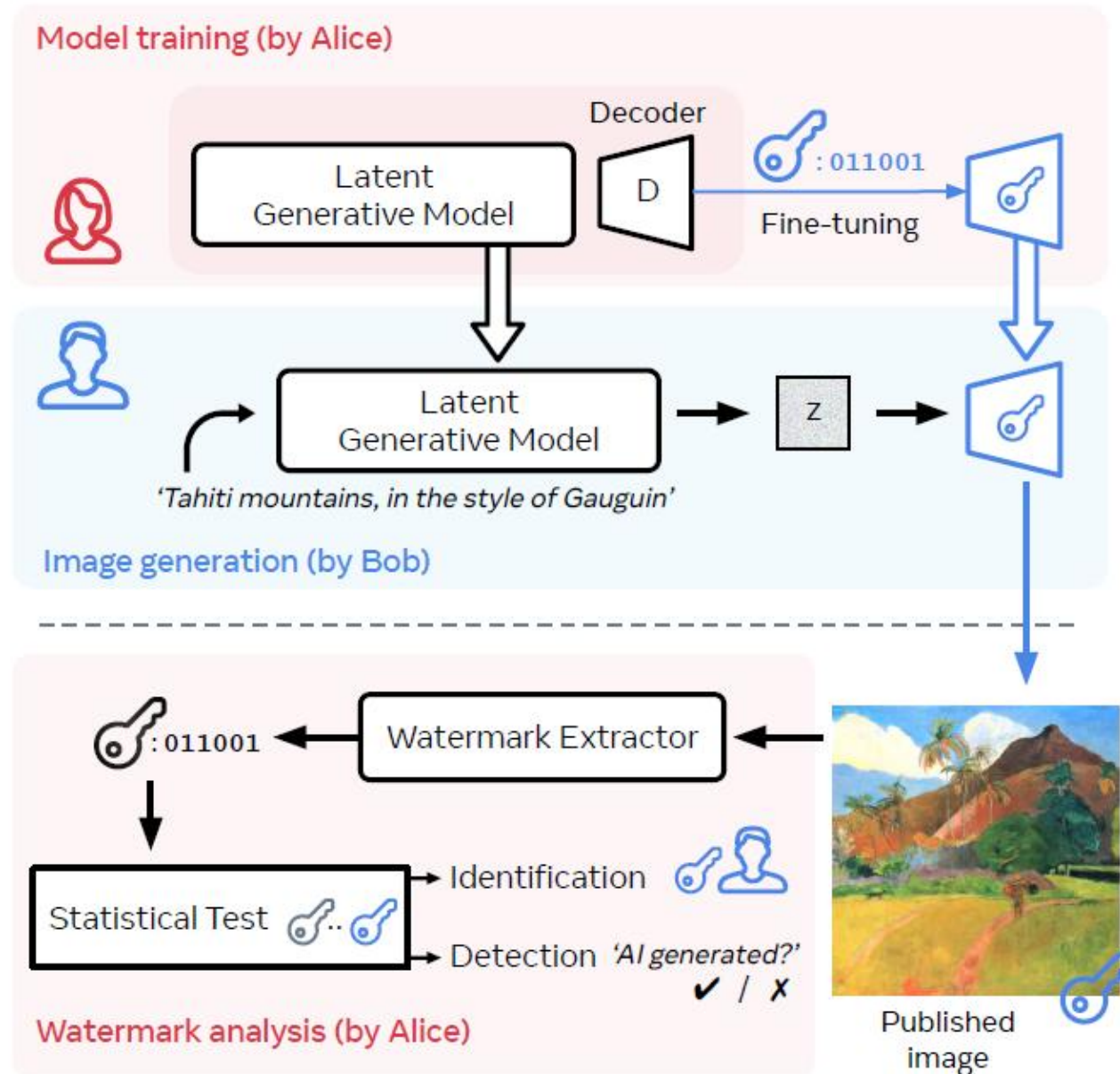


Figure 1. Overview. The latent decoder can be fine-tuned to preemptively embed a signature into all generated images.

The Stable Signature: Rooting Watermarks in Latent Diffusion Models

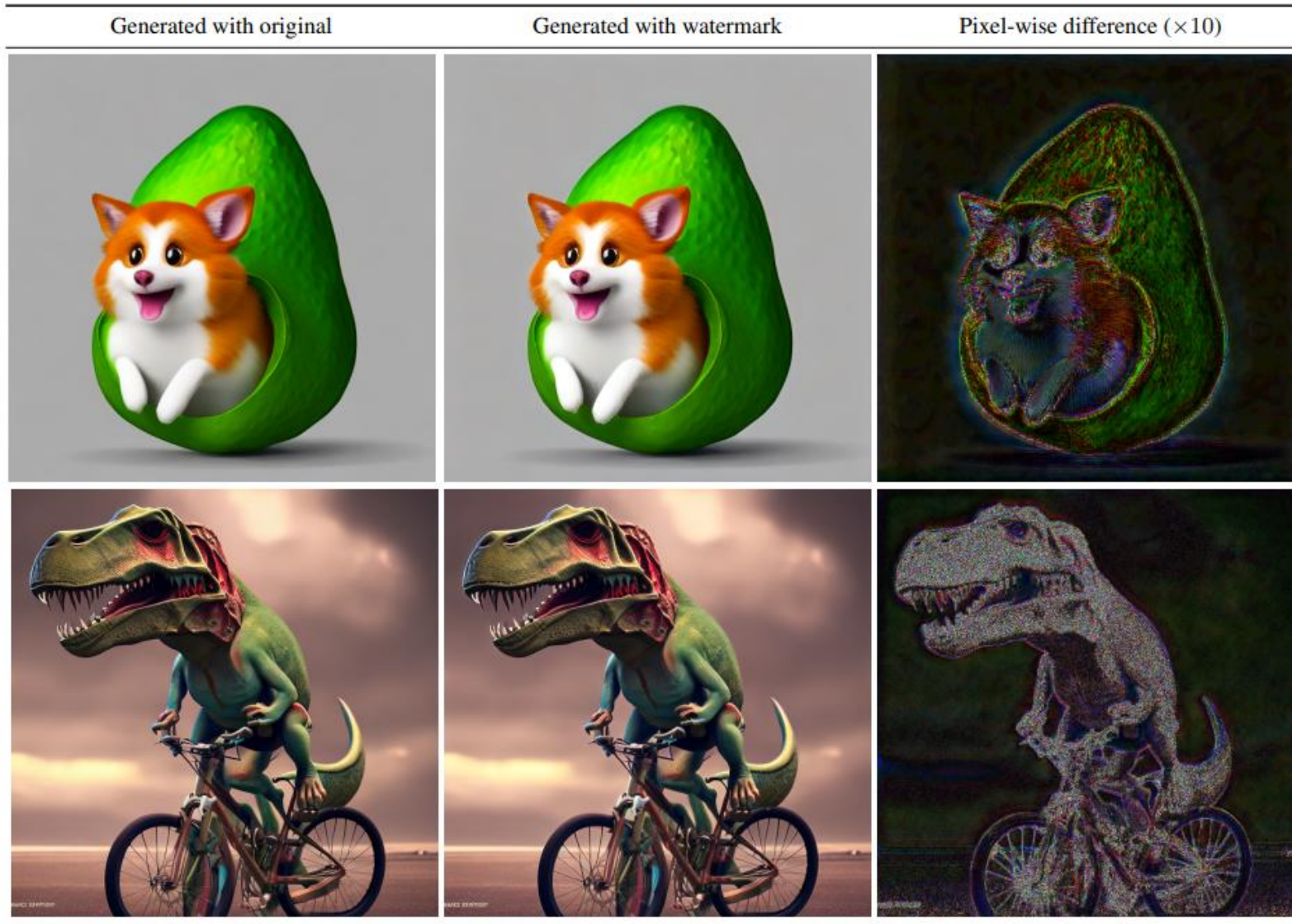


Figure 6. Images generated from text with or without the watermarked generative models. The PSNR is 35.4dB in the first row and 28.6dB in the second. Images generated with Stable Signature look natural because modified areas are located where the eye is not sensitive. More examples are available in Appendix C.

The Stable Signature: Rooting Watermarks in Latent Diffusion Models

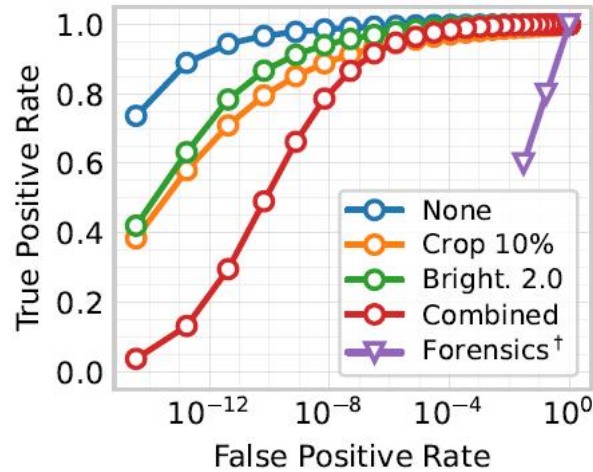


Figure 3. **Detection results.** TPR/FPR curve of the detection under different transformations. Forensics[†] indicates passive detection (without watermark) [12].

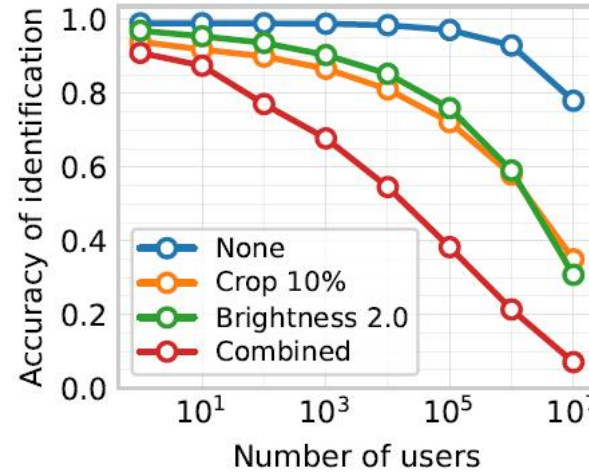


Figure 4. **Identification results.** Proportion of well-identified users. Detection with $FPR=10^{-6}$ is run beforehand, and we consider it an error if the image is not flagged.

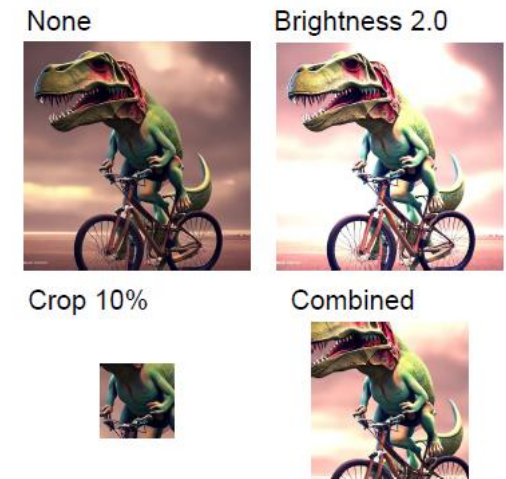


Figure 5. Transformations evaluated in Sec. 4 & 5. ‘Combined’ is made of crop 50%, brightness adjustment 1.5 and JPEG 80 compression.

The Stable Signature: Rooting Watermarks in Latent Diffusion Models

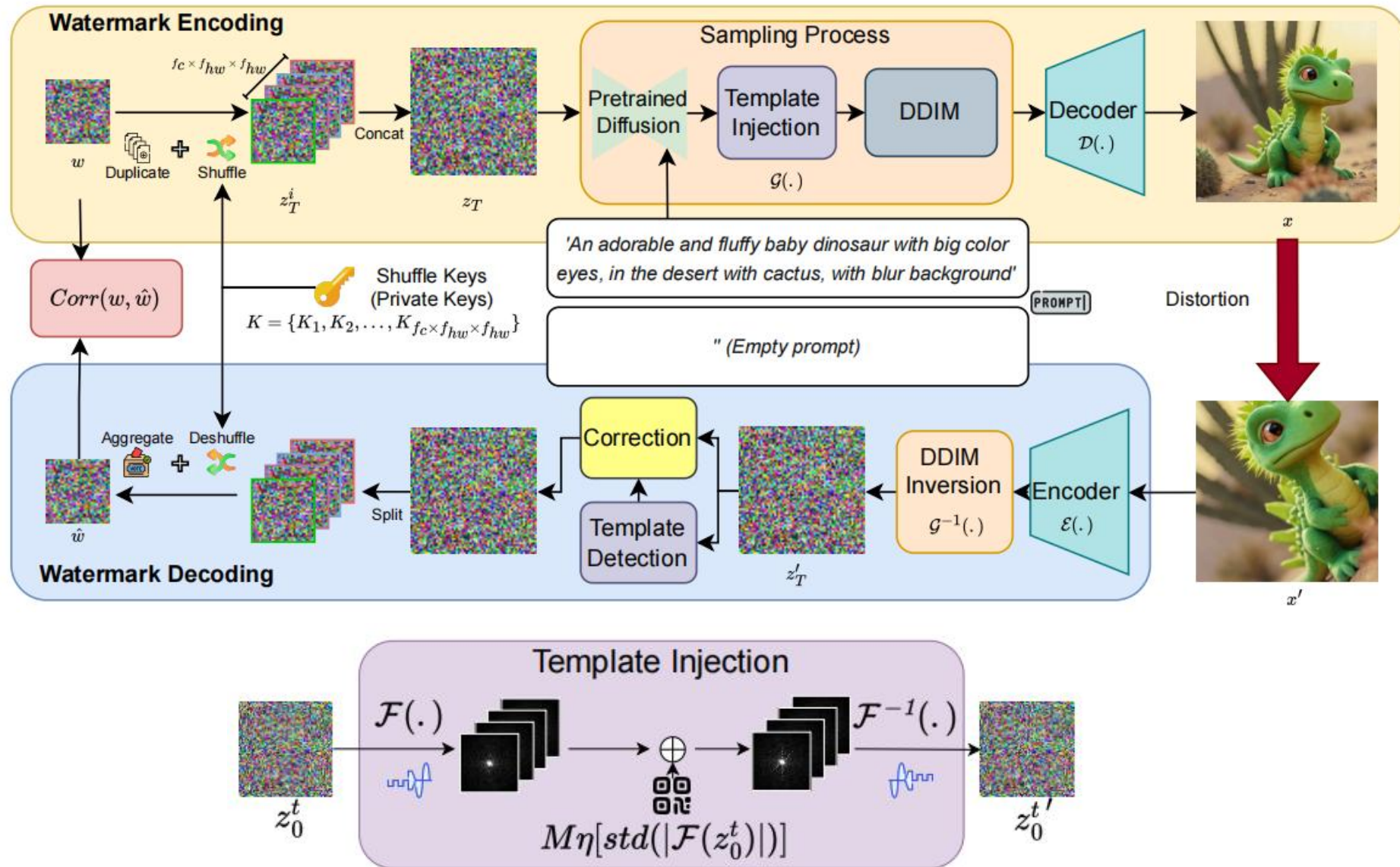


Table 1. Generation quality and comparison to post-hoc watermarking (**post-hoc** , an alternative way to watermark generated images is to process them after the generation) on 512×512 images and 48-bit signatures. PSNR and SSIM are computed between generations of the original and watermarked generators.

			PSNR / SSIM $\uparrow$	FID $\downarrow$	Bit accuracy $\uparrow$ on:			
					None	Crop	Brigh.	Comb.
Tasks	Text-to-Image	LDM [64]	30.0 / 0.89	19.6 (−0.3)	0.99	0.95	0.97	0.92
	Image Edition	DiffEdit [13]	31.2 / 0.92	15.0 (−0.3)	0.99	0.95	0.98	0.94
	Inpainting - Full - Mask only	Glide [56]	31.1 / 0.91	16.8 (+0.6)	0.99	0.97	0.98	0.93
			37.8 / 0.98	9.0 (+0.1)	0.89	0.76	0.84	0.78
	Super-Resolution	LDM [64]	34.0 / 0.94	11.6 (+0.0)	0.98	0.93	0.96	0.92
WM Methods	<i>Post generation</i>							
	Dct-Dwt [14]	0.14 (s/img)	39.5 / 0.97	19.5 (−0.4)	0.86	0.52	0.51	0.51
	SSL Watermark [24]	0.45 (s/img)	31.1 / 0.86	20.6 (+0.7)	1.00	0.73	0.93	0.66
	FNNS [41]	0.28 (s/img)	32.1 / 0.90	19.0 (−0.9)	0.93	0.93	0.91	0.93
	HiDDeN [104]	0.11 (s/img)	32.0 / 0.88	19.7 (−0.2)	0.99	0.97	0.99	0.98
	<i>Merged in generation</i>							
	Stable Signature	0.00 (s/img)	30.0 / 0.89	19.6 (−0.3)	0.99	0.95	0.97	0.92

MaXsive: High-Capacity and Robust Training-Free Generative Image Watermarking

Po-Yuan Mao, Cheng-Chang Tsai and Chun-Shien Lu. MaXsive: High-Capacity and Robust Training-Free Generative Image Watermarking in Diffusion Models, MM '25, October 27–31, 2025, Dublin, Ireland.



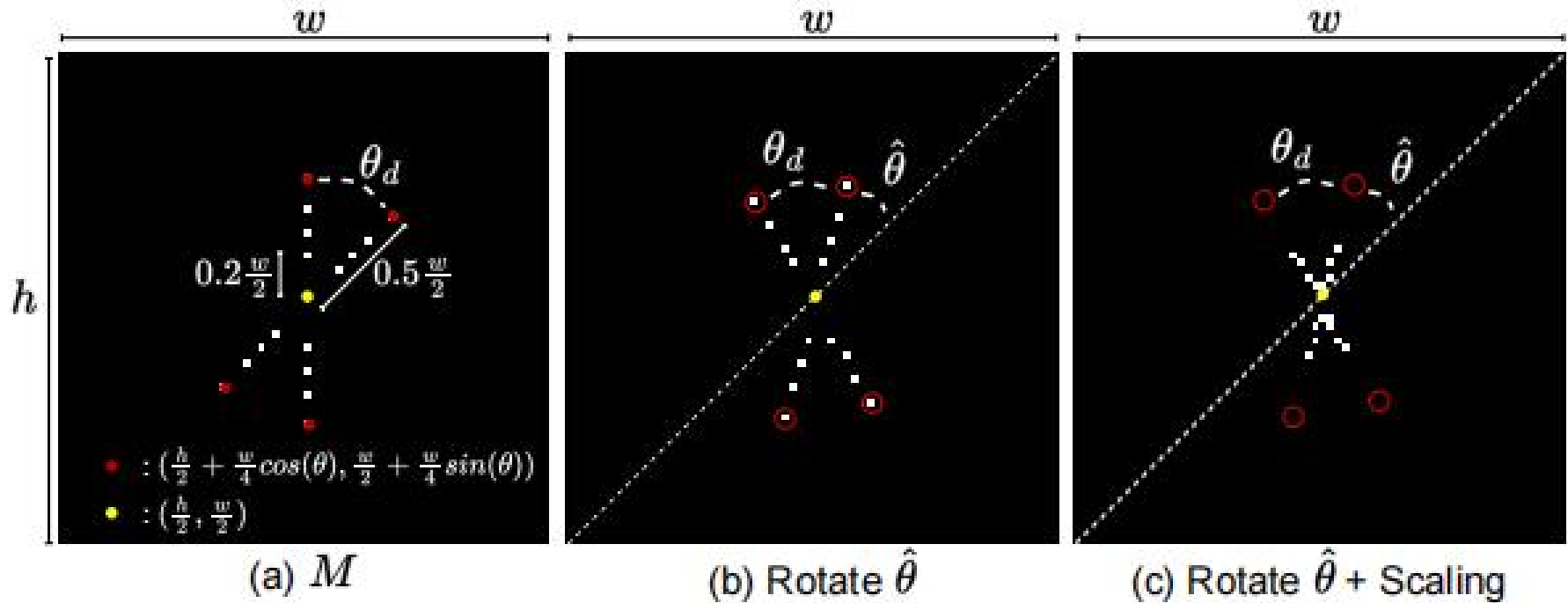


Figure 3: Illustration of Transformation: Rotation and Rotation with Scaling. (a) shows the defined template pattern. The yellow point indicates the center of the image, while the red points mark the outermost selected positions. (b) and (c) illustrate transformation behavior after applying rotation, and rotation combined with scaling, respectively. The red circle highlights a reference position used to verify whether the image has been scaled.

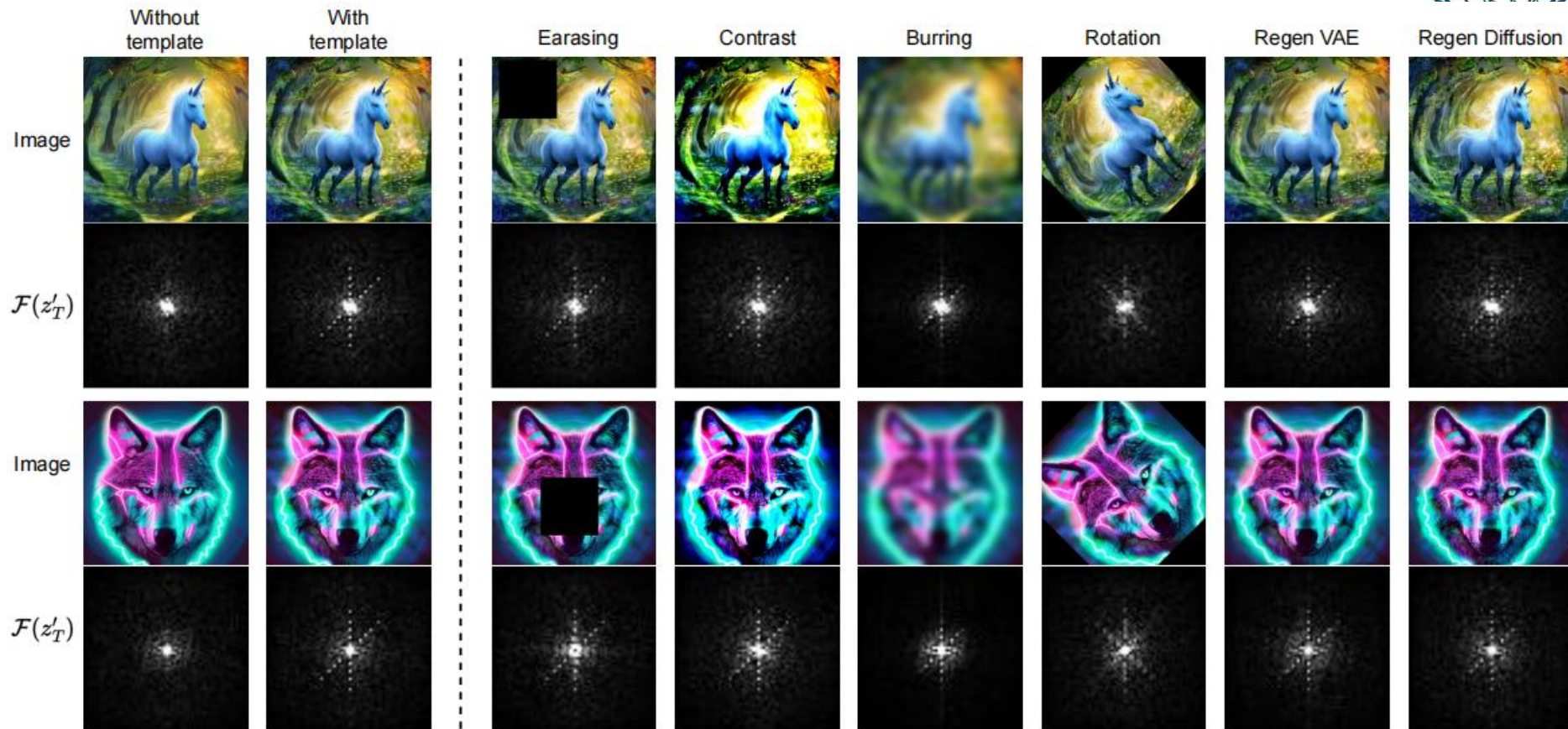


Figure 6: Visualization of the template space (second and fourth rows) under various distortions. To the left of the dashed line, images generated by Stable Diffusion 2.1 share the same prompt and initial noise, with template injections applied in the second column. The corresponding distorted template images are presented to the right of the dashed line.

给生成图像添加水印——基于扩散 (Diffusion) 模型的生成

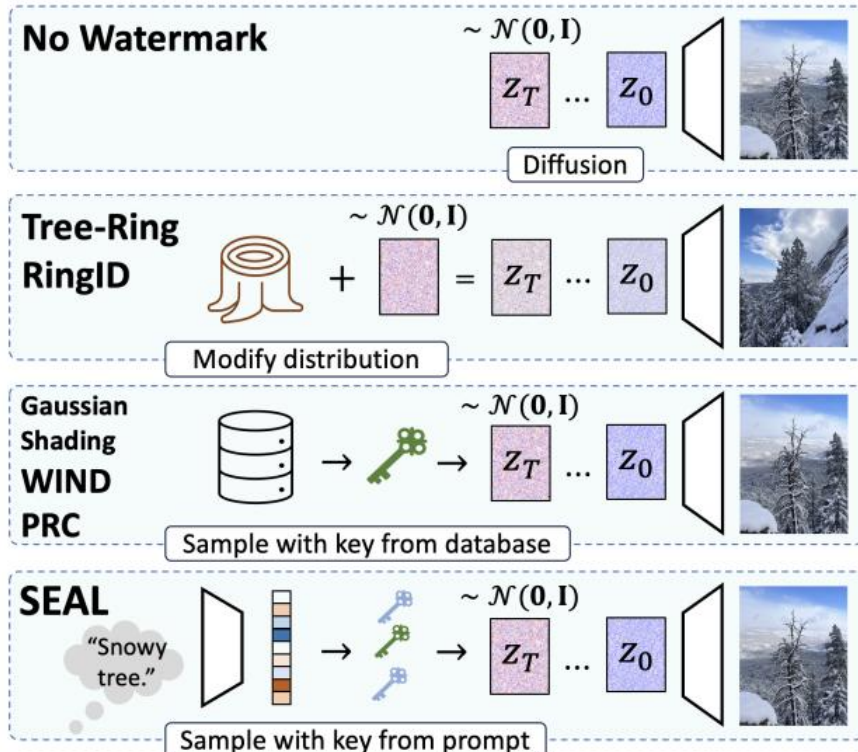
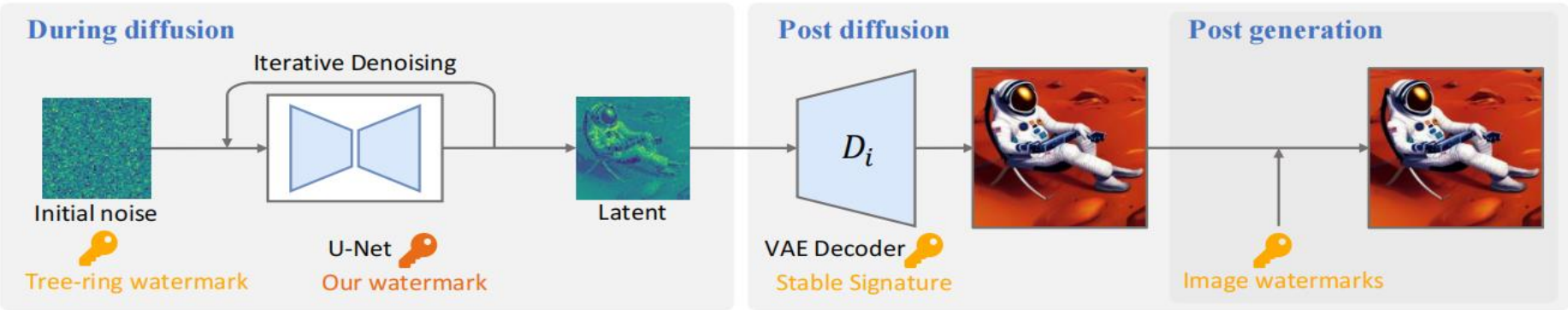
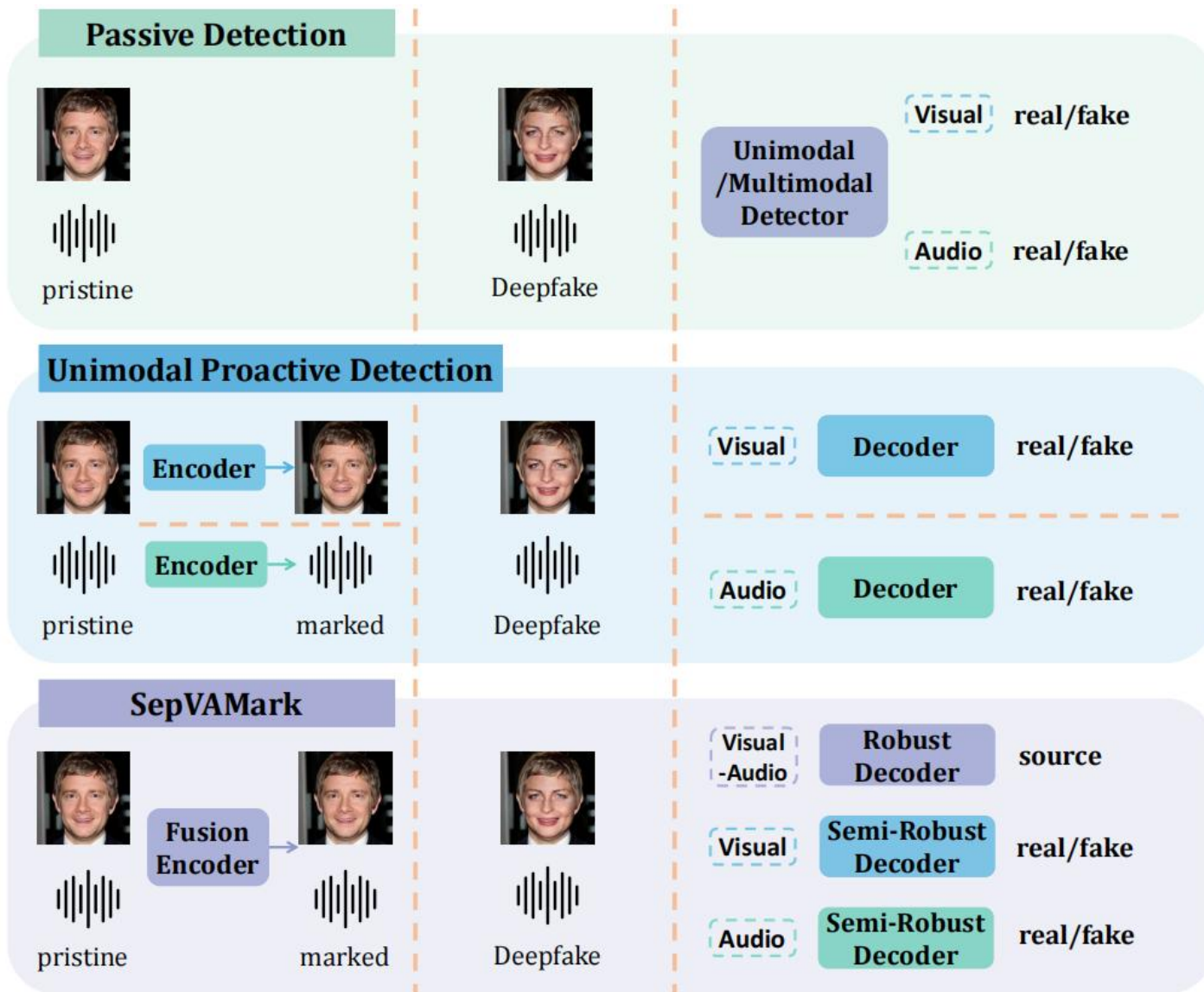


Figure 1. *Illustration of different watermarking frameworks using the initial noise of diffusion models. No Watermark:* A diffusion model maps pure Gaussian noise to an image. **Tree-Ring:** A pattern is added to the initial noise, modifying the distribution of generated images in a detectable way. **Key-Based Watermarking:** A key is sampled to generate distortion-free images linked to the key. **Ours (SEAL):** The initial noise is conditioned on multiple keys derived from the image's semantic embedding, with each key influencing a different patch.

Arabi K, Witter R T, Hegde C, et al. SEAL: Semantic Aware Image Watermarking[C]. Proceedings of the IEEE/CVF International Conference on Computer Vision. Piscataway: IEEE, 2025:16196-16205.

Proactive Deepfake Defense: Watermarking



CHUAN ZHANG, ZIHAN LI, ZIHAO XU, et al. SepVAMark: Deep Separable Visual-Audio Fusion Watermarking for Source Tracing and Deepfake Detection, MM '25, October 27–31, 2025, Dublin, Ireland

Peipeng Yu, Jinfeng Xie, Chengfu Ou, et al. High-Fidelity Face Content Recovery via Tamper-Resilient Versatile Watermarking, <https://arxiv.org/abs/2603.23940v1>

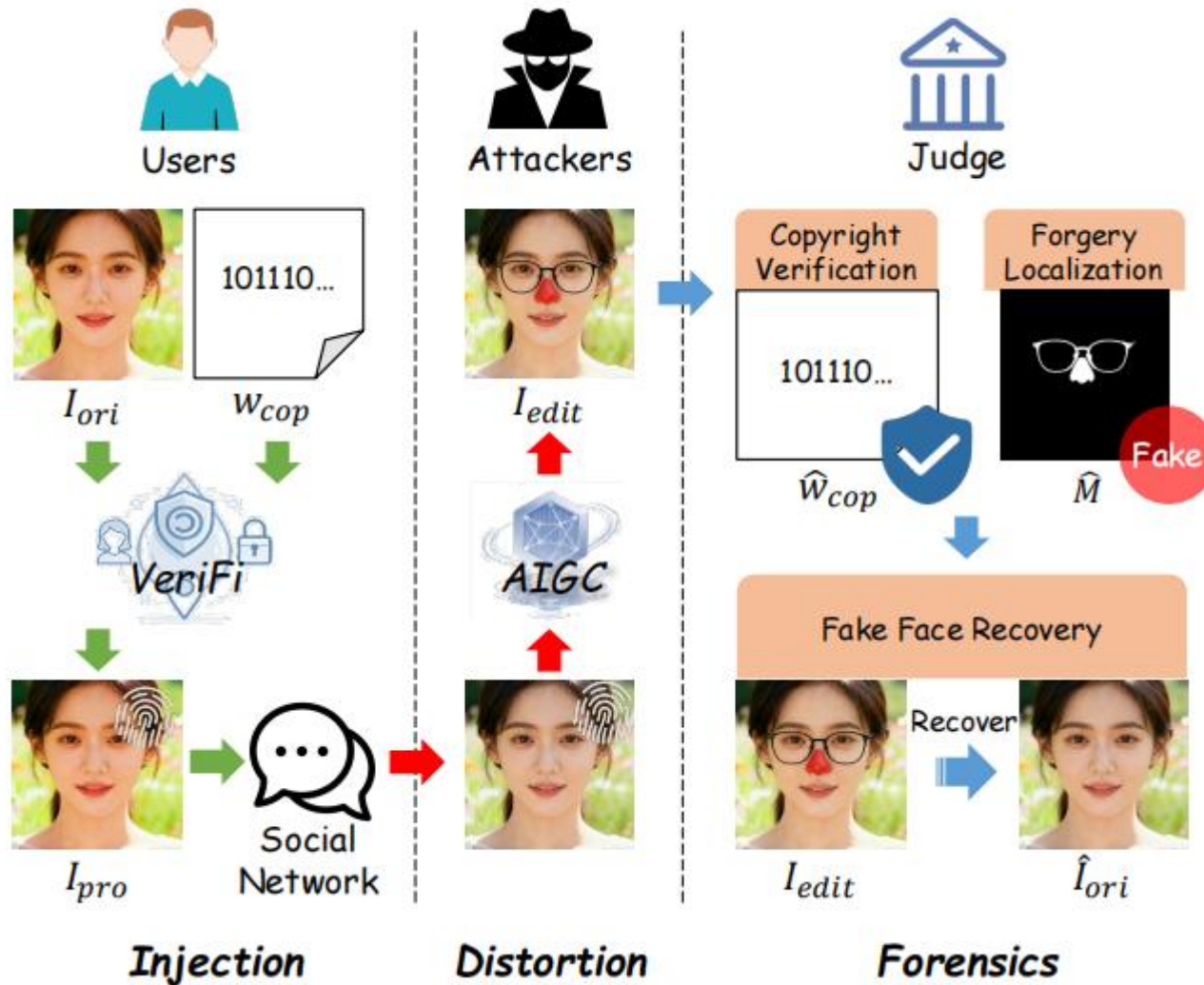


Figure 1: Overview of our VeriFi framework. Our versatile watermarking architecture simultaneously achieves robust copyright protection, precise forgery localization, and high fidelity fake face recovery, providing comprehensive forensic capabilities essential for real-world deepfake defense scenarios.

Proactive Deepfake Defense: Watermarking



Peipeng Yu, Jinfeng Xie, Chengfu Ou, et al. High-Fidelity Face Content Recovery via Tamper-Resilient Versatile Watermarking, <https://arxiv.org/abs/2603.23940v1>

Tamper-Resilient Versatile Watermarking Framework

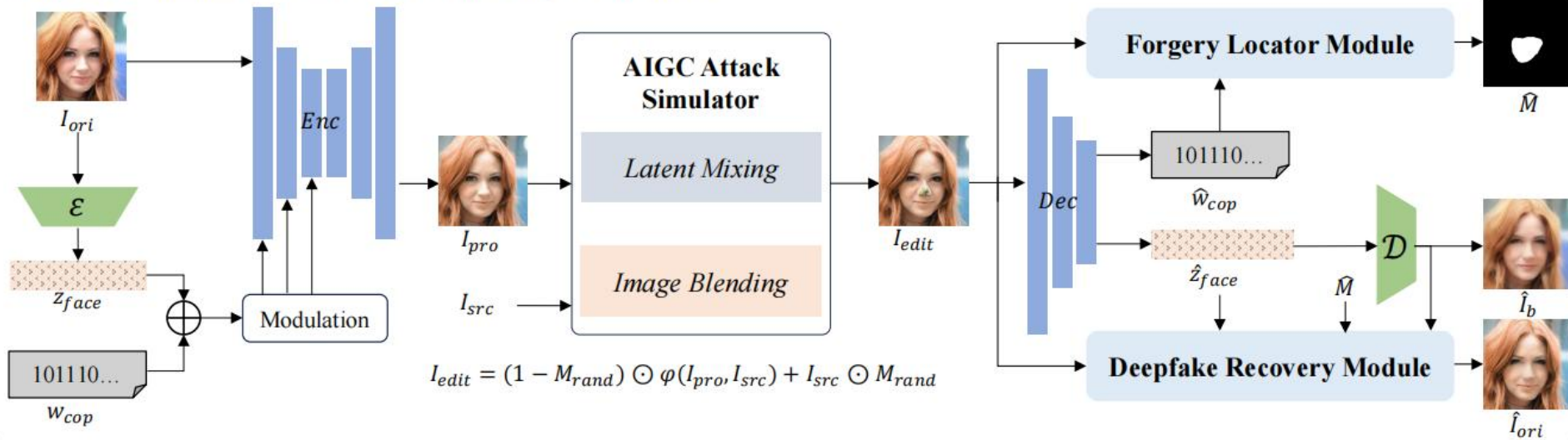
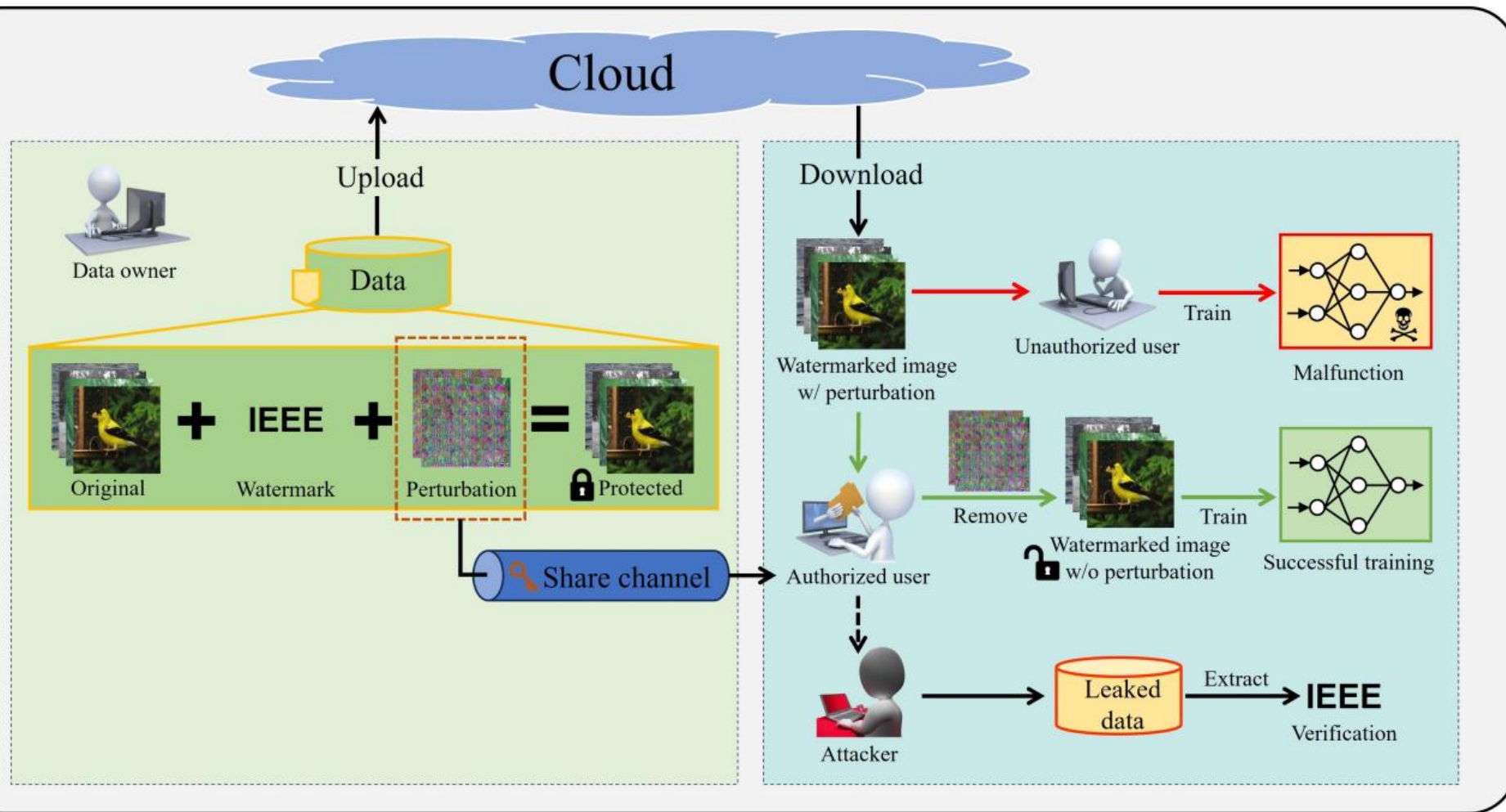
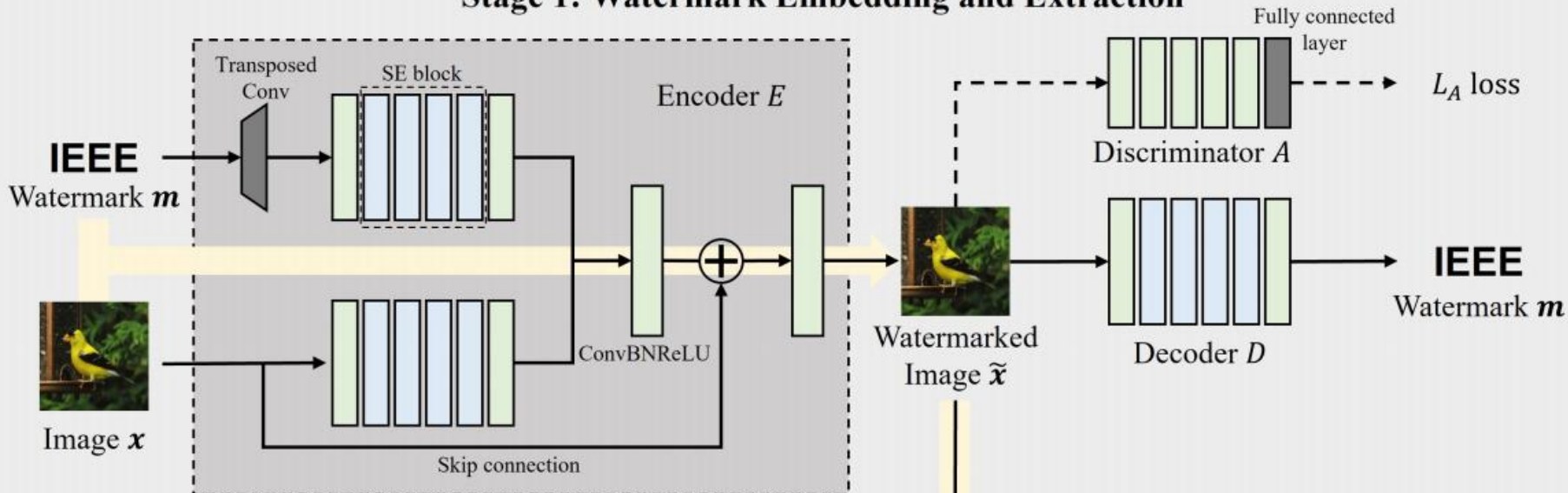


Figure 2: Overview of our VeriFi. (1) Unified Recovery-Copyright Watermark Embedder Enc inserts an ownership code w_{cop} and a compact facial signature z_{face} . (2) AIGC Attack Simulator performs Latent Mixing and Poisson blending to mimic realistic deepfake attacks during training. (3) Watermark Extractor Dec recovers $\hat{w}_{cop}$ and $\hat{z}_{face}$, produces a content proxy $\hat{I}_b$, and uses $\hat{w}_{cop}$ to guide the Forgery Locator to predict the manipulation map $\hat{M}$. (4) Watermark-Guided Deepfake Recovery Network uses a dual-stream Transformer with spatially gated cross-attention to fuse the edited image, $\hat{I}_b$, and $\hat{M}$ for selective restoration.

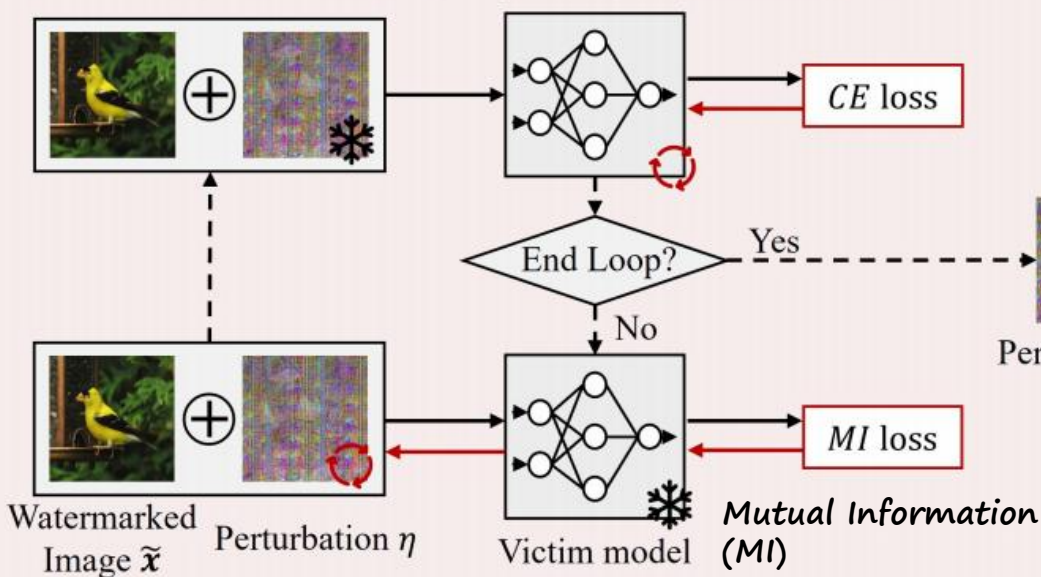


Application scenario involving data owner, authorized user, unauthorized user, and attacker.

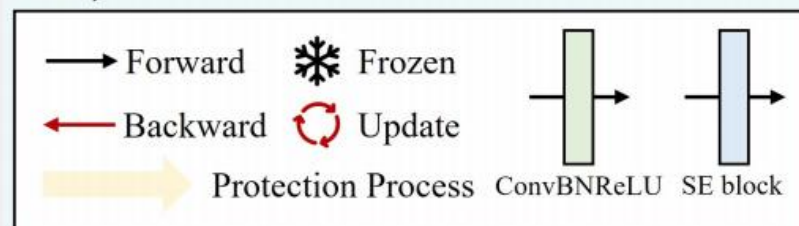
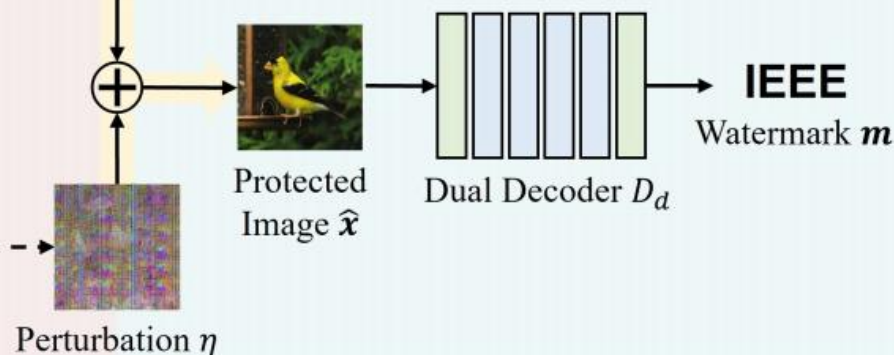
Stage 1: Watermark Embedding and Extraction



Stage 2: Poisoning Attack by Minimizing MI



Stage 3: Dual Watermark Extraction



Clean

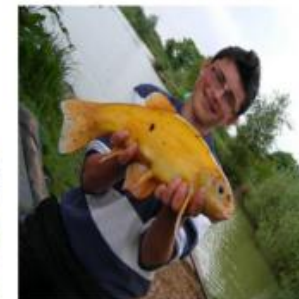
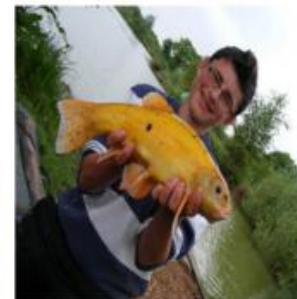
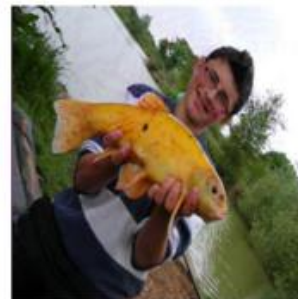
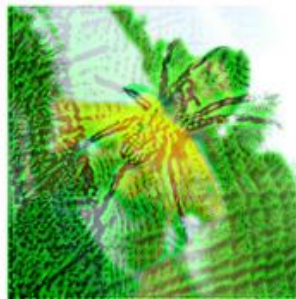
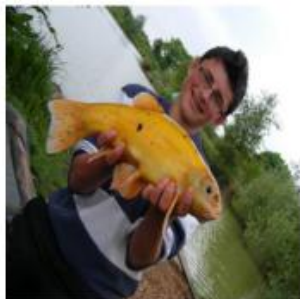
Adv-watermark
w/ perturbation

GEAA
w/ perturbation

GEAA
w/o perturbation

Ours
w/ perturbation

Ours
w/o perturbation



PSNR/SSIM

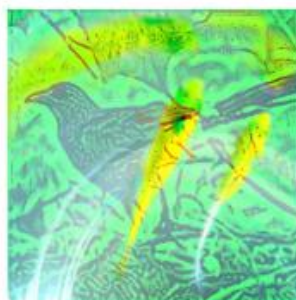
28.46dB/0.9733

13.98dB/0.6961

33.13dB/0.9477

41.03dB/0.9748

47.17dB/0.9960



PSNR/SSIM

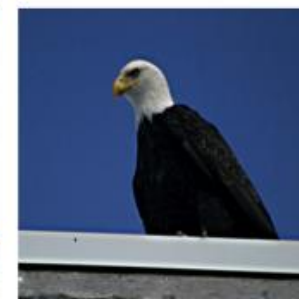
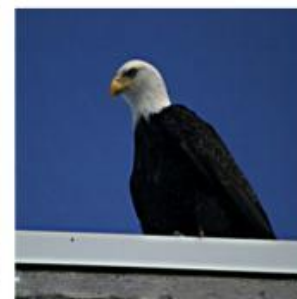
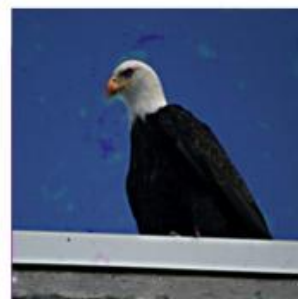
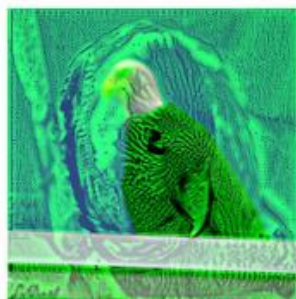
29.79dB/0.9715

13.98dB/0.7163

30.04dB/0.9194

42.02dB/0.9720

51.39dB/0.9966



PSNR/SSIM

26.69dB/0.9336

11.98dB/0.6623

34.07dB/0.9350

42.71dB/0.9644

48.35dB/0.9954



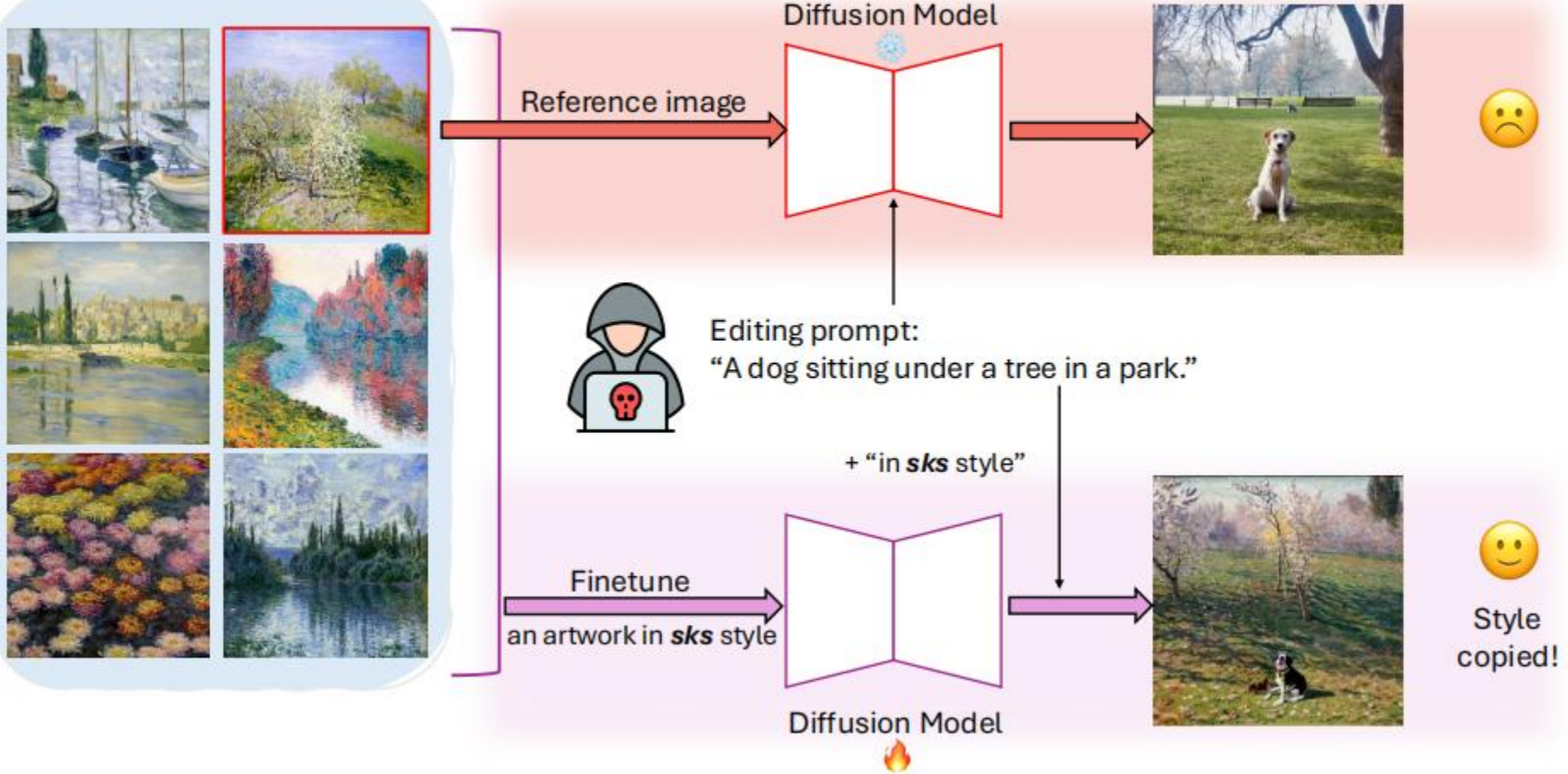
TABLE VI

GENERALIZABILITY COMPARISON OF UNLEARNABLE PERTURBATION. RESULTS MARKED WITH * REFER TO THE ORIGINAL PAPER.

Dataset	Method	ResNet18	VGG19	MobilenetV2	GoogLeNet	Dense121	Avg.
CIFAR10	Clean	94.49 $\pm$ 0.09	90.23 $\pm$ 0.36	84.86 $\pm$ 0.37	87.42 $\pm$ 0.26	94.46 $\pm$ 0.16	90.29
	Emax [14]	9.20 $\pm$ 0.63	20.29 $\pm$ 3.70	19.80 $\pm$ 0.65	28.97 $\pm$ 1.66	9.82 $\pm$ 0.65	17.62
	Emin [20]	16.51 $\pm$ 2.39	18.56 $\pm$ 0.81	16.94 $\pm$ 1.16	21.73 $\pm$ 0.75	15.37 $\pm$ 1.72	17.82
	Tattack [14]	11.70 $\pm$ 1.83	22.91 $\pm$ 7.52	17.53 $\pm$ 3.80	66.20 $\pm$ 16.46	11.28 $\pm$ 0.72	25.92
	Tattack(sample-wise)* [14]	23.79 $\pm$ 0.27	28.49 $\pm$ 0.65	20.53 $\pm$ 0.02	23.05 $\pm$ 0.16	20.90 $\pm$ 0.22	23.35
	Autoregressive [15]	25.98 $\pm$ 1.20	37.52 $\pm$ 0.93	17.88 $\pm$ 1.51	18.58 $\pm$ 1.73	18.12 $\pm$ 1.34	23.56
	Gradient alignment* [18]	68.15 $\pm$ 0.55	66.58 $\pm$ 0.58	66.71 $\pm$ 0.60	71.02 $\pm$ 0.15	70.12 $\pm$ 0.12	68.52
	Ours	10.26 $\pm$ 0.15	11.74 $\pm$ 0.23	11.27 $\pm$ 0.45	13.51 $\pm$ 0.35	10.45 $\pm$ 0.40	11.45
ImageNet-10	Clean	72.77 $\pm$ 0.93	67.60 $\pm$ 0.60	72.37 $\pm$ 1.02	74.47 $\pm$ 1.56	73.67 $\pm$ 1.47	72.18
	Emax [14]	26.37 $\pm$ 0.53	20.93 $\pm$ 0.36	24.20 $\pm$ 0.82	27.30 $\pm$ 0.82	35.90 $\pm$ 1.72	26.94
	Emin [20]	17.23 $\pm$ 0.37	10.73 $\pm$ 0.51	14.77 $\pm$ 0.99	22.33 $\pm$ 1.91	42.93 $\pm$ 2.93	21.60
	Tattack [14]	19.93 $\pm$ 2.89	21.07 $\pm$ 0.68	17.50 $\pm$ 2.20	23.97 $\pm$ 1.17	29.93 $\pm$ 2.23	22.48
	Ours	13.50 $\pm$ 0.46	20.50 $\pm$ 1.61	16.63 $\pm$ 2.05	17.60 $\pm$ 1.65	21.10 $\pm$ 2.51	17.87
ImageNet-100	Clean	71.39 $\pm$ 0.31	65.09 $\pm$ 0.35	76.80 $\pm$ 0.39	77.13 $\pm$ 0.07	73.50 $\pm$ 0.33	72.78
	Emax [14]	11.37 $\pm$ 0.62	4.19 $\pm$ 2.26	8.65 $\pm$ 0.13	12.15 $\pm$ 0.23	12.30 $\pm$ 0.60	9.73
	Emin [20]	9.09 $\pm$ 0.35	2.55 $\pm$ 2.17	2.65 $\pm$ 1.37	1.52 $\pm$ 0.17	24.98 $\pm$ 1.48	8.16
	Tattack [14]	8.45 $\pm$ 0.26	4.90 $\pm$ 0.28	6.61 $\pm$ 0.05	9.59 $\pm$ 0.27	10.75 $\pm$ 0.53	8.06
	Ours	8.75 $\pm$ 0.05	1.11 $\pm$ 0.05	6.62 $\pm$ 0.89	0.94 $\pm$ 0.33	14.39 $\pm$ 0.51	6.36

Qiuyu Tang, Joshua Krinsky, Aparna Bharati. StyleProtect: Safeguarding Artistic Identity in Finetuned Diffusion Models, CVPR 2026, 10759-10769

Claude Monet's artworks



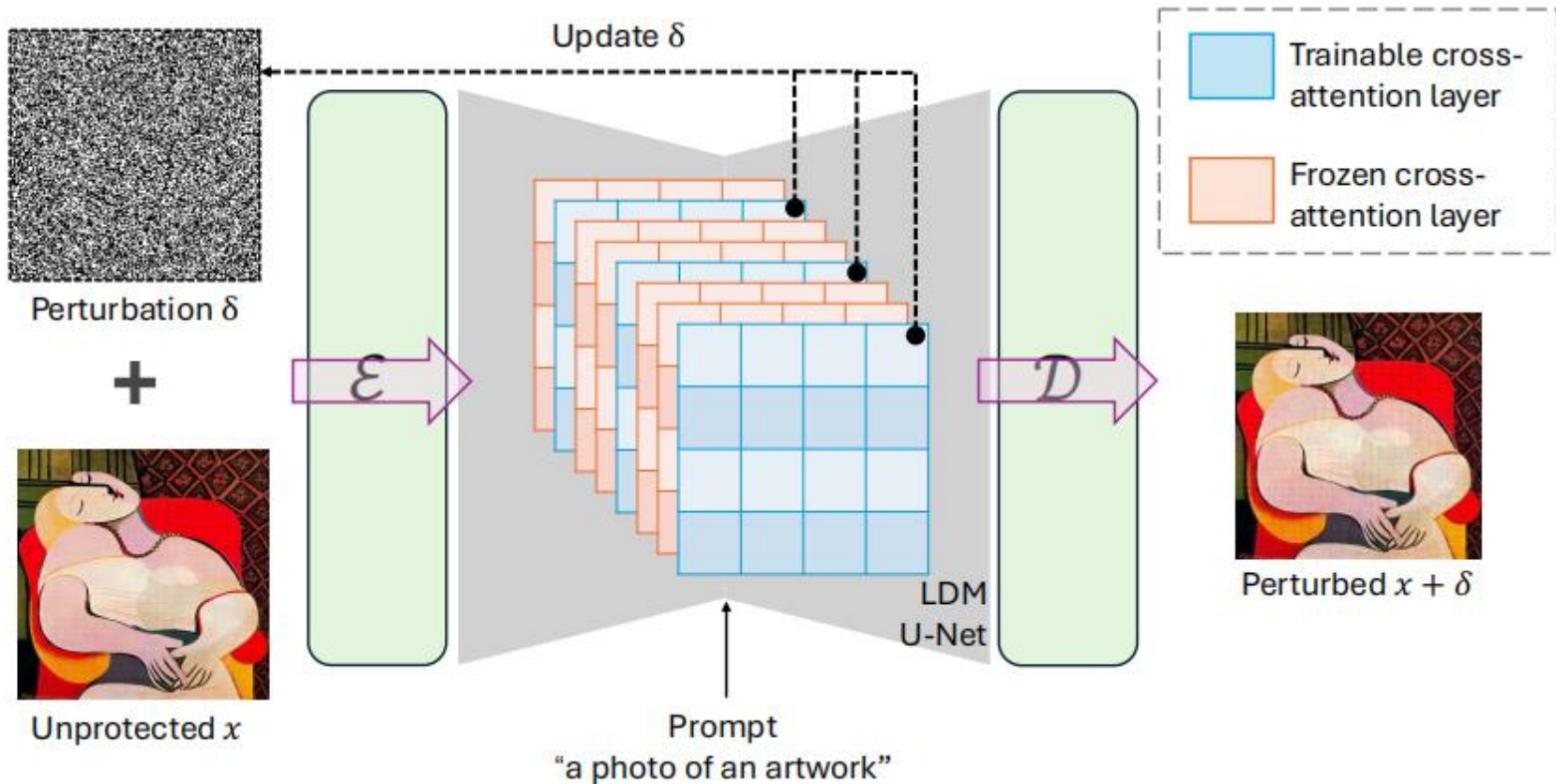


Figure 3. Overview of our pipeline for *StyleProtect*. The proposed method updates the perturbation while finetuning only style relevant cross-attention layers. At each step, the selected layers are trained to enhance style representation, but the perturbation gradually causes the model to lose the aforementioned capabilities.

Original Styles

Edits conditioned on
"A boy riding a bicycle on a sunny street"

Style1



Unprotected



StyleProtect



Style2



Qiuyu Tang, Joshua Krinsky, Aparna Bharati. **StyleProtect: Safeguarding Artistic Identity in Finetuned Diffusion Models**, CVPR 2026, 10759-10769

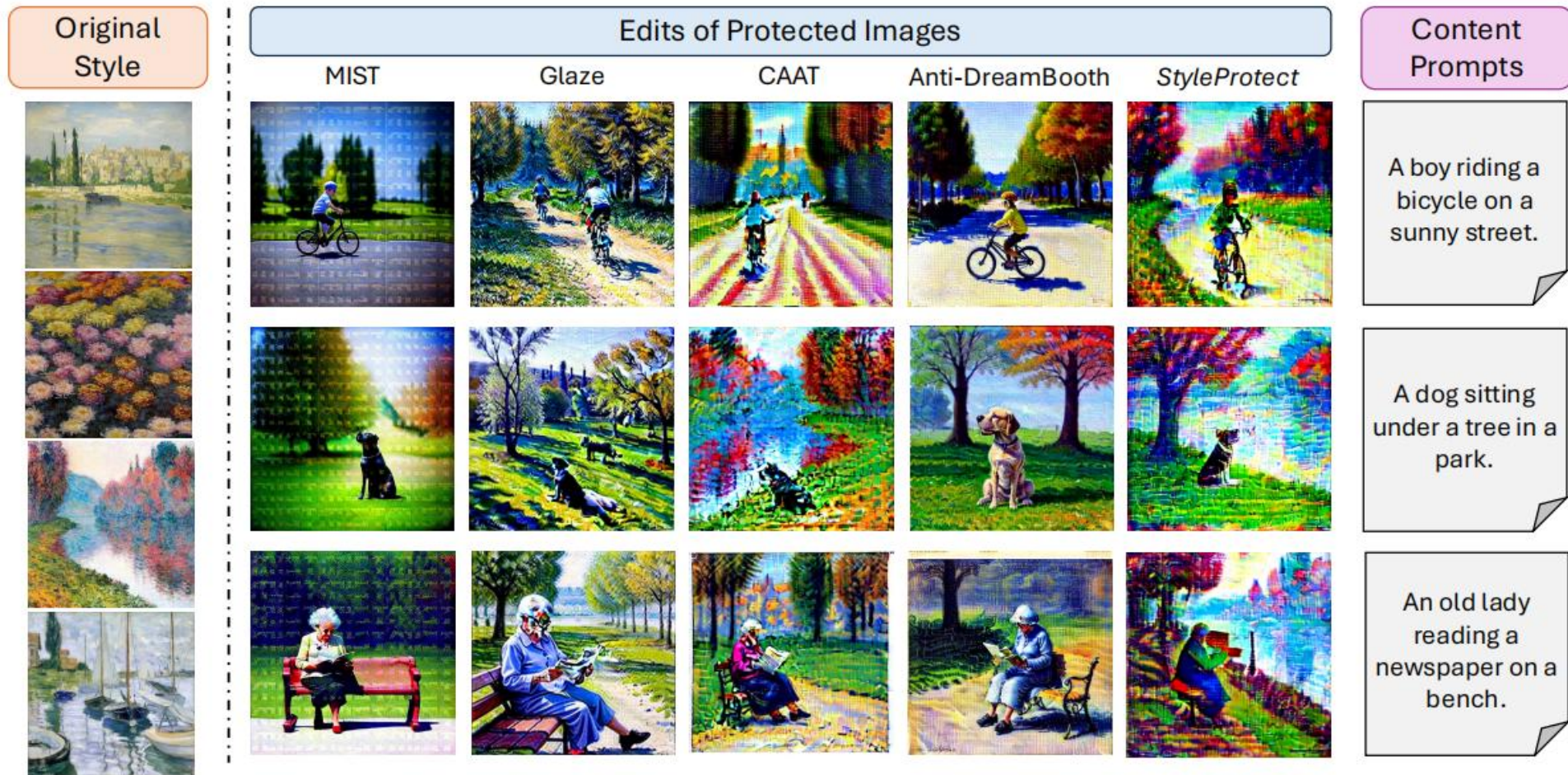


Figure 4. Edit attempts to replicate Monet’s style with DreamBooth under different protections. *StyleProtect* shows the strongest disruption.

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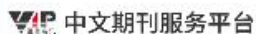
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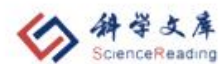
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7	Nature	期刊
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10	OPTICA光学数据库	期刊, 会议论文
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二者要配套

如果说生产实习报告是一棵枝繁叶茂的**大树**，课程设计报告则是其中**一根枝杈**或**一枚树叶**的细致展现；

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题目：数字图像水印

- 一、数字水印的基本概念
- 二、数字图像水印的分类、功用和评价
- 三、数字图像水印技术的发展历程
 - (1) 传统的数字图像水印技术
 - I. 空间域水印
 - II. 变换域（如DFT, DCT, DWT）水印
 - (2) 深度学习下的数字图像水印技术
 - I. 传统水印算法的深度学习改进
 - II. 基于生成模型的数字图像水印
- 四、对数字图像水印的攻击及防御
- 五、未来展望

可参阅相关主题的文献综述、survey或 review

题目：一种用于XXXX的数字图像水印（算法）

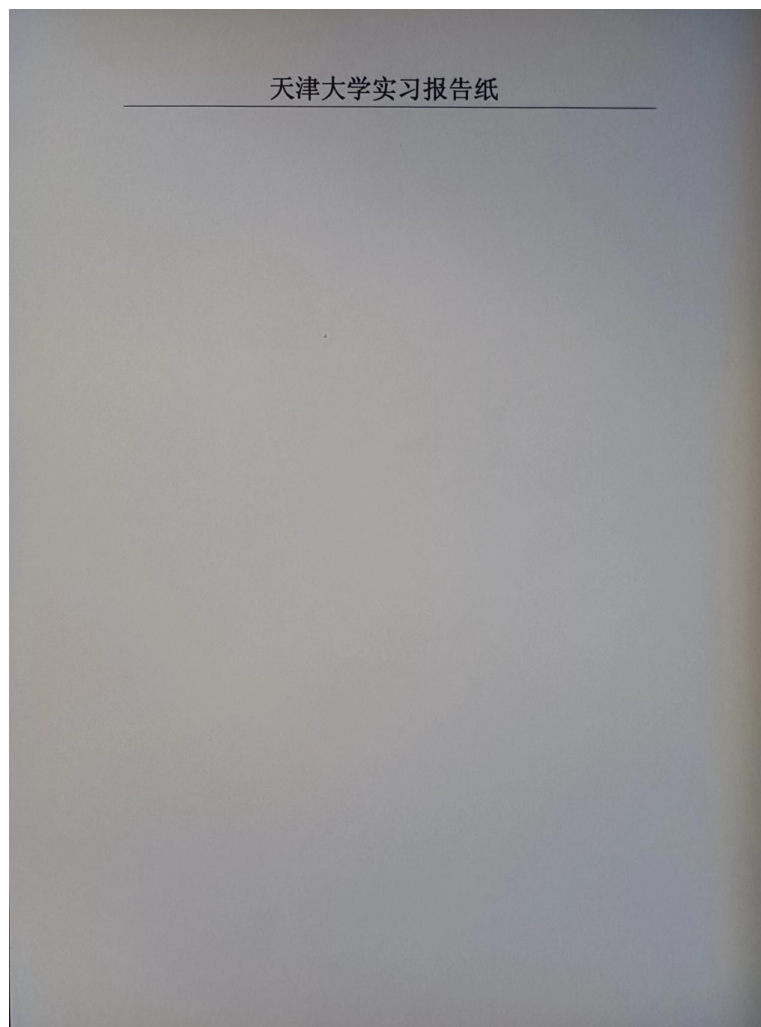
- 一、算法总体框架
- 二、各模块实施细节
- 三、实验结果和分析
- 四、个人心得和体会

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学号:120201801

学院:机械工程学院

专业:机械设计及其自动化

年级:2024级

成绩:

落球法测定液体的粘度

一、实验目的

1. 通过观察小球在液体中的运动过程,了解液体的内摩擦现象
2. 掌握用落球法测定液体的粘度的具体内容

二、实验仪器与设备

粘度测量装置、游标卡尺、停表、温度计、密度计、米尺、移液显微镜等

三、实验原理

如图6-1所示,小球在液体中下落时,受到了3个竖直方向的力,即浮力 PVg (P 是液体的密度, V 是小球的体积),小球的重力 PVg (P 是小球的密度)和粘性力 F (其方向与小球运动方向相反)。在无限广延的液体中,如果液体粘度较大,小球的直径较小,下落运动过程中不产生旋涡,则根据斯托克斯定律 (G. G. Stokes, 1819-1903) 定律,小球所受的粘性力

$$F = 3\pi\eta vd \quad (6-1)$$

式中 η 是液体的粘度, d 是小球的直径, v 是小球的速度。

开始时,小球下落速度较小,粘性力也较小,因而小球做加速运动。随着小球速度的增加,粘性力变大,最后,上述三力达到平衡,即

$$PVg = 3\pi\eta vd + PVg \quad (6-2)$$

于是,小球开始做匀速直线运动 (此时运动速度称为终极速度)。将小球体积 $V = \frac{4}{3}\pi d^3/6$ 代入上式,整理后可得液体的粘度

$$\eta = \frac{P - P_0 g d^2}{18v} \quad (6-3)$$

实验时,将测得液体盛在直径为 D 的量筒中,因而小球在下落过程中不满足液体无限广延的条件,这时实际测量的速度 v_0 和理想条件下的速度 v 存在如下关系:

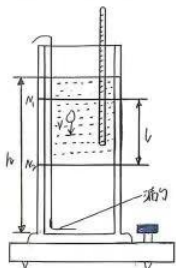


图1 粘度测定

$$v = v_0 (1 + 2.4 \frac{d}{D}) (1 + 1.6 \frac{d}{D}) \quad (6-4)$$

式中 η 是液体的粘度,当 $h \gg d$ 时,有

$$\eta = \frac{P - P_0 g d^2}{18 (1 + 2.4 \frac{d}{D})} \quad (6-5)$$

式中 L 是图6-1中 M_1, M_2 的距离,由式(6-5)可以看出,只要测得 P, P_0, d, D 和 t 量,即可求出液体的粘度 η 。

当小球的直径较大,液体不是无限广延,液体的粘度较小时,小球在液体中的终极速度 v 未达到较大值。可用奥陶斯-果尔斯基公式描述该运动状态对斯托克斯定律的影响,即

$$F = 3\pi\eta vd (1 + \frac{3}{8} Re - \frac{19}{1280} Re^2 + \dots) \quad (6-6)$$

其中, Re 为雷诺数,是表征液体运动状态的量为 1 的无量纲量,并且

$$Re = vd\rho/\eta \quad (6-7)$$

当 $Re < 0.1$ 时,可认为式(6-6)、式(6-7)成立;当 $0.1 < Re < 1$ 时,应考虑式(6-6)中一级修正项的影响;当 $Re > 1$ 时,还需考虑式(6-6)中的高次修正项。

为保证小球在液体中下落时不产生旋涡,其终极速度不能太大,选用的小球直径应适当小一些。本实验由于采用直径为 1mm 的小钢球,在我们测量的温度范围内雷诺数远小于 0.1 ,故可直接用式(6-5)计算粘度。

四、实验步骤

1. 用移液显微镜测出小球直径 d ,进行6次测量,计算平均值及测量不确定度,将实验数据填入表格中
2. 测量量筒上下两标记的高度差 L ,记录数据,同时使用游标卡尺测量量筒内径 D ,记录数据
3. 记录实验装置中的小球密度 P ,并使用密度计测量液体密度 P_0 ,记录数据
4. 实验开始前测一次温度,并取两次平均值作为实验的液体温度

教师签字:

年 月 日

报告中的程序流程图和实验结果图、表等可以
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